

# The Porous Decoupling: Vertical Rivalry and Swing-State Leakage

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## Abstract

After 2018, the United States moved from subsidy- and public-goods-based engagement toward active coercion of China—Entity Lists, export controls, and market-access weaponization—while customs trade statistics nevertheless registered a sharp decoupling. We argue that both the policy turn and the gap between customs and value-added measures of decoupling are equilibrium responses to the same primitive: bottleneck complementarity between Chinese and non-Chinese intermediates in critical sectors. We model the US–swing-state alignment bargain as a three-stage game over a triangular supply chain in which the downstream Hegemon must either subsidize or coerce a swing state that depends on the upstream Rival for essential inputs. The model yields two main predictions. First, a *Regime Switch*: the Hegemon’s optimal policy splits at a sharp threshold in the Rival’s productivity, with coercion intensity rising continuously from zero only beyond that threshold. Second, a *Paradox of Competitiveness*: more productive Rivals are not courted with larger subsidies; they are coerced more aggressively, because voluntary alignment becomes fiscally prohibitive precisely when the Rival is most efficient. US Entity-List enforcement carries the regime-switch fingerprint: fifteen years of constant-slope extra-bilateral coercion against third countries, on top of which a sharp 2018 China-bilateral kink emerges in exactly the form the model predicts. An aggregate-panel dose–response loads in the direction the paradox predicts. A multi-sector extension corroborates the same primitive in cross-sectional value-added data. Customs-based metrics therefore systematically overstate real decoupling in exactly the sectors where decoupling is most sought—by roughly \$22 billion per year of Chinese content reaching the US through swing-state intermediaries.

**Keywords:** Geoeconomics, Vertical Supply Chains, Coercion, US-China Rivalry, Entity List.

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# 1 Introduction

Between 2018 and 2024, the United States imposed Section-301 tariffs averaging roughly 25 percentage points on Chinese imports in critical sectors—semiconductors, precision machinery, batteries, and electric-vehicle components—and, alongside the tariffs, opened a new policy front of active coercion: Entity Lists, export controls, and market-access weaponization. The post-war US toolkit toward strategically important partners had been built around public goods and subsidies; after 2018 it added a coercive register that targeted both China directly and the third-country machinery China relies on. At the same time, two measures of decoupling diverged. Direct imports from China in critical sectors fell sharply on customs data, while imports from a small set of swing states (chiefly Vietnam, Mexico, and Malaysia) rose by a comparable amount. Tracing the *origin* of the value embodied in those swing-state exports—their *foreign value-added* (FVA) share—tells a different story: Chinese content has continued to reach the US indirectly, settling near 6% of swing-state critical-sector exports by 2020–2022, or roughly \$22 billion per year.<sup>1</sup> On customs data the US is decoupling; on value-added data it is recoupling through intermediaries. We ask why the US instrument set shifted, why the two measures of decoupling disagree, and answer with a single model that connects both phenomena to the production-side primitive that drives them.

We argue that the discrepancy is not a measurement artifact or a transitional friction. It is the steady-state outcome of a bargain between the US (the downstream *Hegemon* of the supply chain) and a *swing state*. The mechanism rests on complementarity: in critical sectors, non-Chinese intermediates are poor substitutes for Chinese ones. Boehm et al. (2019) and Atalay (2017) document this empirically. A swing state cannot cheaply rebalance its input mix no matter how generously the US pays to align. The shortfall in compliance is therefore absorbed by continued sourcing of Chinese inputs, which are then re-exported in finished goods. The same complementarity yields two main-text testable predictions: (a) a *Regime*

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<sup>1</sup>Derivation from fitted FVA shares applied to 2022 swing-state-to-US gross exports is given in Appendix C.4.

*Switch* (Proposition 1) whose empirical fingerprint is a kink in coercive enforcement at a critical productivity threshold; and (b) a *Paradox of Competitiveness* (Proposition 2) whose empirical counterpart is a dose–response in coercive intensity to upstream-productivity gains. A multi-sector extension corroborates the same primitive with a cross-sectional moment in value-added data.

We test these predictions on the policy data the model speaks to most directly: US Entity-List enforcement, 2003–2024 (Section 4.1). The data carry the regime-switch fingerprint cleanly. Listings of foreign entities outside China grow at a constant rate over the full sample—a fifteen-year extra-bilateral perimeter interpretable as the enforcement primitives that fix the Hegemon’s fiscal and capacity costs of coercion. Listings of China-based entities grow at the same constant rate until 2018 and then exhibit a sharp kink, with the post-2018 China-bilateral slope an order of magnitude above the pre-2018 trend. The two-rate structure—constant extra-bilateral perimeter beneath, kinked China-bilateral overlay above—is the direct shape predicted by Proposition 1. A complementary aggregate-panel dose–response of US imports from China on the lagged China-listings stock interacted with sector bucket loads in the direction Proposition 2 predicts; we disclose that the same dose–response is null at HS6 resolution (Section 4.2) and read it as suggestive rather than identifying.

To explain this gap we model the US–swing-state relationship as a three-stage game<sup>2</sup> over a triangular (Upstream Rival, Swing State, Downstream Hegemon) supply chain. The triangular structure isolates the two principals’ asymmetric sources of market power: China’s leverage is upstream (low wages, high manufacturing productivity, and a near-monopoly on rare-earth elements), while the United States’ leverage is downstream (dollar hegemony and the world’s largest final-goods market). Swing states caught between rely on China for upstream critical inputs while selling finished goods into the US market; chief among these

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<sup>2</sup>The game is static within the policy period: we take the Rival’s productivity  $\theta$  as exogenous and abstract from dynamic reputation, multi-period learning, and endogenous accumulation of  $\theta$ . Dynamic and endogenous-technology extensions are left for future work.

are Vietnam, Malaysia, and Mexico. We term this configuration the *Vertical Trap*. The Hegemon’s strategic problem is that eliminating the kill-switch risk requires the Swing State to abandon its most efficient input supplier. This deviation is costly, and the Hegemon must either subsidize or coerce.

The model delivers two main results and a corroborating cross-sectional implication. The first is a *Regime Switch* (Proposition 1). The Hegemon’s optimal policy splits into two distinct regimes: a *Generous Empire*, in which alignment is purchased through subsidies alone, and a *Coercive Hegemon*, in which subsidies are supplemented by active coercion. The boundary between them is a kink, not a jump — coercion intensity<sup>3</sup> is exactly zero in the first regime and rises continuously from zero in the second.

The second result identifies what determines that boundary — and the surprising direction in which it moves: the *Paradox of Competitiveness* (Proposition 2). The boundary is set by the Rival’s productivity: the subsidy needed for voluntary alignment scales with the Rival’s productivity, and once it exceeds what the Hegemon will fund,<sup>4</sup> coercion takes over. The paradox is the sign: more productive Rivals are not courted with larger subsidies; they are coerced more aggressively.

A multi-sector extension in Appendix C develops a complementary *Leakage Asymmetry* (Propositions 6 and 7) generated by the same complementarity primitive. It predicts the cross-sectional pattern noted in the introduction—the swing-state premium in Chinese value-added widens in industries facing higher US tariffs on direct Chinese imports—and identifies the bottleneck primitive directly from value-added data, without requiring observation of a regime change. The cross-sectional moment corroborates the regime-switch evidence rather than substituting for it.

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<sup>3</sup>Formally, the Hegemon’s coercion intensity is denoted  $\lambda$ ; the security externality is  $\lambda x$ , where  $x \in [0, 1]$  is the Swing State’s alignment level (defined in Section 3). We treat  $\lambda x$  as reduced-form rather than deriving it from a probabilistic “kill-switch” microfoundation.

<sup>4</sup>The fiscal capacity  $\bar{T}$  that bounds “what the Hegemon will fund” and the security-policy cost  $\gamma$  are microfounded in the appendix but taken as primitives in the main model.

**Related literature and contribution.** The paper isolates a single new mechanism—the endogenous corner solution that emerges when bottleneck complementarity meets a binding fiscal constraint—and positions itself against three strands of work.

*Geoeconomics and the post-2018 turn.* The closest theoretical neighbors are Clayton et al. (2024), who formalize a Hegemon’s Dilemma between coercion and economic integration in general equilibrium, and Clayton et al. (2026), who extend this to a three-player setting with a preference-parameter threshold for switching between inducement and coercion. We share their endogenous-cost view of coercion but pin the regime-switch threshold to a production-structure primitive ( $\sigma_c < 1$ ), not preferences. Most models of trade war treat tariffs as distributional conflict (Grossman and Helpman, 1994) or bargaining failure (Bagwell and Staiger, 1999; Handley and Limão, 2017), and the political-backlash literature (Autor et al., 2020; Rodrik, 2018; Farrell and Newman, 2019) documents the symptoms of policy radicalization without microfounding when the Hegemon switches. We locate the post-2018 hawkish turn as a corner solution that emerges when the subsidy-based interior solution becomes fiscally unbounded.

*Value-added trade measurement.* The Koopman–Wang–Wei decomposition (Koopman et al., 2014; Johnson and Noguera, 2012) provides the accounting identity for tracing embodied value-added through global supply chains. The closest empirical neighbors are Mattoo et al. (2023), who attribute roughly one-third of the US–China bilateral decline to rerouting through intermediaries, and Schulze and Xin (2025), who distinguish trade *reallocation* (domestic substitution) from *rerouting* (transshipment) in six Asian connector countries. Relative to this literature, the FVA cross-partial documented above (and developed in Appendix C) enters as a corroborating cross-sectional moment of a structural primitive—bottleneck complementarity—rather than as a stand-alone measurement exercise.

*Production bottlenecks and contracting.* We model the US–China rivalry as a Common Agency Game (Bernheim and Whinston, 1986; Dixit, 1996) between two principals bidding for a Swing-State agent. The Menu Auction benchmark delivers truthful equilibria under

linear transfers; we introduce a technological bottleneck as a non-linear friction. When one principal controls an essential complement, the Rival cannot compete via linear transfers and exclusion becomes the rational second-best, linking our analysis to the foreclosure literature (Rey and Tirole, 2007; Segal, 1999) applied to sovereign states. The bottleneck assumption itself builds on network-IO work showing that low-substitutability nodes amplify shocks (Acemoglu et al., 2012; Baqaee and Farhi, 2024) and on the vertical GVC tradition (Antràs, 2016; Yi, 2003); relative to horizontal strategic-trade-policy (Brander and Spencer, 1985) and the China-shock literature (Fajgelbaum et al., 2020; Alfaro and Chor, 2023), we add the production-side mechanism that makes coercion optimal beyond a critical Rival-efficiency threshold (Propositions 6 and 7).

## 2 Empirical Patterns

Two stylized patterns motivate the model. First, swing states have become structurally more dependent on Chinese inputs over 2000–2024, even as the US has tried to decouple—the *Vertical Trap*. Second, the post-2018 US tariffs cleanly cut direct US imports from China but had no comparable effect on swing-state imports—the *Decoupling Divergence*. Both patterns are constructed from UN Comtrade aggregate non-US, non-China bilateral flows. The complementary evidence on US coercive instruments is deferred to Section 4.1, where it sits next to the regression tests it motivates.

**The Vertical Trap.** Figure 1 plots the swing-state aggregate’s two principal trade shares. The red line—imports sourced from China (upstream dependence)—more than triples, rising from 6% in 2000 to 22% in 2024. The blue line—exports destined for the US (downstream concentration), constructed via mirror data<sup>5</sup>—is essentially flat at 12–13%. Upstream de-

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<sup>5</sup>Comtrade pulls flow=M (imports) only. The blue line uses USA’s reported imports from each swing state, normalized by the swing state’s total imports from the world (a balance proxy for total trade).

pendence rises threefold; downstream concentration does not move.<sup>6</sup>

Two natural alternatives to the structural-integration reading both fail.

*Aggregation across high- and low-stakes sectors.* If the rise were carried by easily substitutable commodity sectors (textiles, basic chemicals, furniture), restricting attention to chokepoint sectors should attenuate it. It does the opposite. Panel (b) restricts to critical and strategic sectors only;<sup>7</sup> the chokepoint subaggregate *sharpens*, with the upstream-downstream crossover occurring earlier ( $\sim 2008$ ) and the post-2018 acceleration wider. The pattern is concentrated where substitution is hardest, not diluted by it.

*Selection in the swing-state set.* If the rise were driven by one or two outliers, dropping the largest contributors should pull the aggregate down sharply. It does not. Appendix D.9 (Figure 14) shows that excluding Vietnam, Mexico, and Korea—the three largest post-2018 contributors—leaves the 2024 aggregate within  $\approx 1$  percentage point of the full series (21.96%  $\rightarrow$  21.11%), and the trajectory is preserved.

The pattern is therefore broadly distributed and concentrated where substitution is hardest—consistent with upstream complementarity in critical inputs. Swing states remain anchored to what observers have called the “Red Supply Chain” despite a decade of geopolitical pressure.<sup>8</sup>

**The Decoupling Divergence.** The structural integration documented above limits the efficacy of US bilateral tariffs. Figure 2 compares US versus swing-state reliance on China along two channels.

Panel (a) plots the gross-trade pattern. After the 2018 Section-301 tariffs (dotted line), US

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<sup>6</sup>Inclusion rule: 17 non-US, non-China economies with non-trivial Chinese-input or US-export ties (KOR, TWN, VNM, MEX, NLD, SGP, MYS, THA, IND, JPN, DEU, IDN, HUN, CZE, POL, PHL, BRA); held fixed across all panels and the regressions in Section 4.

<sup>7</sup>Critical and strategic sectors follow the OECD ICIO ISIC4 taxonomy mapped to the bottleneck buckets in `empirical/processed/icio_sector_bucket.csv`: semiconductors and electronic equipment (D26), mining of metal ores (D07T08), batteries and electrical equipment (D27), machinery (D28), motor vehicles (D29), pharmaceuticals (D21), and aerospace (D30); commodity sectors (textiles, food, basic chemicals) are excluded. See Appendix D.1 for the full mapping.

<sup>8</sup>The same pattern appears in IMF Direction of Trade Statistics (DOTS) bilateral trade shares and in the trade-reallocation evidence of International Monetary Fund (2023, Chapter 4).

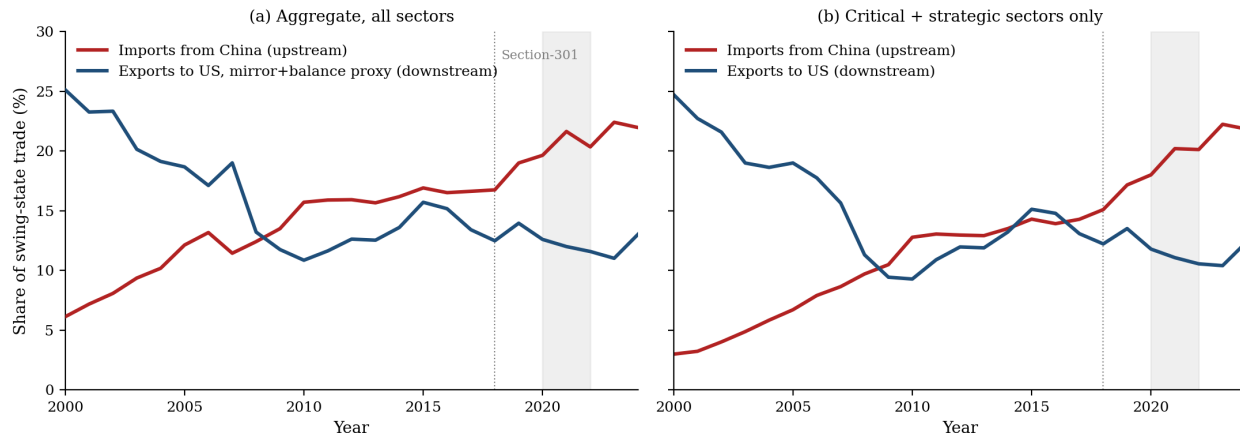


Figure 1: **The Vertical Trap.** Aggregated trade shares of swing states (non-US/non-China). *Panel (a)*: all sectors. Red is the swing-set’s aggregate share of imports sourced from China; blue is the share of US-bound exports (mirror+balance proxy). *Panel (b)*: critical and strategic sectors only (D26 + D07T08 + D27 + D28 + D29 + D21 + D30; see Appendix D.1). Grey shading marks 2020–2022 (COVID inventory rebuild). Dotted vertical: Section-301 (2018). Per-country selection robustness is shown in Appendix D.9 (Figure 14). Source: UN Comtrade, 2000–2024.

direct reliance on China (solid blue) drops by about 5 percentage points (18.2%  $\rightarrow$  13.4%); the swing-state aggregate (solid red) keeps climbing. If the swing-state rise were itself a tariff response—supply diversion induced by Section-301—the post-2018 series should deviate *above* its pre-2018 trajectory. The dashed red line is the OLS extrapolation of 2000–2017 swing-state values; the actual post-2018 series tracks it within one percentage point. The divergence is therefore not a treatment-induced over-shoot in swing states; it is a treatment-induced collapse of the US-direct line, with the swing-state channel essentially unaffected by the tariff.

Panel (b) addresses the natural rebuttal that swing-state imports from China might simply be Chinese goods relabeled and transshipped to evade tariffs. We compute the share of US imports that is Chinese value-added arriving via swing-state intermediaries, using the OECD TiVA Leontief inverse.<sup>9</sup> If the gross-trade rise reflected pure relabeling, real Chinese

<sup>9</sup>For each swing state  $s$  and year  $t$ , the indirect channel is  $\text{share}_t^{\text{ind}} = \sum_s \theta_{s,t}^{\text{FVA-CHN}} \cdot \omega_{s,t}^{\text{US imp}}$ , where  $\theta_{s,t}^{\text{FVA-CHN}}$  is the Chinese-VA fraction in  $s$ ’s gross exports to the US (OECD TiVA Leontief decomposition) and  $\omega_{s,t}^{\text{US imp}}$  is  $s$ ’s share in US total imports. The TiVA panel ends 2022. An equivalent ADB MRIO decomposition (2017–2024) restricted to the ten Asia-Pacific swing states yields the same direction at smaller magnitude (0.3–0.5pp), the level gap reflecting ADB’s narrower country coverage (no European swing states).

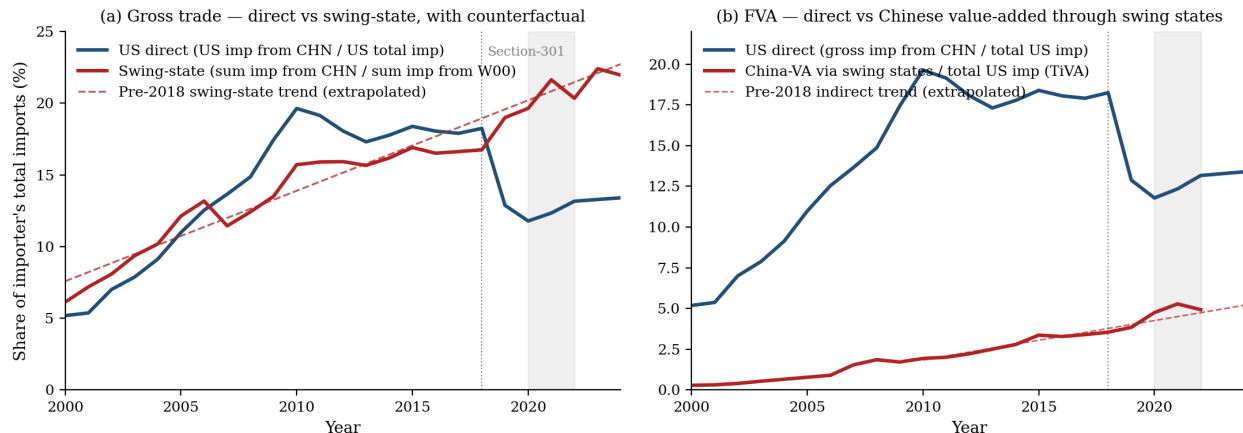


Figure 2: **The Decoupling Divergence.** *Panel (a):* gross trade. Solid blue is US direct China-import share; solid red is the swing-state aggregate. Dashed red is the pre-2018 OLS extrapolation of the swing-state series. *Panel (b):* foreign value-added decomposition. Solid blue is US direct (gross, repeated for reference). Solid red is the share of US imports that is Chinese VA arriving via swing-state intermediaries (TiVA Leontief, 2000–2022). Dashed red is the pre-2018 OLS extrapolation of the indirect-channel series. The post-2018 divergence between solid blue (falling 5pp) and solid red (rising 1.4pp) survives the FVA reconstruction, ruling out pure relabeling. Grey shading: 2020–2022 (COVID). Source: UN Comtrade + OECD TiVA.

content in US imports would be shrinking—only the country-of-shipment label would have moved to the swing states—and the FVA reconstruction would mirror the falling direct line. Instead, the indirect channel *rises* from 3.6% to 4.9% over 2018–2022 while the direct channel falls. The divergence in gross trade survives the value-added decomposition: consistent with real complementarity, inconsistent with pure rerouting.

These two patterns motivate the structural primitive that Section 3 formalizes: bottleneck complementarity between Chinese and non-Chinese intermediates in critical sectors, generating both the upstream dependence (the Vertical Trap) and the failure of bilateral coercion to dislodge it (the Decoupling Divergence). Section 4 returns to the data after the model with descriptive evidence on the US coercive policy environment (Section 4.1) and a complementary dose–response test of the paradox mechanism (Section 4.2). Cross-sectional leakage tests of the multi-sector extension are reported in Appendix C.

See Appendix D.1 for construction details and Appendix D.9 (Figure 15) for the cross-validation panel.

### 3 Model

We formalize the bottleneck primitive that delivers both patterns of Section 2: upstream complementarity that resists substitution (the Vertical Trap) and the failure of bilateral coercion to dislodge it (the Decoupling Divergence). Section 3.1 fixes the environment. Sections 3.2–3.4 derive each player’s best response, solving the three-stage game by backward induction; the Stackelberg leadership Hegemon C  $\rightarrow$  Hegemon U  $\rightarrow$  Swing State (Assumption 1) makes this the natural method, since the Swing State’s Stage-3 problem admits a closed-form best response that the upstream stages then take as given. Section 3.5 establishes the *Paradox of Competitiveness*: an exogenous rise in upstream efficiency triggers downstream coercion in equilibrium. Section 3.6 quantifies the resulting welfare loss. Section 3.7 closes with extensions that do not alter the core results and points to a multi-sector extension developed in Appendix C, which generates a complementary cross-sectional moment (the leakage asymmetry) tested on ICIO/MRIO value-added data there. All proofs are deferred to Appendix B; microfoundations of  $R(x)$ ,  $\bar{T}$ , and  $\gamma$  are in Appendix A.

#### 3.1 The Model Environment

We model a global economy characterized by Vertical Geoeconomic Rivalry. Unlike standard trade models that depict competition between symmetric nations, we treat the global order as a vertical supply chain in which power derives from position: control over upstream inputs versus control over downstream market access.

##### 3.1.1 Agents, Timing, and Commitment

**Agents.** The economy consists of three strategic players arranged hierarchically. First, **Hegemon C (Upstream)**—the *Rival* (e.g., China)—acts as a monopolist supplier of a critical intermediate input, with the input price  $w_C$  as its strategic instrument. Its underlying technological productivity is  $\theta > 0$  (an exogenous primitive of the Stage-1 problem); the export price  $w_C$  decomposes into a marginal cost  $\bar{w}/\theta$  and an endogenous strategic markup

$m$ , formalized in Section 3.4. Second, **Hegemon U (Downstream)**—simply the *Hegemon* in the main text (e.g., the USA)—acts as a monopsonist buyer of final goods and provider of essential intellectual property; its strategic instruments are the market access schedule  $R(x)$  (a contingent contract on the realized alignment, discussed in detail below) and the strategic doctrine  $\lambda$ . Finally, **The Swing State (Midstream)** is a representative firm or sector  $k$  (e.g., South Korea) that transforms intermediate inputs into final goods. It chooses the alignment  $x \in [0, 1]$ , which represents the degree of supply chain decoupling.

**Timing.** The interaction unfolds in three stages, assuming a Stackelberg leadership structure that reflects the rigidity of global supply chains. In **Stage 1**, Hegemon C sets the export price of critical inputs  $w_C$  to maximize a weighted sum of commercial profits and geopolitical influence. In **Stage 2**, observing  $w_C$ , Hegemon U determines its strategic doctrine  $\lambda$  (the weight placed on security) and announces a market access schedule  $R(x)$  to induce alignment. Finally, in **Stage 3**, the Swing State observes the input prices (Chinese input  $w_C$  and the Alternative-supplier price  $w_A$  of non-Chinese inputs) and Hegemon U’s policy schedule  $R(x)$ , then chooses the alignment  $x$  to maximize profit.

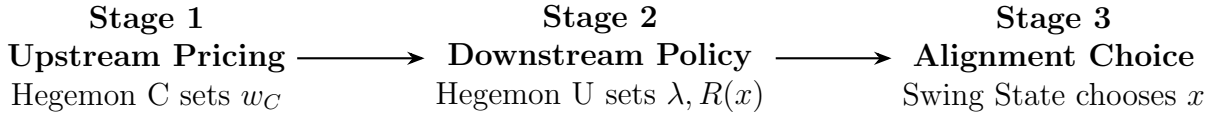


Figure 3: The Timeline of Vertical Rivalry

**Assumption 1** (Commitment and Information). *The extensive form in Figure 3 operates under two informational and institutional restrictions. (i) Sequential commitment: At each stage, the acting player commits to their strategic variable before the next player moves. In particular, Hegemon U commits to the pair  $(\lambda, R(\cdot))$  prior to the Swing State’s alignment choice, and this commitment is binding within the game. (ii) Complete information: All structural parameters are common knowledge. These include the input prices, the production-technology primitives introduced in Section 3.1, and the fiscal-cost and outside-option parameters introduced with Hegemon U’s and the Swing State’s problems below.*

Assumption 1 plays a central role: the credibility of ex-ante commitments in geoeconomics—sanctions threats, security guarantees, access regimes—is itself an institutional achievement, not an automatic feature of the extensive form. We take it as primitive; its shadow cost is formalized when Hegemon U’s problem is specified in Section 3.3.

The schedule  $R(x)$  is a *contingent contract*, not a linear spot price: Hegemon U commits to pay  $R(x)$  to the Swing State conditional on realized alignment.<sup>10</sup> This contractual structure shapes how the alignment  $x$  enters each stage problem in the sections that follow. On the Swing State’s side (Stage 3, Section 3.2),  $x$  is the contracted action, not an input choice solving a cost-minimization problem: the Swing State is an institutional entity that selects an alignment *policy*, and the resource penalty it incurs (formalized in Section 3.1.2) is the cost of *operating* a supply chain at the policy-imposed input mix, not the cost of a firm-theoretic optimization. On Hegemon U’s side (Stage 2, Section 3.3),  $R(x)$  is the contractual transfer paid for delivery at alignment  $x$ , the counterpart to the gross benefit Hegemon U derives from that delivery. Appendix A.4 derives the reduced form  $R(x) = C(x; w_C) + V_{out}$  from the binding participation constraint and documents concrete implementations (preferential tariff, CHIPS-Act-style conditional subsidy, defense-coproduction access).

### 3.1.2 Technology: Supply Chain and Decoupling Cost

The Swing State sources inputs from two distinct pools. The first is the *Red Supply Chain*, inputs  $I_C$  obtained from the Upstream Hegemon at price  $w_C$ . The second is the *Blue Supply Chain*, inputs  $I_A$  obtained from the rest-of-world non-rivalry input pool at the world-market

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<sup>10</sup>Conceptually,  $R(x)$  stands for the broad class of *strategic foreign-subsidy policies* the Hegemon uses to pull the Swing State toward its own bloc—transfers, tariff preferences, and access rights whose value to the Swing State scales with the depth of decoupling from the Rival. Concrete real-world instruments include (i) U.S. CHIPS-Act subsidies to Korean and Taiwanese fabs, conditional on ten-year “guardrails” restricting leading-edge capacity expansion in China; (ii) the Inflation Reduction Act’s \$7,500 EV tax credit, conditional on excluding “foreign entity of concern” (i.e., predominantly Chinese) battery components and critical minerals; and (iii) USMCA automotive rules of origin, where preferential tariff treatment requires a minimum North American value-added share. In each case the dollar value the Swing State receives rises with the demonstrated depth of decoupling from the Rival—this is the empirical content of  $R(x)$  being increasing in  $x$  on the relevant range.

price  $w_A$ .<sup>11</sup> The Swing State chooses a geopolitical alignment  $x \in [0, 1]$ , defined as the physical share of inputs sourced from the Alternative supply:

$$I_A = xI \quad \text{and} \quad I_C = (1 - x)I, \quad (1)$$

where  $I$  is the total aggregate input bundle.  $x$  is a *policy-imposed physical allocation share*, not an input choice solving a cost-minimization problem. Let  $w_C$  and  $w_A$  denote the prices of Chinese and Alternative inputs, respectively. We assume  $w_C < w_A$ , reflecting the Upstream Hegemon’s comparative advantage or subsidization.

We model the Swing State’s *Decoupling Cost* as a function  $C(x; w_C) : [0, 1] \times \mathbb{R}_{++} \rightarrow \mathbb{R}_+$  that measures the resource penalty the Swing State incurs for operating its supply chain at alignment  $x$  given an input price  $w_C$  charged by the Upstream Hegemon.  $C(x; w_C)$  is a primitive object of the model:  $x$  enters  $C$  as a *policy parameter* (the alignment-imposed input mix), not as a control variable of a cost-minimization problem, consistent with the principal–agent framing developed above and formalized in Appendix A.4. The firm-theoretic benchmark, against which we compare the Swing State’s contracted alignment, is the laissez-faire share  $x^{\text{LF}}(w_C)$  a cost-minimizing agent would select.

We impose the following minimal regularity conditions on  $C$ :

**Assumption 2** (Decoupling Cost: Bottleneck Regularity). *The Decoupling Cost  $C(x; w_C)$  is twice continuously differentiable on  $(0, 1) \times \mathbb{R}_{++}$  and satisfies, for every  $w_C \in \mathbb{R}_{++}$ :*

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<sup>11</sup>We model the Blue Supply Chain as a competitive backstop rather than a fourth strategic player. Its participants are price-takers without a strategic instrument, so  $w_A$  enters the model as an exogenous parameter rather than as the equilibrium outcome of an additional optimization. Game-theoretically, the Blue Supply Chain matters only as the Swing State’s outside option—which in turn constrains the strategic markup Hegemon C can charge in Stage 1 (Section 3.4)—and it has no independent payoff to maximize. We also exclude Hegemon U from this pool by construction: including US input producers would open a direct input-sales revenue channel from  $I_A$  into  $V_{US}(x)$ , a general-equilibrium spillover that the partial-equilibrium Stage-2 analysis abstracts from. In the data the dominant non-rivalry producers in our chokepoint sectors (advanced semiconductors, batteries, precision components) are themselves countries in the Swing-State set—Japan, Germany, Taiwan, Korea, the Netherlands. The model’s representative-Swing-State formulation treats  $w_A$  as an exogenous backstop from any individual Swing State’s perspective, thereby abstracting from cross-Swing-State input flows; relaxing this abstraction is left to the multi-country general-equilibrium extension discussed in Section 3.7.

(C1) *Laissez-faire minimum.* There exists a unique  $x^{\text{LF}}(w_C) \in (0, 1)$  at which  $C'(x^{\text{LF}}; w_C) = 0$ ; this is the cost-minimizing benchmark a firm-theoretic agent would select.

(C2) *Strict convexity on the decoupling range.*  $C''(x; w_C) > 0$  for all  $x \in (x^{\text{LF}}(w_C), 1)$ .

(C3) *Explosive marginal decoupling cost.*  $\lim_{x \rightarrow 1^-} C'(x; w_C) = +\infty$ .

(C4) *Chinese-price sensitivity of the gradient.*  $\partial C'(x; w_C) / \partial w_C < 0$  for all  $x \in (x^{\text{LF}}(w_C), 1)$ .

Condition (C1) identifies the laissez-faire reference point. Conditions (C2)–(C3) are the *bottleneck*: beyond the laissez-faire mix, marginal decoupling becomes strictly costly and prohibitively so as the Swing State approaches full decoupling. Condition (C4) is the key comparative-statics primitive: a fall in  $w_C$  (a cheaper Chinese input) steepens the marginal Decoupling Cost on the decoupling range, capturing the model’s central economic insight that “China’s Rise makes the Swing State harder to align.” Together (C1)–(C4) describe a chokepoint technology — rare-earth processing, legacy semiconductors, battery-grade precursors — where the Rival’s upstream input cannot be substituted away cheaply. Except where noted, the analysis that follows uses only Assumption 2.

A concrete parameterization that satisfies Assumption 2 arises when the Swing State’s technology aggregates inputs through a Constant Elasticity of Substitution (CES) form with elasticity  $\sigma$ , distribution parameter  $\alpha \in (0, 1)$ , and  $\sigma < 1$  (gross complements). Applying Shephard’s lemma to the CES aggregator evaluated at the alignment mix  $(1 - x, x)$  yields the closed form

$$C(x; w_C) = \frac{(1 - x) w_C + x w_A}{\left[ \alpha (1 - x)^{\frac{\sigma-1}{\sigma}} + (1 - \alpha) x^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}}, \quad (2)$$

with laissez-faire share

$$x^{\text{LF}}(w_C) = \frac{(1 - \alpha)^\sigma w_C^\sigma}{\alpha^\sigma w_A^\sigma + (1 - \alpha)^\sigma w_C^\sigma}.$$

Under  $\sigma < 1$  and  $w_C < w_A$ , properties (C1)–(C4) follow by direct computation (see the proof of Lemma 1 below). We invoke this CES parameterization in Section 5 for the quantitative

illustration and in Section 3.7 for the empirical defense of the bottleneck restriction, but not elsewhere: all analytic results use Assumption 2 alone.

### 3.2 Stage 3 (Swing State): Alignment Choice

We solve the model via backward induction, beginning with the final stage. The Swing State (Agent) takes the input prices  $\{w_C, w_A\}$  and the Downstream Hegemon’s policy schedule  $R(x)$  as given. It chooses the alignment  $x \in [0, 1]$  to maximize its total profit.

If the Swing State rejects Hegemon U’s contract, it reverts to the default arrangement—sourcing inputs from the Rival (“integration with China”) with no US-granted market access. Let  $V_{out}$  denote the Swing State’s reservation profit under this fallback. We normalize  $V_{out} = 0$  throughout, since an additive constant shifts the policy schedule  $R(x)$  one-for-one and is comparative-statically irrelevant; we therefore suppress  $V_{out}$  symbolically in the equations of Section 3.3 below. The Swing State accepts Hegemon U’s contract only when its optimized profit is weakly positive.

The Swing State’s profit function  $\Pi(x)$  is defined as market access revenue minus the Decoupling Cost,

$$\max_{x \in [0, 1]} \Pi(x) = R(x) - C(x; w_C) \quad (3)$$

Assuming the policy schedule  $R(x)$  is differentiable, the optimal alignment  $x^*$  is characterized by the first-order condition:

$$R'(x^*) = C'(x^*) \quad (\text{Incentive Compatibility}) \quad (4)$$

This condition implies that to induce any non-zero alignment  $x^* > 0$ , the Downstream Hegemon must provide a marginal market access benefit exactly equal to the marginal cost of decoupling.

The behavior of the Swing State is driven by the shape of the Decoupling Cost  $C(x; w_C)$  on the decoupling range  $(x^{\text{LF}}(w_C), 1)$  where the principal’s contract pushes alignment above

the laissez-faire benchmark.

**Lemma 1** (The Bottleneck Gradient). *Under Assumption 2, for every  $w_C$  and every  $x \in (x^{\text{LF}}(w_C), 1)$ :*

1. *Monotonicity.*  $C'(x; w_C) > 0$ ; *decoupling beyond the laissez-faire benchmark always incurs a strictly positive marginal cost.*
2. *Explosive marginal cost.*  $\lim_{x \rightarrow 1^-} C'(x; w_C) = +\infty$ ; *fully replacing the upstream input is asymptotically infeasible.*
3. *Input-price sensitivity.* *A fall in the upstream price steepens the Decoupling-Cost gradient:  $\partial C'(x; w_C)/\partial w_C < 0$ .*

*Proof.* All three claims are immediate consequences of Assumption 2. Monotonicity follows from (C1)–(C2):  $C'(x^{\text{LF}}; w_C) = 0$  and  $C$  is strictly convex on  $(x^{\text{LF}}, 1)$ , so  $C'(x; w_C) > 0$  on that range. Explosive marginal cost is (C3). Input-price sensitivity is (C4).  $\square$

For the CES parameterization (2), direct differentiation gives  $C'(x; w_C) = (w_A - w_C)/A(x) - [(1 - x)w_C + xw_A] \cdot A'(x)/A(x)^2$ , where  $A(x) = [\alpha(1 - x)^{(\sigma-1)/\sigma} + (1 - \alpha)x^{(\sigma-1)/\sigma}]^{\sigma/(\sigma-1)}$ . Under  $\sigma < 1$  and  $w_C < w_A$ : (C1) is the interior first-order condition yielding  $x^{\text{LF}}(w_C)$ ; (C2) follows because the CES aggregator under  $\sigma < 1$  is strictly concave in  $x$  on  $(x^{\text{LF}}, 1)$ , so the unit-cost wedge generates strictly convex  $C$ ; (C3) follows because  $A(x) \rightarrow 0$  as  $x \rightarrow 1^-$  (the  $\alpha(1 - x)^{(\sigma-1)/\sigma}$  term diverges with exponent  $(\sigma - 1)/\sigma < 0$ , while the aggregator exponent  $\sigma/(\sigma - 1) < 0$  sends  $A$  itself to zero), so  $C$  and  $C'$  diverge; (C4) follows because  $\partial C/\partial w_C = (1 - x)/A(x)$  is larger at low  $x$  than at high  $x$ , so a fall in  $w_C$  reduces the cost of low- $x$  operating points by more than high- $x$  ones, steepening the marginal Decoupling Cost on the decoupling range.

**Strategic Implication:** The lemma implies that “China’s Rise” ( $w_C \downarrow$ ) does not merely make the Swing State richer; it makes the Swing State *harder to align*. The Downstream Hegemon faces a steeper marginal cost curve  $C'(x)$  to implement the same target  $x^*$ , requiring a more aggressive (steeper) policy schedule  $R(x)$ .

### 3.3 Stage 2 (Downstream Hegemon): Policy and Doctrine

In Stage 2, the Downstream Hegemon (US) designs the policy schedule  $R(x)$  and selects its strategic doctrine  $\lambda$  to maximize national welfare, anticipating the Swing State’s reaction in Stage 3.

The Hegemon’s utility  $\mathcal{W}_{US}$  is defined over the realized alignment  $x$ , the transfer payment  $R(x)$ , and the security weight  $\lambda \in [0, 1]$ . We model the choice of  $\lambda$  not as a fixed preference, but as an endogenous policy decision (“Securitization”) that incurs convex institutional costs:

$$\mathcal{W}_{US}(x, R(x), \lambda) = \underbrace{\lambda x \theta}_{\text{Security Utility}} + \underbrace{(1 - \lambda)[V_{US}(x) - R(x)]}_{\text{Net Economic Surplus}} - \underbrace{\frac{\gamma}{2}\lambda^2}_{\text{Cost of Securitization}} \quad (5)$$

where  $V_{US}(x)$  represents the gross economic value generated for the US, such as consumer surplus or IP royalties. We assume  $V'_{US}(x) < 0$ , reflecting the efficiency loss from decoupling from the most efficient upstream supplier. The parameter  $\gamma > 0$  quantifies the institutional resistance to adopting a security-heavy doctrine—fiscal constraints, domestic political polarization, or administrative capacity. Equivalently,  $\gamma$  is the reduced form of the commitment cost implicit in Assumption 1: credibly binding to a deviation from the default market-integration doctrine is itself costly (in the spirit of Rogoff, 1985; Maskin and Tirole, 1999; Aghion and Tirole, 1997).

The multiplicative form  $\lambda x \theta$  admits a kill-switch interpretation: with conditional weaponization probability  $1 - x$  (so alignment  $x$  measures the share of inputs insulated against a Rival embargo) and per-unit damage  $D(\theta) = \theta$ , the expected damage averted by alignment is  $x \cdot D(\theta)$ , weighted by the Hegemon’s endogenous security posture  $\lambda$  (cf. equation (23) in Appendix B.2). The appendix derives the linear scaling  $D(\theta) = \theta$  from Assumption 2: a more productive Rival supplies the same physical unit at lower cost ( $\bar{w}/\theta \downarrow$ ), so the Hegemon’s surplus per input unit, and hence the surplus destroyed by withholding, scales with  $\theta$ . The qualitative comparative-static results (Propositions 1–2) require only  $D' > 0$ ; the linear specification fixes the units of  $\lambda$  and the quantitative

magnitudes (notably the threshold  $\bar{\theta}$ ) inherit the curvature of  $D$ , calibrated in Section 5. Crucially, the same primitive  $\theta$  enters Hegemon C's Stage-1 marginal cost  $\bar{w}/\theta$  in (12), so the two roles—cost-reducer for the supplier, damage-magnifier for the consumer—are derived from a single technology parameter rather than co-positied.

The Principal cannot choose  $x$  directly; it must induce  $x$  through the schedule  $R(x)$ . To induce a target alignment  $x$ , the US must satisfy two constraints:

1. *Incentive Compatibility (IC)*:  $R'(x) = C'(x)$ .
2. *Participation Constraint (PC)*:  $R(x) \geq C(x; w_C)$  (under the  $V_{out} = 0$  normalization of Section 3.2).

Since the US minimizes the transfer  $R(x)$ , the PC binds at the optimum, giving the *Implementation Cost Function*

$$R(x) = C(x; w_C), \tag{6}$$

where  $C(x; w_C)$  is the Decoupling Cost introduced in Section 3.1.2 (Assumption 2). Following the standard principal–agent approach with observable actions (Grossman and Hart, 1983), we treat this cost as the effective price of alignment. When the IC condition binds with indifference at the target  $x^*$ , we adopt the standard forcing-contract convention: Hegemon U can impose an arbitrarily small off-path penalty that selects  $x^*$  as the Agent's unique best response.

Substituting the implementation cost into the objective function and letting  $J(x; w_C) \equiv V_{US}(x) - C(x; w_C)$  denote the *Joint Economic Surplus*, the US problem reduces to

$$\max_{x, \lambda} \quad \mathcal{W}_{\text{reduced}} = \lambda x \theta + (1 - \lambda) J(x; w_C) - \frac{\gamma}{2} \lambda^2. \tag{7}$$

The subscript “reduced” marks that  $\mathcal{W}_{\text{reduced}}$  is (5) after the binding PC has been substituted in: the transfer  $R(x)$  no longer appears explicitly because it has been replaced by  $C(x; w_C)$ , leaving the joint surplus  $J(x; w_C)$  inside the economic-component term and the security/cost

terms unchanged. Within Section 3.3 we suppress the  $w_C$  argument of  $J$  in the FOCs below, since Stage 2 takes  $w_C$  as parametric; the dependence is restored in Sections 3.4–3.5 where Hegemon C’s pricing channel is operative.

We characterize the equilibrium by maximizing the reduced objective function (7) with respect to the alignment target  $x$  and the security doctrine  $\lambda$ . The solution is determined by the intersection of two reaction functions.

First, consider the optimal induced alignment  $x^*(\lambda)$ . Taking the security weight  $\lambda$  as given, the US chooses the target  $x^*$  to equate the marginal political gain with the marginal economic loss:

$$\frac{\partial \mathcal{W}}{\partial x} = \lambda \theta + (1 - \lambda) J'(x^*) = 0 \quad \implies \quad J'(x^*) = -\frac{\lambda \theta}{1 - \lambda} \quad (8)$$

This condition implies that as the security doctrine becomes more hawkish ( $\lambda \rightarrow 1$ ), the US pushes the target  $x^*$  further into the economically inefficient region where the marginal loss of joint surplus  $J'(x)$  is high. Upstream productivity  $\theta$  amplifies this effect: a more productive Upstream Hegemon (higher  $\theta$ ) shifts  $x^*$  upward at every level of  $\lambda$ .

Second, consider the endogenous regime choice  $\lambda^*(x)$ . Differentiating with respect to  $\lambda$ , we obtain the optimal doctrine rule:

$$\frac{\partial \mathcal{W}}{\partial \lambda} = x \theta - J(x) - \gamma \lambda = 0 \quad \implies \quad \lambda^*(x) = \min \left\{ 1, \max \left\{ 0, \frac{x \theta - J(x)}{\gamma} \right\} \right\}. \quad (9)$$

The bracketed expression makes the box constraint  $\lambda \in [0, 1]$  explicit: the lower corner  $\lambda^* = 0$  binds when the Strategic Gap is non-positive (Generous Empire, characterized below), while the upper corner  $\lambda^* = 1$  binds when  $x\theta - J(x) \geq \gamma$ . Throughout Sections 3.5–3.6 we maintain  $\gamma$  large enough that the upper corner is slack on the relevant range of  $\theta$ ; the calibration in Section 5 verifies this directly. A formal sufficient condition is  $\gamma > \theta_{\max} - \min_x J(x)$  over the parameter range under consideration.

**The Strategic Gap.** Define the *Strategic Gap* as the numerator of (9),

$$G(x; w_C) \equiv x\theta - J(x), \quad (10)$$

i.e., the threat-weighted Security Benefit minus the Joint Economic Surplus. The optimal doctrine is then  $\lambda^*(x) = G(x; w_C)/\gamma$  on the interior, clipped to zero at the corner. The Strategic Gap is the central comparative-statics object of the model and is referenced repeatedly in Section 3.5: a rising Upstream Hegemon (lower  $w_C$ , higher  $\theta$ ) widens it through two channels — it raises the perceived threat intensity  $\theta$  *and* compresses the Joint Economic Surplus by raising the implementation cost  $R(x) = C(x; w_C)$  of re-routing supply chains.

**The regime threshold  $\bar{\theta}$ .** The Strategic Gap pins down a threshold productivity level above which the corner  $\lambda^* = 0$  ceases to bind. Define

$$\bar{\theta} \equiv \inf\{\theta > 0 : \max_{x \in [0,1]} G(x; w_C^*(\theta)) > 0\}, \quad (11)$$

i.e., the smallest upstream productivity at which the Strategic Gap, evaluated at Hegemon C's Stage-1 best-response price  $w_C^*(\theta)$  (derived in Section 3.4), becomes strictly positive at some  $x$ . Below  $\bar{\theta}$ , the security benefit is dominated by the joint surplus and Hegemon U is content with  $\lambda^* = 0$ ; above  $\bar{\theta}$ , an interior  $\lambda^* > 0$  becomes strictly optimal. Existence of  $\bar{\theta}$  on  $(0, \infty)$  follows from continuity of  $\max_x G(x; w_C^*(\theta))$  in  $\theta$  together with two boundary properties: as  $\theta \rightarrow \infty$ , the security benefit dominates and  $\max_x G \rightarrow +\infty$ ; for  $\theta$  small enough, the security benefit is bounded above by  $\theta \cdot 1$  while the joint-surplus term remains uniformly positive on the relevant range, so  $\max_x G < 0$ . The threshold is then well-defined as the infimum at which the supremum first turns positive.

**Proposition 1** (The Regime Switch Mechanism). *The equilibrium is the fixed point  $(\lambda^*, x^*)$  of these two reaction functions. The nature of the equilibrium depends on whether upstream productivity  $\theta$  falls below or above the threshold (11):*

- *Generous Empire ( $\theta \leq \bar{\theta}$ ): The Strategic Gap is non-positive, so  $\lambda^* = 0$  and Hegemon U relies on market integration alone.*
- *Coercive Hegemon ( $\theta > \bar{\theta}$ ): The Strategic Gap is strictly positive and  $\lambda^*$  rises continuously from the corner, with a kink at  $\bar{\theta}$  where the corner constraint becomes slack; above the threshold,  $\lambda^*$  increases smoothly toward coercion.*

The equilibrium  $(\lambda^*, x^*)$  is well-defined. Appendix B.1 verifies that the Stage-2 objective is strictly concave in  $(x, \lambda)$  whenever  $\gamma$  is sufficiently large—i.e., when the institutional cost of securitization dominates the cross-effect. This concavity yields a unique interior fixed point that is locally stable under best-response dynamics. Economically, the “ $\gamma$  sufficiently large” condition means that doctrine and alignment do not amplify each other into tipping points; when  $\gamma$  is too small, multiplicity arises and the economy can jump discontinuously between peaceful and coercive equilibria.

### 3.4 Stage 1 (Upstream Hegemon): Pricing Strategy

In the first stage, Hegemon C sets the terms of trade for the critical input. To distinguish between technological fundamentals and strategic intent, we decompose the input price into production costs and industrial policy.

Let  $w_C$  be the final export price of the input. We decompose this price into an exogenous technology component and an endogenous policy component,

$$w_C = \underbrace{\frac{\bar{w}}{\theta}}_{\text{Marginal Cost}} + \underbrace{m}_{\text{Strategic Markup}} \quad (12)$$

where  $\bar{w}$  denotes the baseline factor price. The parameter  $\theta > 0$  represents the *Exogenous Technological Productivity*; an increase in  $\theta$  captures the structural phenomenon of “China’s Rise” or technological catch-up. The variable  $m$  represents the *Endogenous Industrial Policy*, which captures the state’s strategic intent. A positive value ( $m > 0$ ) indicates a standard

profit-maximizing monopolistic markup, whereas a negative value ( $m < 0$ ) indicates a strategic state subsidy (or dumping) designed to capture market share.

The productivity primitive  $\theta$  in (12) is the same  $\theta$  that scaled the Security Utility in Stage 2 (equation (5), Section 3.3). The two roles—cost-reducer here in Stage 1 and threat-magnifier in Stage 2—are economically the same:  $\theta$  measures how sharp the upstream chokepoint is, and the depth of the bottleneck determines both how cheaply Hegemon C can supply and how dangerous a withdrawal of that supply would be. Under complete information (Assumption 1), Hegemon U observes  $\theta$  directly and need not infer it from  $w_C$ . The microfoundation for the Stage-2 scaling is the probabilistic kill-switch interpretation in Appendix B.2: the potential damage that alignment averts is increasing in upstream productivity, so the security benefit  $\lambda x$  in Hegemon U’s utility is scaled by  $\theta$ .

Normalizing final output  $Y = 1$ , Hegemon C chooses the markup  $m$  (and thus the price  $w_C$ ) to maximize a weighted sum of commercial profit and geopolitical influence:

$$\max_m \mathcal{W}_C = \underbrace{m \cdot Q_C(x(w_C); w_C)}_{\text{Net Commercial Return}} + \underbrace{\phi(1 - x(w_C))}_{\text{Geopolitical Influence}} \quad (13)$$

where  $Q_C(x; w_C)$  denotes the Swing State’s physical demand for Chinese inputs at alignment  $x$  and upstream price  $w_C$ , a primitive of the production technology of Section 3.1.2. In the CES special case (2),  $Q_C = (1 - x)/A(x)$ . The objective highlights the trade-off: lowering  $m$  (subsidies) incurs a fiscal cost but increases the strategic value  $\phi(1 - x)$  by binding the Swing State to the Red Supply Chain.

Hegemon C faces a derived demand curve  $Q_C(x(w_C); w_C)$  for its input. Its total elasticity  $\varepsilon_D \equiv -d \ln Q_C / d \ln w_C$  combines two channels: (i) the *direct demand response* through the Swing State’s input substitutability, governed by the convexity of the Decoupling Cost (Assumption 2, property (C2); in the CES special case (2), parameterized by  $\sigma$ ); and (ii) a (*potential*) *regime-feedback channel*: by Stackelberg leadership (Assumption 1), Hegemon C anticipates Stage 2 best-responses, so a sufficiently aggressive cut in  $w_C$  can trigger a coercive

regime switch ( $\lambda^* \uparrow$ ) that forces  $x$  upward and contracts Chinese export volume  $Q_C$  despite the lower price. The regime-feedback channel operates only on parameter ranges where the Stage-2 fixed point is differentiable in  $w_C$  (Appendix B.1); on the Generous-Empire interior ( $\lambda^* = 0$  throughout a neighbourhood of  $w_C^*$ ),  $\varepsilon_D$  collapses to channel (i) alone.

A first-order necessary condition then gives a modified monopoly pricing rule of the form

$$\frac{w_C^* - (\bar{w}/\theta)}{w_C^*} = \underbrace{\frac{1}{\varepsilon_D}}_{\text{Monopoly Markup}} - \underbrace{\frac{\Phi_{\text{strat}}}{w_C^*}}_{\text{Geopolitical Discount}}, \quad (14)$$

where  $\bar{w}/\theta$  is Hegemon C's own (firm-theoretic) marginal cost of producing the input—a primitive of *Hegemon C's* technology, distinct from the Swing State's Decoupling Cost  $C(x; w_C)$  defined in Section 3.1.2;  $\varepsilon_D$  is the composite demand elasticity defined above; and  $\Phi_{\text{strat}} > 0$  collects the marginal strategic value of additional Red-Chain market share through the influence motive  $\phi(1 - x)$  and the regime-feedback channel just described. Because influence matters ( $\phi > 0$ ), Hegemon C optimally sets a price *lower* than the standard profit-maximizing monopoly level, effectively engaging in predatory pricing to secure the vertical supply chain.

We label (14) an *implicit characterization* rather than a closed-form solution: the term  $\Phi_{\text{strat}}$  depends on  $dx^*/dw_C$  through both the Stage 3 demand response and the Stage 2 doctrine response, and a fully reduced expression requires inverting the joint Stage 2 fixed point. We do not need that reduced form for any subsequent result; only the comparative-static signs  $dw_C^*/d\theta < 0$  (from  $\theta \uparrow \Rightarrow$  marginal cost  $\downarrow$  and larger  $\Phi_{\text{strat}}$  as influence becomes more valuable) and the predatory-pricing implication enter Sections 3.5–3.6. The full Stage 1 fixed-point derivation, including conditions under which  $w_C^*$  is unique, is deferred to Appendix B.5 as part of the Paradox of Competitiveness proof.

### 3.5 General Equilibrium and the Paradox of Competitiveness

We now close the model. Combining the three best responses derived above—pricing (Hegemon C), doctrine (Hegemon U), and alignment (Swing State)—we trace how an exogenous rise in Chinese productivity ( $\theta \uparrow$ ) reverberates through the vertical supply chain and reshapes the global security architecture.

**Proposition 2** (The Paradox of Competitiveness). *An exogenous rise in Upstream Productivity ( $\theta \uparrow$ , “China’s Rise”) strictly increases the Downstream Security Doctrine on the coercive interior:  $d\lambda^*/d\theta > 0$  for  $\theta > \bar{\theta}$ . The more efficient the Rival, the more hawkish the Hegemon.*

*Proof sketch.* The shock transmits through four links. First, higher  $\theta$  lowers Hegemon C’s marginal cost  $\bar{w}/\theta$ . Second, Hegemon C re-optimizes: since both commercial profit and the geopolitical influence motive ( $\phi > 0$ ) favor capturing market share, the optimal equilibrium price falls,  $dw_C^*/d\theta < 0$  (see (12)). Third, the cost-of-decoupling gap  $\Delta C(x) = C(x) - C(x^{\text{LF}})$  widens because the gradient sensitivity is negative ( $\partial C'/\partial w_C < 0$ , Assumption 2(C4)/Lemma 1): a fall in  $w_C$  steepens the marginal Decoupling Cost on the decoupling range, so the Red-Chain-intensive base drops by more than the Blue-Chain-intensive tail. Fourth, the Strategic Gap in (9) widens and  $\lambda^*$  rises:  $d\lambda^*/d\theta = (\partial\lambda^*/\partial w_C)(dw_C^*/d\theta) > 0$ , both factors negative. Full derivation in Appendix B.5.  $\square$

The transition to coercive trade policy is not necessarily driven by ideological shifts or “unfair” industrial policy ( $m$ ). Even if Hegemon C behaves as a benign market actor (holding  $m$  constant), a pure productivity shock is sufficient to trigger the regime switch.

We term this dynamic the *Tragedy of Efficiency*: the more efficient China becomes, the harder it is for the US to compete through positive inducements such as subsidies or market access. Once the required subsidy exceeds US fiscal or political capacity, the optimal strategy flips from integration (low  $\lambda$ , high  $R$ ) to coercion (high  $\lambda$ , negative  $R$ ). The US then turns to high- $\lambda$  coercion not as a defensive industrial policy, but as a tool to nullify the Rival’s

efficiency advantage at the swing state's margin: raising the strategic cost of Red Chain alignment until the Blue Chain regains its relative attractiveness.

### 3.6 Welfare: The Tragedy of Efficiency

A claim of tragedy requires a welfare statement. Define the joint economic surplus of the three players as

$$\mathcal{S}(x; \theta) \equiv V_{US}(x) + \pi_C(x; w_C(\theta)) - C(x; w_C(\theta)), \quad (15)$$

where  $V_{US}(x)$  is Hegemon U's gross value,  $\pi_C = m \cdot Q_C(x)$  is Hegemon C's commercial profit (from Section 3.4), and  $C(x; w_C)$  is the Swing State's Decoupling Cost. The formal decomposition is Proposition 5 in Appendix B.5. The integrated planner's *first-best alignment*  $x^{FB}(\theta)$  maximizes  $\mathcal{S}$  while ignoring the security externality,<sup>12</sup> and is *declining* in  $\theta$ : a more efficient Upstream Rival raises the opportunity cost of decoupling.<sup>13</sup> The equilibrium  $x^*(\theta)$  moves in the opposite direction (Proposition 2). Define the *efficiency wedge*:

$$\Delta(\theta) \equiv \mathcal{S}(x^{FB}(\theta); \theta) - \mathcal{S}(x^*(\theta); \theta) \geq 0. \quad (16)$$

**Proposition 3** (The Tragedy of Efficiency). *Under Assumption 2 and in the coercive regime ( $\lambda^* > 0$ ), the efficiency wedge  $\Delta(\theta)$  is strictly increasing in Upstream productivity  $\theta$ . That is, improvements in Upstream technology generate larger global welfare losses relative to the integrated first-best.*

*Proof sketch.* In the coercive regime,  $x^*(\theta)$  rises with  $\theta$  while  $x^{FB}(\theta)$  falls; the gap widens monotonically, and strict concavity of  $\mathcal{S}$  at  $x^{FB}$  delivers the result. The full welfare

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<sup>12</sup> $\mathcal{S}$  also drops Hegemon C's geopolitical influence motive  $\phi(1-x)$  from Section 3.4 and the doctrine cost  $\frac{\gamma}{2}\lambda^2$  from (5). Under the kill-switch interpretation (Appendix A),  $\lambda x\theta$  is a transfer between Hegemons U and C and nets out in joint welfare;  $\frac{\gamma}{2}\lambda^2$  vanishes at the planner's optimum  $\lambda^{FB} = 0$ ;  $\phi(1-x)$  is excluded on normative grounds, consistent with the calibration choice  $\phi = 0$  in Table 1.

<sup>13</sup>A true first-best would also impose marginal-cost pricing  $w_C = \bar{w}/\theta$ . Here we treat the Stage-1 markup  $w_C(\theta)$  as parametric, so  $\mathcal{S}$  is a second-best benchmark for a planner with authority over swing alignment  $x$  alone, not over Hegemon C's pricing.

decomposition—including the curvature and second-order-markup conditions under which the commercial component is strictly decreasing in  $\theta$ —is in Appendix B.5.  $\square$

*Interpretation.* Proposition 3 is the formal sense in which “Tragedy of Efficiency” is literal. Under horizontal rivalry, a pure productivity gain in the Upstream Rival would be unambiguously welfare-improving. Here, instead, it triggers an equilibrium response in Downstream doctrine that destroys commercial surplus faster than the gain accrues. The Swing State bears the bulk of the loss, because the coercive corner truncates its choice set. Appendix B.9 contrasts the vertical bottleneck with two horizontal-rivalry benchmarks (subsidy war, bidding war) and shows that linear price-out cost in those benchmarks keeps  $\lambda^* = 0$ , so the coercive regime is specific to the vertical structure.

### 3.7 Robustness and Extensions

The single-sector baseline of Sections 3.1–3.6 embeds three modelling choices that warrant scrutiny: take-it-or-leave-it pricing by Hegemon U at Stage 2, exogenous assignment of the upstream/downstream vertical roles, and a deterministic Security Utility specification. We relax each in turn below and additionally address a robustness concern about the consistency between the discrete regime switch of Proposition 1 and the smooth aggregate dynamics of Figure 4. A separate *multi-sector extension*, which generates two cross-sectional moments (a flow-margin leakage ratio and a steady-state cross-partial on Chinese value-added), is developed in Appendix C; its empirical counterparts corroborate the same  $\sigma_c < 1$  primitive that drives the regime switch but are not load-bearing for the main-text validation. Each item is summarized here so the reader can audit the dependence of Propositions 1–3 on the corresponding choice; full derivations are in Appendix B.

**Reconciling the discrete regime switch with smooth aggregate patterns.** Proposition 1 predicts a discrete kink in  $\lambda^*(\theta)$  at  $\bar{\theta}$ , while Figure 4 shows a continuous post-2018 acceleration in coercive listings. Three non-exclusive aggregation channels reconcile the two

without modifying the within-sector micro-prediction: heterogeneous sectors crossing their type-specific  $\bar{\theta}_k$  sequentially, Bayesian uncertainty about  $\sigma$  that converts the discrete cutoff into a posterior-weighted average, and institutional adjustment costs on coercive capacity that smooth its time path. Appendix B.10 formalizes each channel and shows that any one is sufficient to deliver smooth aggregate  $\lambda$  from the discrete underlying mechanism.

**Endogenous Swing-State bargaining power.** Replacing the take-it-or-leave-it Stage 2 with Nash bargaining over  $(\lambda, R)$  with Swing-State weight  $\beta \in [0, 1]$  sharpens rather than dampens the Paradox: the regime threshold  $\bar{\theta}(\beta)$  is strictly decreasing in  $\beta$  (Proposition 4, Appendix B.3). A higher- $\beta$  Swing State extracts more of the joint surplus, tightening Hegemon U's Strategic Gap budget and triggering the switch from Generous Empire to Coercive Hegemon at a lower upstream productivity. The model thus rationalizes the empirical pattern of subsidies coexisting with coercion for high- $\beta$  allies—Korea, Taiwan, the Netherlands—without invoking discretionary policy choice.

**Endogenous vertical position.** The baseline treats the upstream/downstream assignment  $C \rightarrow S \rightarrow U$  as a primitive of the environment. Adding a Ricardian specialization pre-stage in which countries sort into upstream or downstream production based on relative productivities  $(\theta_C, \theta_U)$  and endowments  $(E_C, E_U)$  delivers the baseline configuration as the unique equilibrium whenever  $\theta_C > \theta_U$  and  $E_U > E_C$  (Appendix B.4). The upstream-China/downstream-US asymmetry is therefore an outcome of comparative-advantage forces rather than an exogenous label, and the bottleneck primitive of Assumption 2 is what jointly drives vertical sorting and the subsequent coercive equilibrium.

**Probabilistic kill-switch microfoundation.** The Stage-2 Security Utility  $\lambda x \theta$  admits a probabilistic interpretation in which  $1 - x$  is the conditional probability of a Rival weaponization event and  $\theta$  is the per-unit damage conditional on weaponization (Appendix B.2, equation (23)). Under this microfoundation  $\lambda$  is the Hegemon's expected damage averted per

unit of alignment, and the comparative static  $d\lambda^*/d\theta > 0$  becomes the formal statement that upstream efficiency endogenously raises Hegemon U’s perceived weaponization probability—the security-dilemma logic recovered from optimizing primitives rather than imposed as an axiom.

None of the four arguments overturn the qualitative Paradox of Competitiveness or the Tragedy of Efficiency; collectively they sharpen the calibration of  $\bar{\theta}$ , the policy interpretation of  $\lambda$ , and the empirical mapping between within-sector predictions and aggregate observables.

## 4 Empirical Validation: The Coercive Regime Switch

This section validates the central theoretical claim of the paper: a regime switch in the Hegemon’s doctrine (Proposition 1), triggered by upstream productivity advance and amplified by the paradox mechanism (Proposition 2). Section 4.1 documents the policy environment that activates  $\lambda$  in the model, distinguishing the bilateral kink that signatures the regime switch from the constant-slope third-country machinery that implements it. Section 4.2 reports a complementary dose–response test of Proposition 2, and Section 4.3 synthesizes what the two pieces of evidence buy. The cross-sectional leakage tests of Propositions 6–7, which require the multi-sector extension developed in Appendix C, are reported there.

### 4.1 The Coercive Policy Environment

US policy toward China and its supply-chain intermediaries has undergone a structural transformation since the mid-2000s. We document this using the US Bureau of Industry and Security (BIS) Entity List, which records firms denied access to US technology and licensing. Two patterns matter: the coercive perimeter has been extra-bilateral for at least fifteen years, and the post-2018 China-bilateral expansion sits on top of (not in place of) that pre-existing third-country machinery.

Network coercion preceded the 2018 trade war by more than a decade. The stock of

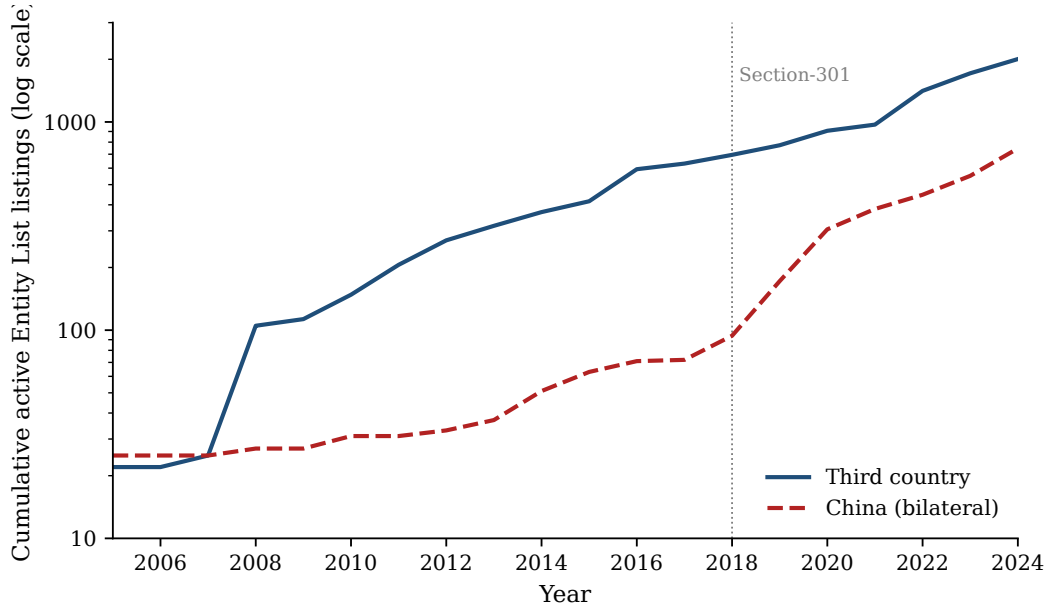


Figure 4: **The Network War.** Cumulative count of active US Entity List listings on a log scale, split by country of the listed entity (third-country vs. China-bilateral). The third-country line has a roughly constant log-slope throughout the sample; the China line is nearly flat pre-2018 and kinks upward sharply after Section-301. Compound growth rates cited in the body.

third-country listings has exceeded the stock of direct China listings in every year since the mid-2000s, with ratios ranging from roughly 5:1 in 2010 to 9:1 in 2017. Figure 4 plots the cumulative count of active entities on the Entity List on a log scale, which makes growth rates rather than absolute levels readable. Two features of the plot matter. First, the third-country line (solid blue) has a roughly constant log-slope throughout the sample—its compound growth rate is  $\approx 23\%$  per year in 2010–2017 and  $\approx 19\%$  per year in 2018–2024, essentially unchanged by Section-301. Second, the China line (dashed red) is nearly flat from 2005 to 2017 ( $\approx 13\%$  per year) and then kinks sharply upward after 2018 ( $\approx 41\%$  per year). Post-2018 the bilateral stock converges toward the third-country stock, narrowing the level ratio to roughly 3:1 by 2024. The infrastructure for third-country coercion was therefore already in place before the tariff shock; what 2018 triggered was a bilateral intensification on top of a pre-existing extra-bilateral coercive machine.<sup>14</sup>

<sup>14</sup>We do not claim a structural break test; the description is observational.

Two further qualitative shifts in the listing pattern are consistent with the coercion mechanism, though they do not identify it. First, the Entity List’s sectoral focus has narrowed toward high-technology targets: the share of new entries containing “Semiconductor,” “AI,” or “Quantum” rose from under 10% in the early 2010s to over 20% by 2024. Second, administrative effort per listing has risen. The average number of aliases per sanctioned entity spiked to nearly 2.7 in 2023, suggesting growing evasion behavior by sanctioned entities. We report the underlying series in Appendix Figure 12; these patterns motivate but are not load-bearing for the formal analysis.

Read through the model, the two Entity-List series map differently onto the coercion variable  $\lambda^*$ . The China-bilateral line (which kinks at 2018) is the direct Regime-Switch signature:  $\lambda^*(\theta)$  crosses  $\bar{\theta}$  once Section-301 raises  $\theta$  past the Generous-Empire corner. The third-country line (constant log-slope throughout the sample) is not  $\lambda^*$  itself but the administrative infrastructure that enforces it—the primitives  $\gamma$  and  $\bar{T}$  in the model. Its steady growth is therefore consistent with the model, not evidence against the 2018 kink.

## 4.2 Paradox Dose–Response

A complementary test of the paradox mechanism (Proposition 2) regresses log US imports from China on the lagged Entity-List China-listings stock  $EL_{t-1}^{\text{CHN}}$  interacted with sector bucket. The sector-asymmetric loading is sharp on the 3-sector aggregate panel ( $-0.217$ ,  $p < 0.001$ ): coercive intensity correlates with the China–Critical gap in exactly the direction the paradox predicts. At HS6 resolution the same dose–response is null—the post-2018 critical-bucket slope is  $+0.021$  (SE 0.061), and strategic, commodity, and flow variants are null throughout. We therefore treat the HS6 null as the honest benchmark and read the aggregate-panel slope as suggestive rather than identifying. The paradox mechanism is consistent with the aggregate evidence and not ruled out at HS6, but the dose–response is not the empirical workhorse of this section. Appendix Figure 13 plots the underlying sector-aggregate series.

### 4.3 Synthesis

Two pieces of evidence anchor the empirical reading of the model. The Network-War pattern (Section 4.1) provides the regime-switch signature: a constant-slope third-country coercive perimeter interpretable as the enforcement primitives  $\gamma$  and  $\bar{T}$ , on top of which a 2018 China-bilateral kink in  $\lambda$ -loading is the direct Regime-Switch fingerprint of Proposition 1. The Paradox Dose–Response (Section 4.2) provides a complementary aggregate-panel test of Proposition 2: coercive intensity scales with the China–Critical gap in the sign the paradox predicts, though the HS6 null requires that we read the slope as suggestive rather than identifying. The leakage tests of Propositions 6–7 are reported in Appendix C; they sharpen the cross-sectional implications of the same  $\sigma_c < 1$  primitive that drives the regime switch and so corroborate, rather than substitute for, the regime-switch reading developed here.

Having identified the regime switch and the paradox dose–response empirically, we now visualize the model that generates them.

## 5 Numerical Illustration and Robustness

To visualize the Regime Switch (Proposition 1) and the Paradox of Competitiveness (Proposition 2) that flows from it, we simulate the model under parameters calibrated to the bottleneck-sector literature. We set  $\sigma = 0.35$ , intermediate between two anchors: the sector-level materials elasticity of approximately 0.1 estimated by Atalay (2017), and the macro-level range 0.5–0.7 used in the production-network calibrations of Baqaee and Farhi (2024). In addition, Boehm et al. (2019) document a short-run elasticity between imported and domestic inputs that is indistinguishable from Leontief, placing an even lower floor on plausible  $\sigma$  at business-cycle horizons. All three estimates place  $\sigma$  comfortably below unity and satisfy the Bottleneck Condition (Assumption 2). The distribution parameter is set symmetrically at  $\alpha = 0.5$ . We then simulate “China’s Rise” by exogenously decreasing the upstream input price  $w_C$  while holding the alternative price  $w_A$  and the US market value  $V_{US}$  constant.

| Parameter                          | Value      | Source / role                                                                 |
|------------------------------------|------------|-------------------------------------------------------------------------------|
| $\sigma$ (CES elasticity)          | 0.35       | Between Atalay (2017) ( $\approx 0.1$ ) and Baqaee and Farhi (2024) (0.5–0.7) |
| $\alpha$ (Chinese-input weight)    | 0.5        | Symmetric baseline                                                            |
| $w_A$ (alternative price)          | 1          | Numeraire                                                                     |
| $V_{out}$ (Swing outside option)   | 0          | Normalization                                                                 |
| $\bar{w}$ (factor-price numeraire) | 1          | Normalization in (12)                                                         |
| $V_0$ (US baseline value)          | 0.84       | Calibrated to threshold $\bar{w}_C = 1.6$                                     |
| $\beta$ (slope of $V_{US}$ )       | 1.0        | Calibrated jointly with $V_0$                                                 |
| $\gamma$ (security-policy cost)    | 2.5        | Calibrated for smooth $\lambda^*$ ; passes local Hessian check                |
| $\phi$ (China geopolitical weight) | 0          | Isolates productivity shock channel                                           |
| $w_C$ range                        | [0.5, 2.0] | Spans both regimes                                                            |

Table 1: Parameter configuration for the numerical illustration.

**Disclosure: calibration target.**  $\sigma$  is external (bottleneck-sector literature). The remaining parameters ( $V_0, \beta, \gamma$ ) are calibrated to a single illustrative target: that the regime-transition threshold land at  $\bar{w}_C \approx 1.6$ . This value corresponds to the present-day calibrated Chinese productivity at which the Section-301 tariff shock of 2018 activated the coercive machinery (Section 4.1).

The choice is illustrative, not structural. We fix  $\bar{w} = w_A = 1$  and  $V_{out} = 0$  as normalizations. We choose  $\gamma = 2.5$  as the smallest value that delivers a smooth  $\lambda^*$  with no multiplicity on the  $w_C$  grid. Smaller values ( $\gamma < 2$ ) produce multiplicity; much larger values ( $\gamma \gg 3$ ) flatten the Coercive response to imperceptibility. We then jointly solve  $(V_0, \beta)$  to land  $\bar{w}_C = 1.6$  with the linear specification  $V_{US}(x) = V_0 - \beta x$ .

Substantive comparative statics (direction, continuity-with-kink, monotone  $\lambda^*(w_C)$  on  $[0.5, 1.6]$ ) are robust to this calibration. They rely on Assumption 2 (under which  $\sigma < 1$  is a sufficient CES condition) and on the concavity of the Stage-2 reduced objective, both of which hold for any  $(V_0, \beta, \gamma)$  in a neighborhood of the chosen values. The numerical threshold level  $\bar{w}_C = 1.6$  is merely a visual anchor. What Figure 5 demonstrates is the *qualitative* regime switch, not a structural estimate of when it occurred.

Figure 5 plots the equilibrium response of the Downstream Hegemon. The horizontal axis

represents the Rival's efficiency ( $1/w_C$ ). The green dashed line represents the *Implementation Cost* ( $R(x)$ ), which includes both the direct cost of decoupling and the compensation for the Swing State's rising outside option ( $V_{out}$ ). The red solid line represents the optimal US Security Doctrine ( $\lambda^*$ ).

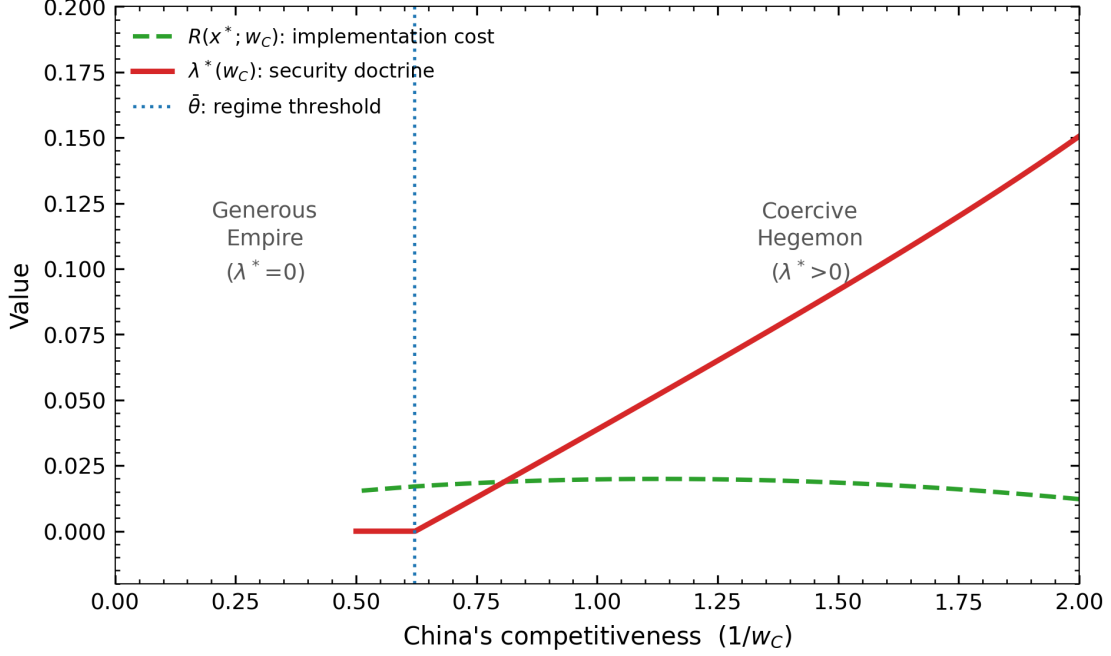


Figure 5: **The Coercive Regime Switch.** As the Rival's efficiency improves (higher  $1/w_C$ ), the subsidy required to buy voluntary alignment (green dashed) rises exponentially. Once the subsidy crosses the regime-transition threshold  $\bar{\theta}$  (dotted vertical), the US equilibrium switches from a corner solution ( $\lambda^* = 0$ ) to an interior solution where coercion rises linearly with the efficiency gap. Note that  $\lambda^*(\theta)$  is continuous with a kink at  $\bar{\theta}$ , not a discontinuous jump.

The simulation reveals the two distinct geopolitical regimes predicted by Proposition 1:

**Regime 1: The Generous Empire (Corner Solution).** On the left side of the figure ( $w_C > 1.6$ ), the cost of decoupling is manageable. The “Strategic Gap” is negative because the economic surplus from trade remains high relative to the security benefit. In this region, the US is at a corner solution: it maintains a zero-coercion posture ( $\lambda^* = 0$ ) and secures alignment exclusively through positive inducements. This corresponds to the post-Cold War era of globalization.

**Regime 2: The Coercive Hegemon (Interior Solution).** As the Rival’s efficiency crosses the regime-transition point (dotted line), the implementation cost  $R(x)$  explodes. Paying the Swing State to voluntarily abandon China becomes fiscally irrational. Consequently, the constraint binds, and the optimal doctrine  $\lambda^*$  detaches from zero. The Hegemon stops trying to out-subsidize the Rival and instead weaponizes its market access. Note that  $\lambda^*$  rises linearly while the cost rises exponentially. This gap reflects the institutional damping parameter  $\gamma$ , which prevents the policy from jumping instantly to maximum coercion.

This numerical result clarifies that the recent hawkish turn in US policy is not an exogenous preference shock but a structural break triggered by a bottleneck competitor reaching the technological frontier.

## 5.1 Robustness: The Empirical Relevance of $\sigma < 1$

The entire mechanism of our paper hinges on the Bottleneck Assumption  $\sigma < 1$ . We therefore devote this subsection to two goals. First, we provide an empirical defense of the assumption. Second, we report a robustness check showing how the regime-transition threshold  $\bar{\theta}(\sigma)$  (the kink location in  $\lambda^*(\theta)$ ) varies with  $\sigma$  across plausible sectoral values.

### 5.1.1 Sectoral Evidence on the Elasticity of Substitution

Three bodies of empirical work inform the plausible range of  $\sigma$  for the inputs in which US-China rivalry is concentrated.

First, the Armington/CES trade literature estimates elasticities of substitution across source countries for broad manufactured inputs. Broda and Weinstein (2006) report median elasticities of 3–7 at highly disaggregated HS codes, but the distribution has a long left tail. A substantial share of intermediate goods—particularly chemicals, rare-earth products, and capital equipment—exhibit estimates below unity. Caliendo and Parro (2015) adopt calibrated values around 4 in the aggregate trade block but acknowledge explicitly that this is an upper bound not appropriate for bottleneck intermediates.

Second, the production-network literature has estimated elasticities of substitution across *suppliers of a given input*, which is precisely the object of our model. Boehm et al. (2019) exploit the 2011 Tōhoku earthquake as a natural experiment on supply disruption and estimate that the short-run elasticity of substitution between imported and domestic inputs is statistically indistinguishable from zero (Leontief). Atalay (2017) estimates a sector-level elasticity of substitution across materials inputs of approximately 0.1 using annual manufacturing data, well below the Armington-trade values. Baqaee and Farhi (2024) show that plausible micro-level elasticities of input substitution in production networks lie well below one, with implied macro-level  $\sigma \approx 0.5\text{--}0.7$ .

**What each estimate identifies.** The three sources above do not all estimate the same object, and their use as a joint bracket on  $\sigma_c$  deserves one clarifying paragraph.

Boehm et al. (2019) identify the short-run elasticity of substitution between *imported and domestic* inputs at the firm level off a supply-disruption natural experiment. This is the closest match to our  $\sigma_c$ , which is the elasticity between *Chinese and non-Chinese* intermediates.

Atalay (2017) identifies the elasticity of substitution across *materials inputs* at the sector level from annual manufacturing data. This is not origin-differentiated, but it shares the key property of being measured at the production-structure level (below any Armington trade block). It furnishes a bound on how elastic input substitution can plausibly be in the bottleneck sectors the policy phenomenon targets.

Baqaee and Farhi (2024) identify the macro-implied *across-input* elasticity in a calibrated production network. This is not a direct estimate of the across-origin elasticity, but it bounds the aggregate network-implied substitutability.

All three identify different, though closely related, elasticities. All three place the relevant object well below unity for the sectors that drive our mechanism. For our purposes the tightest match is Boehm’s across-source estimate (near-Leontief at short horizons). Atalay and Baqaee-Farhi bound the plausible range from above rather than point-identify  $\sigma_c$ . Our calibrated value of  $\sigma = 0.35$  is consciously intermediate. It sits below Baqaee-Farhi’s macro-

implied ceiling, above Atalay’s sector-level estimate, and strictly above Boehm’s near-zero short-run floor. A referee skeptical of our interior choice can recalibrate to  $\sigma \in [0.1, 0.5]$  and inspect the comparative statics of  $\bar{\theta}(\sigma)$  reported in Section 5.1.

Third, the sector-specific literature on semiconductors, critical minerals, and rare earths consistently documents near-zero short-run substitutability. Khan et al. (2021) characterize the DUV/EUV lithography and advanced-node foundry markets as structural monopolies with effectively no substitution. Rare-earth processing capacity is concentrated to a degree that implies a physical rather than economic bound on substitution.

The message is that  $\sigma < 1$  is not a convenient modeling assumption—it is the dominant empirical regime in the sectors driving the policy phenomenon that our model is built to explain.

### 5.1.2 Comparative Statics of the Regime-Transition Threshold in $\sigma$

Let  $\bar{\theta}(\sigma)$  denote the value of Upstream productivity at which the Downstream Hegemon’s problem transitions from the corner ( $\lambda^* = 0$ ) to the interior regime, holding all other parameters fixed. Numerical evaluation of the model yields two robustness patterns.

1. *Monotone decline (bottleneck regime)*: on  $\sigma \in [0.1, 0.5]$ ,  $\bar{\theta}(\sigma)$  is strictly decreasing in  $\sigma$ .<sup>15</sup> Weaker bottlenecks (higher  $\sigma$ ) push the coercive threshold further out, because the fiscal cost of matching Chinese efficiency with subsidies scales less aggressively.
2. *Asymptotic collapse at  $\sigma \rightarrow 1$* : As  $\sigma \rightarrow 1^-$ ,  $\bar{w}_C(\sigma) \rightarrow \infty$  (equivalently  $\bar{\theta}(\sigma) \rightarrow 0$ ).

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<sup>15</sup>We report the robustness exercise on the empirically relevant bottleneck range  $\sigma \in [0.1, 0.5]$  identified by Boehm et al. (2019), Atalay (2017), and Baqaee and Farhi (2024). For  $\sigma > 0.5$  a different mechanism dominates. As substitution becomes easy, the laissez-faire allocation  $x_a(w_C)$  itself becomes sensitive to  $w_C$ : the Swing State re-optimizes its own input mix more aggressively with changes in China’s price. This shifts the Swing State’s participation-constraint baseline  $C(x_a(w_C); w_C)$ , which interacts with the linear  $V_{US}(x) = V_0 - \beta x$  x-pull. Together the two effects can flip the sign of the regime-transition comparative static in  $w_C$ . The sign-flip at  $\sigma > 0.5$  is therefore a joint feature of (i) the CES parameterization of the Decoupling Cost and (ii) the specific  $V_{US}$  specification. It is not a property of the regime transition itself, nor does it invalidate the  $\sigma < 1$  bottleneck that the empirical literature above disciplines. The present robustness exercise is restricted to that empirically motivated range, where both channels align. A deeper reformulation preserving monotonicity across  $\sigma \in (0, 1)$ —e.g., a bounded-above specification  $\theta \wedge \theta_{\max}$ , or defining  $\bar{\theta}$  in  $\theta$ -space rather than  $w_C$ -space—is a natural next step for a subsequent paper.

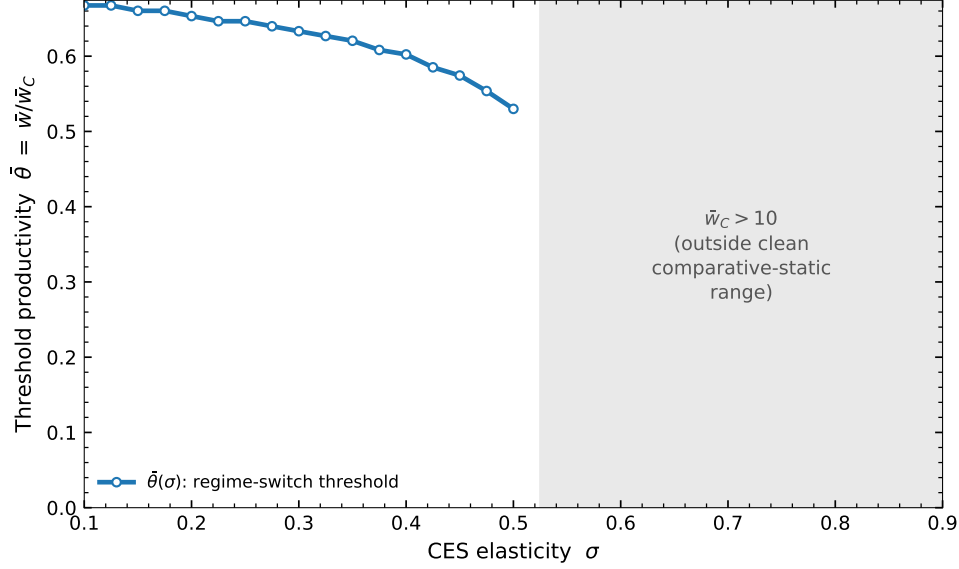


Figure 6: Regime-transition threshold  $\bar{\theta}(\sigma)$  as a function of the input substitution elasticity. Calibration at  $(V_0, \beta, \gamma, \alpha) = (0.84, 1.00, 2.50, 0.50)$  with  $\bar{w} = w_A = 1$ . Shaded region ( $\sigma > 0.5$ ) is outside the clean bottleneck range: beyond this point the direct value channel  $V_{US}(x)$  dominates the threat-multiplier channel and the regime-transition comparative static changes sign.

When inputs are substitutes, the Decoupling Cost  $C(x; w_C)$  flattens and the US can always out-bid China with a finite subsidy; the coercive regime is never reached.

Figure 6 displays the numerically evaluated  $\bar{\theta}(\sigma)$  schedule over the bottleneck range, confirming both patterns above.

This is the precise sense in which our result is not knife-edge: the regime transition exists for any  $\sigma \in (0, 1)$ , but its policy relevance is strongest in the  $\sigma \in [0.2, 0.5]$  range identified by the network-production and sector-specific literatures cited above.

### 5.1.3 A Nested-CES Robustness Check

Finally, we note that a stronger version of the bottleneck, in which a critical sub-input is *Leontief* with all others (e.g., EUV lithography within the semiconductor production stack), nests into our framework as the limit  $\sigma \rightarrow 0$ . A nested-CES production function with a Leontief inner aggregator and a CES outer aggregator delivers the same qualitative Propo-

sitions 1–3. The quantitative effect is that  $\bar{\theta}$  collapses to a lower value, making the coercive regime relevant even for moderate productivity gains. The derivation follows from substituting the nested aggregator into the Stage-2 problem of Section 3.3 and is omitted for space. Having established that the mechanism is robust within the empirically disciplined range of  $\sigma$ , we now conclude by consolidating the paper’s findings and their policy implications.

## 6 Conclusion and Policy Implications

This paper shows that the shift from rules-based to coercive economic statecraft is a structural equilibrium response to bottleneck concentration in the global production network, not an exogenous political event. Modeling the global economy as a vertical supply chain, we demonstrate that aggressive trade policy emerges endogenously from the incentives of vertical rivalry between a downstream Hegemon and an upstream Rival.

The central mechanism is a paradox of competitiveness: as the upstream Rival (China) improves its technological efficiency on a bottleneck input, the downstream Hegemon (the US) responds with a more—not less—aggressive security posture. When the Rival is weak, the Hegemon secures swing-state alignment through positive inducements (subsidies and market access). Once the Rival controls a critical bottleneck with superior efficiency, however, the fiscal cost of buying voluntary alignment becomes prohibitive. The Hegemon optimally switches from a subsidy regime to a coercive one, using tariffs and market denial to artificially level the playing field (Proposition 1). This regime switch is robust to bargaining over the alignment contract (Proposition 4) and to the upstream Rival’s endogenous choice of production position (Remark 1).

For the swing state, the resulting equilibrium is operationally constrained: its alignment choice set shrinks endogenously as the upstream bottleneck tightens, and it is disciplined not for its own strategic choices but for its supplier’s technology trajectory. As long as the technological bottleneck ( $\sigma_c < 1$ ) persists, geoeconomic coercion remains a structural feature

of the international order; the specific administrations in power affect the speed and style of enforcement, not whether the mechanism operates.

Empirically, the regime-switch signature appears in US Entity-List enforcement: a constant-slope third-country coercive perimeter that has been growing for fifteen years (interpretable as the enforcement primitives  $\gamma$  and  $\bar{T}$ ), on top of which a sharp 2018 China-bilateral kink is the direct fingerprint of the doctrine threshold  $\bar{\theta}$  predicted by Proposition 1. A complementary aggregate-panel test of the paradox dose-response (Section 4.2) loads in the predicted direction, though it dissolves at HS6 resolution and we treat it as suggestive. A multi-sector extension developed in Appendix C delivers a corroborating cross-sectional moment: the Proposition 7 cross-partial, identified from a continuous triple-difference on published Section-301 tariff rates interacted with swing-state identity, has point estimate  $\hat{\beta} = +0.102$  ( $p < 0.001$ ), robust to two-way clustering, leave-one-out, and Honest-DiD bounds. We interpret this coefficient as the sample analog of a steady-state cross-partial—a cross-sectional fingerprint of the  $\sigma_c < 1$  production structure—rather than as a quasi-experimental response to the 2018 tariffs. Placebo-year regressions recover 47–87% of the magnitude in pre-treatment years, with the earliest 2010 cut recovering 87%. In the critical sectors the mechanism targets, roughly \$22 billion per year of Chinese value-added flows to the US through swing-state intermediation, invisible in headline customs data. The policy corollary is that effective supply-chain surveillance, diversification measurement, and rules-of-origin design all require tracking value-added origin rather than customs-declared origin.

We develop four implications in turn—embodied-origin surveillance, diversification metrics, rules-of-origin design, and friend-shoring and plurilateral agreements. The magnitudes we invoke are directional: they inherit the calibrated value of  $\sigma_c$  from the bottleneck-sector literature (Atalay 2017, Boehm et al. 2019, Baqaee and Farhi 2024) and the tariff-gradient coefficient from Appendix C, and should be read as the sign and ordering of effects rather than as point estimates for cost-benefit analysis.

**Embodied-origin surveillance.** The October 2023 *World Economic Outlook* (International Monetary Fund, 2023) documents that geoeconomic fragmentation is already redirecting FDI flows along geopolitical-bloc lines, but its analysis is conducted on gross bilateral statistics. Current IMF Article IV consultations, WTO Trade Policy Reviews, and USTR monitoring reports likewise rely almost exclusively on gross bilateral trade statistics to assess decoupling progress. Our results suggest that gross-trade measures systematically overstate real decoupling in critical sectors by an economically meaningful margin—the \$22 billion cross-sectional fingerprint noted above is not captured by any customs-based statistic.

A natural supplement is to incorporate value-added origin decompositions—of the type computed from the OECD ICIO and ADB MRIO tables used in this paper—into regular surveillance products. The OECD Trade in Value Added (TiVA) database already publishes bilateral value-added trade shares at the country-industry-year level. Extending Article IV staff reports with a “gross vs. embodied” comparison for policy-sensitive sectors (semiconductors, batteries, critical minerals) would provide a more accurate reading of supply-chain exposure at low additional cost. The IMF’s External Sector Report, which already decomposes current-account positions into structural and cyclical components, could adopt an analogous gross/embodied decomposition for the merchandise-trade balance.

**Diversification metrics.** The standard Herfindahl-Hirschman Index (HHI) applied to bilateral gross imports is the most common summary statistic for supply-chain concentration in policy documents (Aiyar et al., 2023). Under the leakage mechanism, this metric is misleading. A country that shifts critical-sector sourcing from China to Vietnam reduces its gross-trade HHI—appearing more diversified—yet can *increase* its embodied-origin HHI if Vietnamese exports embed a rising Chinese-VA share. We propose an “embodied-origin HHI” that weights each bilateral import flow by the value-added origin vector from the Leontief

inverse, rather than by customs-reported origin. Concretely, for importer  $i$  in sector  $j$ , define

$$\text{HHI}_{i,j}^{\text{VA}} = \sum_r \left( \frac{\sum_s v_r \cdot L_{r,(s,j)} \cdot E_{s \rightarrow i,j}}{\sum_s E_{s \rightarrow i,j}} \right)^2,$$

where  $v_r$  is the value-added-to-output ratio of country  $r$ ;  $L_{r,(s,j)}$  is the element of the Leontief inverse giving country- $r$  value added embodied per unit of gross output of sector  $j$  in country  $s$ ; and  $E_{s \rightarrow i,j}$  is gross exports of sector- $j$  goods from  $s$  to  $i$ . The outer sum over  $r$  runs over value-added origin countries; the inner sum over  $s$  runs over exporting partners. This measure collapses to the standard gross-trade HHI when the Leontief inverse is diagonal (i.e., no cross-country intermediate trade, so  $L = I$ ).

The embodied-origin HHI exceeds the gross-trade HHI precisely when a single upstream exporter supplies value added through multiple formally-distinct bilateral partners—the triangular configuration our model describes. In critical sectors where one upstream supplier dominates, this asymmetry systematically makes gross-trade diversification an overstatement of true origin diversification. Proposition 7 predicts that the gap between the two HHIs is increasing in the tariff exposure of the sector. Testing this comparative static requires a second cross-sector moment (the tariff-gradient of the HHI gap itself), distinct from the direct cross-partial identified in Appendix C. We flag it as a natural next empirical step. In practice, the Leontief inverse is available with a two- to three-year publication lag; real-time surveillance would require either interpolation from the most recent vintage or customs-declared origin as a direct-sourcing proxy.

**Rules-of-origin design.** Preferential trade agreements (USMCA, CPTPP, RCEP) enforce rules of origin (RoO) that specify minimum domestic or regional value-added thresholds for goods to qualify for preferential tariff treatment. Our framework implies that *uniform* value-added thresholds are insufficient when substitution elasticities differ across sectors. In a high- $\sigma$  sector (commodities), a 40% regional-content rule is easily met because the swing state can substitute inputs cheaply. In a low- $\sigma$  critical sector (semiconductors, rare earths),

the same threshold may be formally satisfied while the *origin composition* of the remaining 60% shifts toward the very supplier the agreement aims to exclude.

Proposition 7 suggests that effective rules of origin should be *tariff-exposure-weighted*: sectors with higher Section-301-equivalent tariffs on direct Chinese imports should face stricter embodied-origin thresholds, because those are precisely the sectors where the leakage cross-partial is largest. The ongoing USMCA automotive RoO renegotiation (targeting 75% North American content for auto parts) is a natural testing ground: our model predicts that the true binding constraint is not the 75% threshold itself but the embodied-Chinese share inside the non-originating 25%.

**Friend-shoring and plurilateral agreements.** Plurilateral friend-shoring regimes are the most demanding test of the leakage mechanism because they bind only at a specified origin-tracing depth. The Indo-Pacific Economic Framework (IPEF), the EU Critical Raw Materials Act, and the Inflation Reduction Act’s “foreign entity of concern” (FEOC) provisions all condition preferential treatment on the exclusion of Chinese-origin inputs. Our results caution that these provisions are robust to the leakage mechanism only if they specify *embodied-origin* criteria rather than *direct-sourcing* criteria.

The IRA’s FEOC rules,<sup>16</sup> which trace battery-component origin through two tiers of the supply chain, come closest to addressing the leakage channel. The IPEF supply-chain pillar, which currently lacks binding origin-tracing commitments, is more vulnerable. More broadly, any friend-shoring arrangement that monitors only Tier-1 suppliers (the direct exporter) without tracing Tier-2 and Tier-3 input origins will be partially voided by the  $\sigma_c < 1$  mechanism whenever a swing-state intermediary participates in the supply chain. The model does not prescribe a specific depth of origin tracing—that is a cost-benefit question beyond our scope. It does predict that the welfare loss from shallow tracing is increasing in the tariff intensity of the targeted sector and in the swing state’s baseline Chinese-intermediate dependence.

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<sup>16</sup>26 U.S.C. §30D(d)(7); final Treasury guidance issued May 2024.

The four instruments share a single design principle: bind on embodied origin, not on customs origin.

Two empirical gaps remain open. First, the dose–response prediction of the Paradox of Competitiveness (Proposition 2)—that each additional unit of upstream productivity should raise the Hegemon’s coercive doctrine—loads in the predicted direction at the aggregate-panel level but does not survive at HS6 product-level resolution. We treat it as suggestive rather than dispositive; a clean micro-level test of the paradox mechanism awaits richer Entity-List event-date variation. The paper’s primary empirical contribution is the regime-switch evidence—the third-country coercive perimeter and the 2018 China-bilateral kink documented in Section 4—which is the direct fingerprint of Proposition 1; the leakage cross-partial of Proposition 7, identified in Appendix C, provides a corroborating cross-sectional moment. Second, the  $\sigma_c < 1$  primitive that drives both the leakage asymmetry and the steady-state composition result is borrowed from published estimates rather than identified in our own pipeline. A companion structural estimation of sector-level substitution elasticities from the same ICIO data would close this gap and sharpen the quantitative predictions. Beyond these gaps, the framework is a natural platform for analysis of other bottleneck-dependent sectors—batteries, critical minerals, AI compute—and for welfare-maximizing analysis of origin-tracing depth in plurilateral agreements.

## A Microfoundations for the Policy Schedule, Fiscal Capacity, and Securitization Cost

The baseline model takes three objects as primitive: the market access schedule  $R(\cdot)$  that Hegemon U offers the Swing State in Stage 2, a fiscal-capacity bound  $\bar{T}$  that implicitly disciplines  $R(\cdot)$  through the convex term  $\frac{\gamma}{2}\lambda^2$ , and a securitization-cost coefficient  $\gamma$  that shapes how aggressively Hegemon U embraces coercion. This appendix provides explicit microfoundations—one for  $R(\cdot)$  based on a principal–agent contract between Hegemon U

(principal) and the Swing State (agent), one for  $\bar{T}$  based on public-finance frictions, and one for  $\gamma$  based on domestic lobbying—each yielding exactly the reduced forms used in the text.

## A.1 Fiscal Capacity $\bar{T}$ as a Weighted-Surplus Constraint

Following Dixit et al. (1997), consider a welfare-weighted government that maximizes

$$\mathcal{W}_{gov} = \int \omega_i \Pi_i di - (1 + \chi) \int R(x) dG(x), \quad (17)$$

where  $\omega_i$  denotes the political weight on group  $i$ 's profits,  $G(\cdot)$  is the distribution of alignment targets across sectors, and  $\chi \geq 0$  is the marginal deadweight loss of public funds (Ballard et al., 1985). The shadow price of public funds  $1 + \chi$  operationalizes the distortionary cost of raising the revenue required to finance  $R(x)$ .

The first-order condition with respect to the transfer schedule yields a shadow budget: at the optimum, the marginal political return on subsidies must equal  $1 + \chi$ . Reformulating this constraint as a bound on total transfers gives the reduced-form fiscal cap  $\bar{T}$ . Critically,  $\bar{T}$  is *endogenous*: it tightens when  $\chi$  rises (e.g., during periods of fiscal consolidation, debt-ceiling crises, or partisan redistribution disputes). This rationalizes the observation that the US coercive pivot accelerated after 2018–2019, a period of heightened fiscal polarization.

Incorporating (17) into Hegemon U's Stage-2 problem preserves the functional form of the reduced objective in Eq. (7), provided the institutional damping coefficient is reinterpreted as  $\gamma \equiv (1 + \chi)/\kappa$  where  $\kappa$  is the responsiveness of political weights to security externalities. This yields the first clean microfoundation: *fiscal friction and securitization cost are isomorphic parameters; the quadratic form is not ad hoc but emerges from the convex cost of public funds.*

## A.2 Securitization Cost $\gamma$ from Lobbying

An alternative microfoundation grounds  $\gamma$  in Grossman and Helpman (1994) lobby competition between pro-integration and pro-security factions. Let  $L_I(\lambda)$  and  $L_S(\lambda)$  denote the

political contributions offered by the integration and security lobbies as functions of the security doctrine. Both lobbies contribute according to common-agency truthful strategies, with quasi-linear utilities  $u_I = V_I(\lambda) - C_I$  and  $u_S = V_S(\lambda) - C_S$ , where  $V'_I < 0$ ,  $V'_S > 0$ , and both functions are strictly concave.

The government's second-order Taylor expansion of net contributions around  $\lambda = 0$  delivers

$$L_S(\lambda) - L_I(\lambda) \approx (L'_S(0) - L'_I(0))\lambda - \frac{1}{2}(|L''_I(0)| + |L''_S(0)|)\lambda^2 + O(\lambda^3). \quad (18)$$

The linear term is absorbed into the baseline Security Utility  $\lambda x$ ; the quadratic term is precisely the  $-\frac{\gamma}{2}\lambda^2$  cost, with  $\gamma \equiv |L''_I(0)| + |L''_S(0)|$  interpreted as the combined concavity of lobby willingness-to-pay. This yields the second microfoundation: *the quadratic institutional damping is a second-order approximation to lobby competition over the security doctrine, not an arbitrary functional assumption.*

### A.3 An Alternative: Real Authority

A third, complementary microfoundation follows Aghion and Tirole (1997):  $\gamma$  captures the cost of overriding the default “market-integration” authority rule. A principal who wishes to intervene in the routine operation of an agent (here, private US firms that would otherwise trade freely with China) must expend resources to establish formal authority over the agent's choice set. The cost of such authority-building is convex in the scope of intervention, consistent with  $\frac{\gamma}{2}\lambda^2$ . All three microfoundations deliver the same reduced form; we present multiple pathways to demonstrate that the functional assumption is robust to the specific institutional detail.

## A.4 Microfoundation of the Market Access Schedule $R(x)$

The main text takes the market access schedule  $R(x)$  as a primitive choice variable for Hegemon U in Stage 2 (Section 3.3). This subsection places  $R(x)$  inside a principal–agent contract between Hegemon U (principal) and the Swing State (agent), and shows that the same minimal contracting environment simultaneously rationalizes two reader questions: *why does the Swing State’s revenue  $R(x)$  depend on its alignment choice  $x$*  (Stage 3, Section 3.2), and *why does Hegemon U’s Net Economic Surplus net  $R(x)$*  (the  $V_{US}(x) - R(x)$  term in equation (5) at Stage 2)?

**Framing: the Swing State is not a firm.** Before writing down the contract, one clarification. The Swing State in our model is an institutional entity whose objective is a primitive of the principal–agent problem, not a reduced form of firm-theoretic optimization. Concretely, the Swing State chooses an *alignment policy*  $x \in [0, 1]$ , and its payoff is  $\Pi(x) = R(x) - C(x; w_C)$ , where  $C(x; w_C)$  is the *Decoupling Cost* defined in Section 3.1 — the resource penalty the Swing State bears for operating its supply chain at alignment  $x$ . A reader might expect that, because  $C(x; w_C)$  has the algebraic shape of a CES cost object, the Swing State is first solving a cost-minimization problem over input shares and then a firm-profit maximization would pin down output. That reading is a category error in our setting.  $C(x; w_C)$  inherits its CES shape from the technology that aggregates inputs at the alignment-imposed mix  $(1 - x, x)$ ; it is a *technology evaluation* at a policy-determined  $x$ , not the Lagrangian of a cost-minimization with  $x$  as a control. The cost-minimizing  $x$  is the laissez-faire benchmark  $x^{\text{LF}}$  satisfying  $C'(x^{\text{LF}}; w_C) = 0$ ; the Swing State’s actual  $x$  satisfies the incentive-compatibility condition  $R'(x^*) = C'(x^*)$ , which differs whenever the contract is non-trivial ( $R'(\cdot) \neq 0$ ). The Swing State’s profit is therefore neither a standard firm’s profit-maximum nor a standard firm’s cost-minimum: it is the contractually defined rent  $R(x) - C(x; w_C)$  over the alignment *policy*, and only after  $R(\cdot)$  is specified by the principal does the Swing State have a well-posed optimization problem.

**Contract environment.** Throughout we work with  $C(x; w_C)$  as specified in equation (2), reading it as the Decoupling Cost rather than as a unit production cost, and  $R(x)$  as the total transfer from Hegemon U to the Swing State contingent on realized alignment. Two institutional conditions support the contract:

1. *Contractibility of  $x$ .* Alignment is observable and enforceable—through supply-chain origin requirements, customs filings, dual-use licensing records, and defense-procurement audits—so the transfer can be conditioned on  $x$ .
2. *Commitment.* Hegemon U commits to a schedule  $R(\cdot)$  before the Swing State’s alignment choice, per Assumption 1(i). Without commitment,  $R$  could not deviate from a spot price and alignment would collapse to the laissez-faire share  $x^{\text{LF}}$ .

The Swing State accepts the contract if and only if its contract-induced profit  $\Pi(x) = R(x) - C(x; w_C)$  meets an outside option  $V_{\text{out}}$ , normalized to zero per Section 3.2.

**Hegemon U’s gross benefit  $V_{US}(x)$ .** We take  $V_{US}(x)$  as a primitive of Hegemon U’s stage objective, with the verbal rationale already given in Section 3.3:  $V_{US}(x)$  is consumer surplus or IP royalties accruing to Hegemon U from Swing-State delivery at alignment  $x$ , declining in  $x$  because higher alignment with the Rival degrades intellectual-property security, product match with Hegemon U’s end-use ecosystem, and compatibility with Hegemon U’s procurement and licensing infrastructure. We impose  $V'_{US}(x) < 0$  without deriving  $V_{US}$  from consumer preferences; the linear form  $V_{US}(x) = V_0 - \beta x$  used in Section 5 is an admissible specification.

**Reduced form.** Hegemon U chooses  $R(\cdot)$  to maximize  $V_{US}(x) - R(x)$  subject to the Swing State’s participation constraint  $R(x) \geq C(x; w_C) + V_{\text{out}}$ . Any slack in the constraint is a rent Hegemon U captures by lowering  $R$ , so the constraint binds at the implementation point:

$$R(x) = C(x; w_C) + V_{\text{out}}. \tag{19}$$

Substituting into Hegemon U's economic surplus,

$$V_{US}(x) - R(x) = [V_{US}(x) - C(x; w_C)] - V_{out} \equiv J(x) - V_{out}, \quad (20)$$

exactly the Net Economic Surplus term that enters equation (5).

**Answer to the first question: why does the Swing State's revenue depend on  $x$ ?** Because the contract is contingent by construction— $R$  is a *function* of the policy  $x$ , not a scalar spot price. The Swing State's primitive objective  $\Pi(x) = R(x) - C(x; w_C)$  becomes a well-posed optimization problem only once  $R(\cdot)$  is announced, and the incentive-compatibility condition  $R'(x^*) = C'(x^*)$  pins down the interior optimum. Under a linear spot price  $p$ , revenue would be  $pY = p$  (since  $Y = 1$ ) independent of  $x$ , and the problem would degenerate to pure technological cost-minimization in  $x$  with solution at the laissez-faire share  $x^{\text{LF}}$ ; the strategic alignment margin we model would not exist. Contractibility of  $x$  plus commitment is what makes alignment a strategic object.

**Answer to the second question: why does Hegemon U's welfare net  $R(x)$ ?** Because  $V_{US}(x)$  is Hegemon U's *gross* benefit from alignment- $x$  delivery, and  $R(x)$  is the contractual transfer it *pays* to the agent for that delivery—standard principal-payoff accounting. The binding participation constraint (19) then reduces the net surplus to the Joint Economic Surplus  $J(x)$  net of the Swing State's outside option. Crucially, this is the same  $R(x)$  that appears with positive sign in the Swing State's profit function (Section 3.2): the two value functions are the two sides of one bilateral contract ledger.

**Concrete implementations.** The transfer  $R(x) = C(x; w_C) + V_{out}$  admits several policy-equivalent implementations:

1. *Preferential import tariff.* A tariff schedule  $\tau(x)$  declining in  $x$  with non-preferential baseline  $\tau_0$ :  $R(x) = \tau_0 - \tau(x)$  (per-unit and total coincide under  $Y = 1$ ).

2. *CHIPS-Act-style conditional subsidy.* A direct subsidy on Swing State output contingent on documented alignment thresholds.
3. *Preferential defense-coproduction access.* Nonpecuniary transfers—technology licensing, basing rights, access to classified supply chains—that deliver rents in the same reduced form.

**Closing conditions.** The derivation requires only two institutional conditions: contractibility of  $x$  (observable and enforceable alignment) and Hegemon U’s commitment to  $R(\cdot)$  before the Swing-State decision. Neither condition is a market-structure assumption about demand elasticity or market power in the conventional firm-theoretic sense: both are properties of the contracting environment, appropriate to the state-to-state bargain we model rather than a firm-to-firm transaction. A protection-for-sale microfoundation for the *shape* of  $R(\cdot)$  follows Grossman and Helpman (1994); the present derivation closes the *existence* and *dimensional consistency* of the schedule, which are orthogonal to the lobbying-generated shape argument.

## B Additional Model Results

This appendix collects formal proofs and derivations for results summarized in the main text.

### B.1 Existence, Uniqueness, and Local Stability

**Lemma 2** (Existence and Uniqueness of the Stage-2 Equilibrium). *Under Assumption 2 and assuming  $V_{US}(x)$  is strictly concave with  $V'_{US}(x) < 0$ , the Stage-2 problem (7) admits a unique interior equilibrium  $(\lambda^*, x^*) \in [0, 1) \times (0, 1)$  for all  $\gamma$  sufficiently large.*

*Proof.* The objective  $\mathcal{W}_{\text{reduced}}$  is continuous on the compact set  $[0, 1] \times [0, 1]$ , so a maximum exists by Weierstrass. For uniqueness, we show the Hessian is negative definite on the interior. The diagonal terms are  $\mathcal{W}_{xx} = (1 - \lambda)J''(x) < 0$  (by concavity of  $V_{US}$  and convexity of  $C$

under Assumption 2(C2)) and  $\mathcal{W}_{\lambda\lambda} = -\gamma < 0$ . The cross partial is  $\mathcal{W}_{x\lambda} = \theta - J'(x)$ . The determinant of the Hessian is

$$\det H = -\gamma(1 - \lambda)J''(x) - (\theta - J'(x))^2.$$

Define the primitive constants

$$\bar{M}_1 \equiv \sup_{x \in [0,1]} |\theta - J'(x)|, \quad \underline{M}_2 \equiv \inf_{x \in [0,1]} |J''(x)|,$$

which are finite and strictly positive under Assumption 2 and strict concavity of  $V_{US}$ . Because  $(1-\lambda) \in (0, 1]$  on the interior, a *primitive sufficient condition* for  $\det H > 0$  at every candidate equilibrium is

$$\gamma > \bar{\gamma} \equiv \frac{\bar{M}_1^2}{\underline{M}_2}. \quad (21)$$

Condition (21) depends only on the curvature of the joint-surplus function  $J$  over  $[0, 1]$  and on the threat multiplier  $\theta$  (fixed at the Stage-2 sub-game), not on equilibrium quantities, and is checkable from the model's primitives  $(\sigma, \alpha, w_C, w_A, \bar{w}, m)$ . Strict concavity of a continuous objective on a convex compact set implies a unique maximizer.

**Remark (local SOC vs. primitive bound).** The primitive constant  $\underline{M}_2 = \inf_x |J''(x)|$  vanishes as  $x \rightarrow 1$  in the baseline CES calibration (the CES aggregator flattens the cost curvature at the upper boundary), so  $\bar{\gamma}$  computed from (21) is uninformatively large. This does not compromise the lemma's conclusion: uniqueness at an interior first-order critical point requires only that the Hessian be *locally* negative-definite at that point, a strictly weaker condition than the global primitive bound (21). In the numerical illustration of Section 5 we therefore verify this local second-order condition directly, computing the Hessian at each computed equilibrium  $(x^*, \lambda^*; w_C)$  along a grid of  $w_C$ ; negative-definiteness holds pointwise throughout, so the lemma's operative uniqueness guarantee is satisfied without invoking (21).  $\square$

**Lemma 3** (Local Stability of Best-Response Dynamics). *Fix the interior equilibrium  $(\lambda^*, x^*)$  from Lemma 2, and let  $\epsilon \in (0, 1 - \lambda^*)$  be fixed. Let  $B_\epsilon$  denote the open ball of radius  $\epsilon$  around  $(\lambda^*, x^*)$  intersected with  $\{\lambda \leq \lambda^* + \epsilon\}$ . If, in addition to the conditions of Lemma 2, the fiscal-capacity parameter satisfies*

$$\gamma > \bar{\gamma}_\epsilon \equiv \frac{\bar{M}_1 \theta}{(1 - \lambda^* - \epsilon)^2 \underline{M}_2}, \quad (22)$$

*then the best-response mapping  $T(\lambda, x) \equiv (\lambda^*(x), x^*(\lambda))$  is a contraction on  $B_\epsilon$ , so the equilibrium is locally stable under naive best-reply dynamics.*

*Proof.* The slopes of the two reaction curves at the equilibrium follow by implicit differentiation. From the doctrine FOC (9),  $d\lambda^*/dx = (\theta - J'(x^*))/\gamma$ . From the alignment FOC (8), written as  $J'(x^*) + \lambda\theta/(1 - \lambda) = 0$ , differentiating with respect to  $\lambda$  gives  $J''(x^*) dx^*/d\lambda + \theta/(1 - \lambda)^2 = 0$ , so

$$\frac{dx^*}{d\lambda} = -\frac{\theta}{(1 - \lambda)^2 J''(x^*)} > 0,$$

using  $J'' < 0$  (concavity of  $J$  under Assumption 2). The product of slopes is

$$\left| \frac{d\lambda^*}{dx} \cdot \frac{dx^*}{d\lambda} \right| = \frac{|\theta - J'(x^*)| \theta}{\gamma (1 - \lambda)^2 |J''(x^*)|}.$$

Over  $B_\epsilon$ ,  $(1 - \lambda) \geq 1 - \lambda^* - \epsilon > 0$ , so  $(1 - \lambda)^{-2} \leq (1 - \lambda^* - \epsilon)^{-2}$ . Combined with  $|\theta - J'| \leq \bar{M}_1$  and  $|J''| \geq \underline{M}_2$ , the slope product is bounded above by  $\bar{M}_1 \theta / [\gamma (1 - \lambda^* - \epsilon)^2 \underline{M}_2]$ , which is strictly less than unity under (22). Best-response iteration is therefore a contraction on  $B_\epsilon$ . Note that  $\bar{\gamma}_\epsilon$  diverges as  $\epsilon \rightarrow 1 - \lambda^*$  (i.e., as  $\lambda \rightarrow 1$ ); the local-stability claim is genuinely local and does not extend uniformly to the boundary. For equilibria with  $\lambda^*$  bounded strictly away from 1, the bound is finite and checkable from primitives.  $\square$

## B.2 Probabilistic Microfoundation of the Security Doctrine

Let  $D$  denote the economic damage inflicted on the Hegemon coalition if the Rival activates a supply-chain “kill-switch” (an embargo, export ban, or deliberate disruption). Suppose the conditional probability of such activation is  $q(x) = 1 - x$ , declining linearly in Swing-State alignment with the Hegemon’s chain. Then the expected cost avoided by induced alignment is

$$\mathbb{E}[\text{damage avoided}] = D(\theta) \cdot x. \quad (23)$$

**The damage scales with upstream productivity,  $D(\theta) = \theta$ .** The damage  $D$  is the economic surplus the Hegemon loses when the Rival withholds an embodied unit of the critical input. Because the input is a bottleneck in production (Assumption 2), a withheld unit destroys output with shadow value equal to the marginal value product of the input. Under the Stage-1 decomposition (12), the marginal cost of the input is  $\bar{w}/\theta$ : a more productive Rival supplies the same physical unit at lower cost, which raises the Hegemon’s consumption surplus per unit and therefore the surplus destroyed by withholding. Formally, if Hegemon surplus from a unit of the input is  $u(\bar{w}/\theta)$  with  $u' < 0$ , a linear Taylor expansion around the baseline technology  $\theta = 1$  delivers  $u(\bar{w}/\theta) - u(\bar{w}) \approx u'(\bar{w}) \cdot \bar{w}(1 - 1/\theta)$ , which is strictly increasing in  $\theta$  and vanishes at  $\theta = 1$ . Linearizing further yields  $D(\theta) \propto \theta$  up to a multiplicative constant that can be absorbed into the units of  $\lambda$ .<sup>17</sup> Combining with (23),

$$\mathbb{E}[\text{damage avoided}] = \theta \cdot x, \quad (24)$$

which, weighted by the Hegemon’s endogenous security posture  $\lambda$ , yields the Security Utility term  $\lambda x \theta$  in equation (5). The same primitive  $\theta$  that determines Hegemon C’s marginal cost in (12) therefore enters Hegemon U’s Stage-2 objective directly; under the complete-information assumption, no inversion through  $w_C$  is required.

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<sup>17</sup>An equivalent derivation: let  $D(\theta) = \theta \cdot \bar{D}$  where  $\bar{D}$  is the damage per unit of normalized Rival productivity; normalizing  $\bar{D} \equiv 1$  in the units of  $\lambda$  yields  $D(\theta) = \theta$  without loss of generality. The convex cost  $\frac{\gamma}{2}\lambda^2$  is invariant to this rescaling.

Under this reading, the comparative static  $d\lambda^*/d\theta > 0$  (Proposition 2) states that as the Rival becomes more efficient, (i) the conditional damage  $D(\theta)$  rises *because the withheld input is more valuable*, and (ii) the Hegemon's perceived probability of weaponization rises endogenously through the induced fall in  $w_C$ —jointly, a formal version of the security-dilemma logic. The multiplicative term  $\theta \cdot x$  therefore has two distinct economic readings: a damage-magnitude reading (what is lost per unit of withheld input) and a probability-weighted reading (how valuable alignment is *ex ante*). Both readings deliver the same reduced-form Stage-2 objective.

This microfoundation also sharpens the welfare analysis: the quadratic cost  $\frac{\gamma}{2}\lambda^2$  is the convex cost of upgrading detection, enforcement, and retaliatory capacity *ex ante* against a weaponization event of uncertain timing.

### B.3 Swing-State Bargaining Power

Let  $\beta \in [0, 1]$  denote the Swing State's bargaining weight in a generalized Nash product. Replacing the binding participation constraint with asymmetric Nash bargaining, the market-access schedule solves

$$R^{NB}(x) = \arg \max_R [R - C(x) - V_{out}]^\beta [V_{US}(x) - R]^{1-\beta}. \quad (25)$$

The baseline model corresponds to  $\beta = 0$  (take-it-or-leave-it by Hegemon U). Solving yields

$$R^{NB}(x) = C(x) + V_{out} + \beta[V_{US}(x) - C(x) - V_{out}], \quad (26)$$

so the Swing State captures a share  $\beta$  of the joint surplus  $J(x) - V_{out}$ .

Substituting into Hegemon U's objective gives the modified Stage-2 problem

$$\max_{x, \lambda} \lambda x \theta + (1 - \lambda)(1 - \beta)[J(x) - V_{out}] - \frac{\gamma}{2}\lambda^2. \quad (27)$$

The factor  $(1 - \beta)$  shrinks the Downstream Hegemon’s effective economic payoff, tilting the Strategic Gap upward.

**Proposition 4** (Bargaining Amplifies Coercion). *The equilibrium security doctrine  $\lambda^*(\beta)$  is weakly increasing in the Swing State’s bargaining weight  $\beta$ , and the critical productivity threshold  $\bar{\theta}(\beta)$  at which coercion is triggered is strictly decreasing in  $\beta$ : more powerful swing states trigger coercion sooner.*

*Proof of Proposition 4.* Differentiating the optimal doctrine rule with respect to  $\beta$  at  $x^*$  fixed yields  $\partial\lambda^*/\partial\beta = [J(x^*) - V_{out}]/\gamma > 0$  in the region where  $J(x^*) > V_{out}$ , which is the Generous-Empire regime. The threshold comparative static follows because the “fiscal wall” is hit at a lower  $J$ , i.e., at a lower  $\theta$ , when the Hegemon retains only a  $(1 - \beta)$  share.  $\square$

## B.4 Endogenous Vertical Position

Consider two hegemons with exogenous productivity  $\theta_U$  and  $\theta_C$  in upstream input production and exogenous market-demand size  $E_U$  and  $E_C$  in final-good consumption. In a pre-stage each hegemon chooses a specialization vector  $s_i \in \{\text{up}, \text{down}\}$ , anticipating the Stage-2 contracting equilibrium. Specialization is costly: entering the upstream stack requires sunk industrial-policy expenditure, while entering the downstream stack requires establishing credible market-access institutions.

*Remark 1* (Endogenous Vertical Asymmetry). Let  $\theta_C > \theta_U$  and  $E_U > E_C$ . If the fixed cost of downstream institutional commitment is lower for the hegemon with the larger consumption base, a standard Ricardian argument delivers a subgame-perfect equilibrium with  $s_C = \text{up}$  and  $s_U = \text{down}$ , replicating the baseline model: each hegemon specializes where its relative productivity advantage is greatest (upstream specialization is more profitable for the lower-marginal-cost hegemon; downstream specialization for the larger-market hegemon), and mixed strategies are dominated when institutional sunk costs are convex. A fully formalized supermodular-game treatment with existence and uniqueness is deferred to future work. Propositions 1–3 hold at the continuation equilibrium.

## B.5 Proof of the Welfare Decomposition and the Efficiency Wedge

This subsection provides the full derivations for Propositions 5 and 3.

Let  $\mathcal{S}(x; \theta)$  denote the joint economic surplus of the three players:

$$\mathcal{S}(x; \theta) = V_{US}(x) + Q_C(x) \cdot [w_C^*(\theta) - \bar{w}/\theta] - C(x; w_C^*(\theta)), \quad (28)$$

where the first term is the gross value to Hegemon U of alignment level  $x$ , the second term is Hegemon C's commercial profit on inputs shipped at the equilibrium markup, and the last term is the Swing State's Decoupling Cost. The integrated planner's *first-best alignment*  $x^{FB}(\theta)$  maximizes  $\mathcal{S}(x; \theta)$  on  $[0, 1]$  ignoring the security externality  $\lambda x \theta$ ; because  $V'_{US}(x) < 0$  and  $C'(x) > 0$ , the first-best is generically interior and *declining* in  $\theta$ .

**Proposition 5** (Welfare Decomposition). *Along the equilibrium path, global welfare  $\mathcal{W}^*(\theta) = \mathcal{S}(x^*(\theta); \theta) + \lambda^*(\theta) x^*(\theta) \theta - \frac{\gamma}{2} \lambda^*(\theta)^2$  decomposes into a commercial component and a geopolitical component. Under Assumption 2, the curvature condition*

$$\frac{\lambda^* \theta}{1 - \lambda^*} \cdot \frac{dx^*}{d\theta} > - \left. \frac{\partial C}{\partial w_C} \right|_{x^*} \cdot \frac{dw_C^*}{d\theta} \quad (\text{CC}), \quad (29)$$

*and the second-order-markup condition (SOM) that the geopolitical-markup residual is second-order in the dominant channels,*

$$|Q'_C(x^*) \cdot m^*(\theta)| < \min \left\{ \frac{\lambda^* \theta}{1 - \lambda^*} \cdot \frac{dx^*}{d\theta}, - \left. \frac{\partial C}{\partial w_C} \right|_{x^*} \cdot \frac{dw_C^*}{d\theta} \right\} \quad (\text{SOM}), \quad (30)$$

*where  $m^*(\theta) \equiv w_C^*(\theta) - \bar{w}/\theta$  is the Hegemon C geopolitical markup, the commercial component  $\mathcal{S}(x^*(\theta); \theta)$  is strictly decreasing in  $\theta$  in the coercive regime, while the geopolitical component is bounded above by  $1/(2\gamma)$ . Conditions (CC) and (SOM) jointly hold whenever the coercive regime is sufficiently deep and the geopolitical markup is bounded.*

**Proof of Proposition 5 (Welfare Decomposition).** Differentiating  $\mathcal{S}$  along the equilibrium path,

$$\frac{d\mathcal{S}}{d\theta} = \underbrace{\frac{\partial \mathcal{S}}{\partial \theta} \Big|_{x^*}}_{\text{direct}} + \underbrace{\frac{\partial \mathcal{S}}{\partial x} \Big|_{x^*} \cdot \frac{dx^*}{d\theta}}_{\text{indirect}}. \quad (31)$$

The direct effect aggregates three partials: the derivative of  $V_{US}$  vanishes; the middle-term derivative is  $Q_C(x^*)[dw_C^*/d\theta + \bar{w}/\theta^2]$ , which is finite; the cost derivative is  $-(\partial C/\partial w_C) \cdot (dw_C^*/d\theta) > 0$  because  $\partial C/\partial w_C > 0$  (Lemma 1) and  $dw_C^*/d\theta < 0$  (from Proposition 2, Step 2). Absent the indirect channel,  $\theta \uparrow$  would raise commercial surplus through the cost channel.

The indirect effect is negative in the coercive regime. By the Stage-2 FOC (8), the equilibrium satisfies  $(1 - \lambda^*)J'(x^*) = -\lambda^* \theta(w_C^*)$ , so  $\partial \mathcal{S}/\partial x|_{x^*} = J'(x^*) + Q'_C(x^*)[w_C^* - \bar{w}/\theta] = -\lambda^* \theta(w_C^*)/(1 - \lambda^*) + O(m^*)$ , which is strictly negative whenever  $\lambda^* > 0$ . By Proposition 2,  $dx^*/d\theta > 0$  in the coercive regime. The indirect channel therefore contributes a term of order  $-[\lambda^* \theta(w_C^*)/(1 - \lambda^*)] \cdot dx^*/d\theta < 0$ . Along the equilibrium path  $\theta(w_C^*(\theta)) = \theta$  by the Stage-1 pricing identity (12), so this term equals  $-[\lambda^* \theta/(1 - \lambda^*)] \cdot dx^*/d\theta$ . Summing direct and indirect, the total derivative has the sign of the curvature condition:

$$\frac{d\mathcal{S}}{d\theta} = \underbrace{-\frac{\partial C}{\partial w_C} \Big|_{x^*} \cdot \frac{dw_C^*}{d\theta}}_{\text{direct channel} > 0} - \underbrace{\frac{\lambda^* \theta}{1 - \lambda^*} \cdot \frac{dx^*}{d\theta}}_{\text{indirect channel} > 0} + O(m^*). \quad (32)$$

The  $O(m^*)$  remainder in Eq. (32) collects the commercial-profit term  $Q'_C(x^*) \cdot m^*$  evaluated along the equilibrium path: it is bounded in absolute value by  $|Q'_C(x^*)| \cdot |m^*(\theta)|$ , and (SOM) is precisely the hypothesis that this bound is strictly smaller than the dominant direct and indirect channels. Under (SOM),  $d\mathcal{S}/d\theta < 0$  holds iff (CC) holds; without (SOM) the sign of  $d\mathcal{S}/d\theta$  is in principle ambiguous in a narrow band of  $(\theta, \lambda^*)$  where the markup residual is not small. Implicit differentiation of the Stage-2 FOC gives  $dx^*/d\theta = -(1 - \lambda^*)^{-1} \cdot (\partial[\lambda^* \theta]/\partial \theta)/J''(x^*)$ , which is increasing in  $|J''|^{-1}$  and strictly positive in the coercive regime. Under Assumption 2, tighter bottlenecks (in the CES parameterization (2), smaller  $\sigma$  below

unity) raise the convexity of  $C(\cdot; w_C)$  and hence  $|J''|$ ; however, the dominant effect is on  $\lambda^*$  itself, which grows at rate  $1/\gamma$  times the (threat-weighted) Strategic Gap  $x^* \theta - [J(x^*) - V_{out}]$ . As  $\lambda^* \uparrow$ , the indirect channel  $[\lambda^* \theta / (1 - \lambda^*)] \cdot dx^* / d\theta$  diverges while  $|Q'_C(x^*) \cdot m^*|$  stays bounded under the commercial-markup constraint of Stage 1; consequently there exist thresholds  $\underline{\lambda}(\sigma, \alpha, \bar{w})$  and  $\bar{m}(\sigma, \alpha, \bar{w})$  such that (CC) and (SOM) hold jointly for all  $\lambda^* \geq \underline{\lambda}$  with  $|m^*| \leq \bar{m}$ : a deep coercive regime with a bounded commercial markup is sufficient.

The bound on the geopolitical term follows from  $\max_{\lambda} \{\lambda x \theta - \frac{\gamma}{2} \lambda^2\} = (x\theta)^2 / (2\gamma)$  at  $\lambda = x\theta / \gamma$ , which at  $x \in [0, 1]$  and bounded  $\theta$  is finite.  $\square$

**Proof of Proposition 3 (Tragedy of Efficiency, Quantified).** Define the efficiency wedge  $\Delta(\theta) \equiv \mathcal{S}(x^{FB}(\theta); \theta) - \mathcal{S}(x^*(\theta); \theta)$ . Because  $x^{FB}(\theta)$  is defined as the argmax of  $\mathcal{S}(\cdot; \theta)$ , the envelope theorem gives  $d\mathcal{S}(x^{FB}; \theta) / d\theta = \partial \mathcal{S} / \partial \theta|_{x^{FB}}$ . Comparing against the  $d\mathcal{S}(x^*; \theta) / d\theta$  computed above,

$$\frac{d\Delta}{d\theta} = \underbrace{[\partial \mathcal{S} / \partial \theta|_{x^{FB}} - \partial \mathcal{S} / \partial \theta|_{x^*}]}_{\text{direct-channel gap}} - \underbrace{\frac{\partial \mathcal{S}}{\partial x}|_{x^*} \cdot \frac{dx^*}{d\theta}}_{\text{indirect channel}}. \quad (33)$$

The direct-channel gap is second-order in  $(x^* - x^{FB})$  and small. The indirect channel is strictly positive in the coercive regime: both factors are negative, so their product contributes positively to  $d\Delta / d\theta$ . Since  $x^*(\theta)$  rises and  $x^{FB}(\theta)$  falls with  $\theta$  (the first by Proposition 2, the second because  $V'_{US} < 0$  and  $dw_C^* / d\theta < 0$  reduces the marginal cost of integration), the gap  $x^* - x^{FB}$  widens monotonically, and strict concavity of  $\mathcal{S}(\cdot; \theta)$  at  $x^{FB}$  implies  $d\Delta / d\theta > 0$ .  $\square$

## B.6 Innocuousness of the Multi-Sector Extension for Single-Industry Characterizations

Appendix C.1 enriches the baseline of Sections 3.1–3.6 on three margins: (i) sector types  $k \in \{c, m\}$  with type-specific CES elasticities; (ii) a parallel direct  $C \rightarrow U$  sourcing channel alongside the  $C \rightarrow S \rightarrow U$  re-export route; and (iii) an industry-level border tariff  $\tau_j$  as

an additional Hegemon-U instrument. We verify here that each enrichment is innocuous for the baseline characterizations—Proposition 1 (regime threshold), Proposition 2 (Paradox of Competitiveness), and Proposition 3 (Tragedy of Efficiency)—in the sense that all three propositions continue to hold as within-type statements once the enrichments are imposed.

**(i) Multi-sector with sectoral separability.** Under the separability conditions stated in Appendix C.1 (each type has its own CES input mix;  $\lambda_k$  is set type by type with no joint constraint;  $\tau_j$  and  $R_k(x_k)$  operate within type), the three-stage game decouples across types. Within type  $k$ , the Swing State’s Stage-3 problem, Hegemon U’s Stage-2 problem, and Hegemon C’s Stage-1 problem are structurally identical to the single-industry game with type-specific primitives  $(\sigma_k, \alpha_k, \theta_k, w_{A,k}, V_{0,k}, \beta_k, \gamma_k)$  replacing the unsubscripted ones. Propositions 1–3 therefore apply within each type with this substitution; the regime threshold becomes  $\bar{\theta}_k$ , and the paradox and tragedy comparative statics are signed within each  $k$ .

**(ii) Parallel sourcing channels.** Allowing a direct  $C \rightarrow U$  flow alongside the indirect route adds a second demand curve at Hegemon C’s Stage 1 and a second sourcing decision at Hegemon U’s Stage 2. We assume Hegemon U’s utility separates additively across channels,

$$V_U = V_U^{\text{indirect}}(x_S; \lambda) + V_U^{\text{direct}}(q_{C \rightarrow U}; \tau_j),$$

and Hegemon C’s profit separates additively,  $\pi_C = \pi_C^{\text{indirect}}(w_C) + \pi_C^{\text{direct}}(p_{C \rightarrow U})$ . Under additive separability, the FOC for  $\lambda$  at the U–Swing interface, the binding-PC schedule  $R(x)$ , and Hegemon C’s strategic markup  $m$  for the indirect channel are independent of  $(q_{C \rightarrow U}, p_{C \rightarrow U}, \tau_j)$ ; the direct channel’s own FOCs decouple from the indirect-channel subproblem. Channel separability is the formal counterpart to the institutional fact that the U–Swing schedule  $R(x)$  and the U–China tariff  $\tau_j$  are administered through distinct legal instruments (defense-coproduction access and CHIPS-Act-style conditional transfers vs. Section-301 ad-valorem duties at the customs border) and reach different participants. Cross-

channel general-equilibrium spillovers—e.g.,  $U$ 's marginal utility from one channel shifting with quantity in the other—would couple the two FOCs; ruling them out is the specific cost of preserving the within-type characterizations and is an explicit modelling choice rather than a derivation.

**(iii) Additional instrument  $\tau_j$ .**  $\tau_j$  enters Hegemon  $U$ 's instrument set as an additional degree of freedom alongside  $(\lambda, R)$ . Because the baseline has no direct  $C \rightarrow U$  flow, the baseline optimum  $(\lambda^*, R^*)$  remains feasible in the enriched problem at any value of  $\tau_j$  (including the corner where the direct channel is shut down): adding  $\tau_j$  strictly relaxes Hegemon  $U$ 's optimization, weakly improves the value, and does not bind the within-type FOC for  $\lambda$ . The regime threshold, paradox, and tragedy results are inherited as within-type statements.

Together, (i)–(iii) imply that the single-industry characterizations of Sections 3.5–3.6 survive the Appendix C.1 enrichments intact: each proposition holds within each type  $k$  under the type-specific primitives. The leakage asymmetry of Appendix C.1 is genuinely additional structure—it operates on the cross-type and cross-industry margins that the single-industry baseline cannot speak to—and does not invalidate any baseline result.

## B.7 Proof of the Leakage Asymmetry (Flow Margin)

This subsection provides the full derivation for Proposition 6 (Leakage Asymmetry, Flow Margin), expanding the sketch in the main text.

Consider sector  $k$  with CES cost minimization over Chinese intermediates  $I_k^C$  and non-Chinese intermediates  $I_k^A$  with elasticity  $\sigma_k$ . The Swing State's cost minimization delivers the conditional demand

$$I_k^C = \alpha_k^{\sigma_k} (p_k^C / P_k)^{-\sigma_k} Y_k, \quad I_k^A = (1 - \alpha_k)^{\sigma_k} (p_k^A / P_k)^{-\sigma_k} Y_k, \quad (34)$$

where  $P_k = [\alpha_k^{\sigma_k} (p_k^C)^{1-\sigma_k} + (1 - \alpha_k)^{\sigma_k} (p_k^A)^{1-\sigma_k}]^{1/(1-\sigma_k)}$  is the CES price index and  $p_k^C, p_k^A$  are

the effective prices faced at the swing-state border.

A rise in the Hegemon’s security doctrine  $\lambda$  operates through the menu schedule  $R(x)$  to raise the *U-border* shadow price of direct Chinese inputs; crucially,  $\lambda$  does *not* raise the C–S interface price  $p_{k,S}^C$  at which the Swing State sources. Consequently, U’s direct imports from C fall by the full elasticity  $\sigma_k$  multiplied by the induced price gap:

$$\frac{\partial \ln |\Delta M_k^{C \rightarrow U}|}{\partial \lambda} \propto \sigma_k^{(U)} \cdot \frac{\partial \ln p_{k,U}^C}{\partial \lambda} > 0. \quad (35)$$

Swing-state imports from C, in contrast, respond only indirectly—through the derived demand for swing-state exports of  $k$ -content to U. In the elastic regime ( $\sigma_k > 1$ ), the Swing State substitutes cheaply toward non-Chinese intermediates in response to any incremental price pressure, so its imports from C fall sharply, leaving little re-exported Chinese content downstream. In the bottlenecked regime ( $\sigma_k < 1$ ), substitution toward the non-Chinese supplier is bounded above by  $\sigma_k \cdot |d \ln(p_k^C/p_k^A)|$ , which is small; the Swing State continues sourcing from C and absorbs the displaced U–C trade into its own supply chain, which is then re-embodied in Swing→U exports.

Formally, writing the leakage ratio as  $L_k(\lambda) \equiv (\partial \Delta M_k^{C \rightarrow S} / \partial \lambda) / (\partial |\Delta M_k^{C \rightarrow U}| / \partial \lambda)$ ,

$$L_k = s_{k,S}^C (1 - s_{k,S}^C) (1 - \sigma_k) \cdot \phi_k, \quad (36)$$

where  $s_{k,S}^C \in (0, 1)$  is the Swing State’s expenditure share on Chinese intermediates and

$$\phi_k \equiv \frac{X_k^{S \rightarrow U}}{Y_k} \in [0, 1]$$

is the *re-export share* of swing-state output in sector  $k$  that reaches the US market. By construction,  $\phi_k \geq 0$ , and  $\phi_k > 0$  for any sector in which the Swing State actually re-exports to U—the empirically relevant case in our OECD ICIO and ADB MRIO panels, where swing-state exports to U in critical sectors are strictly positive for every industry-year cell. Under

this mild nondegeneracy,  $s_{k,S}^C(1 - s_{k,S}^C) \in (0, 1/4]$  and  $\phi_k > 0$  are both strictly positive, so the sign of  $L_k$  is governed by the sign of  $(1 - \sigma_k)$ : for  $\sigma_k < 1$ ,  $L_k > 0$  and increasing in the Swing State's baseline Chinese share  $s_{k,S}^C$ ; for  $\sigma_k > 1$ ,  $L_k < 0$  and the ratio becomes negative—the Swing State substitutes away from China faster than U's direct-import shock. This establishes the strict inequality

$$L_k|_{\sigma_c < 1} > L_k|_{\sigma_m > 1},$$

which is precisely the claim of Proposition 6. □

## B.8 Proof of the Leakage Asymmetry (Composition Margin)

This subsection provides the full derivation for Proposition 7 (Leakage Asymmetry, Composition Margin).

For origin  $s \in \{S, N\}$  and industry  $j$ , the CES expenditure share on Chinese intermediates is

$$s_{s,j}^C = \frac{\alpha_{s,j}^{\sigma_c} (p_{s,j}^C)^{1-\sigma_c}}{\alpha_{s,j}^{\sigma_c} (p_{s,j}^C)^{1-\sigma_c} + (1 - \alpha_{s,j})^{\sigma_c} (p_{s,j}^A)^{1-\sigma_c}}, \quad (37)$$

where  $\alpha_{s,j}$  is origin  $s$ 's baseline Chinese-input distribution parameter in industry  $j$ , with  $\alpha_{S,j} > \alpha_{N,j}$  by assumption (the Swing State is more Chinese-input-dependent at baseline). The Foreign Value-Added share  $FVA_{s,j}^C$  in origin  $s$ 's gross exports to U in industry  $j$  is, to leading order,  $s_{s,j}^C$  times a constant reflecting the Leontief multiplier.

A higher tariff  $\tau_j$  raises the effective price of *direct* Chinese exports at the U border, but not at the C- $s$  border. Therefore  $\tau_j$  does not appear in  $p_{s,j}^C$  directly; instead, it operates through the general-equilibrium reallocation of demand from the direct channel (C→U) to the indirect channel (C→ $s$ →U). Letting  $\psi_s(\tau_j)$  denote origin  $s$ 's demand for  $k$ -content from U given  $\tau_j$ , and assuming  $\psi'_s > 0$  (higher  $\tau_j$  raises the U-demand for indirect content), the derived-demand effect on  $p_{s,j}^C$  is  $\partial p_{s,j}^C / \partial \tau_j = \eta_s \psi'_s(\tau_j) > 0$ , where  $\eta_s$  is the Chinese-input-share pass-through.

Differentiating (37), the cross-partial is

$$\begin{aligned} \frac{\partial^2 s_{s,j}^C}{\partial \tau_j \partial \mathbf{1}\{s = S\}} &= \left. \frac{\partial s_{s,j}^C}{\partial \tau_j} \right|_{s=S} - \left. \frac{\partial s_{s,j}^C}{\partial \tau_j} \right|_{s=N} \\ &= s_{S,j}^C(1 - s_{S,j}^C)(1 - \sigma_c) \cdot \eta_S \psi'_S - s_{N,j}^C(1 - s_{N,j}^C)(1 - \sigma_c) \cdot \eta_N \psi'_N. \end{aligned} \quad (38)$$

Under  $\sigma_c < 1$ , the common factor  $(1 - \sigma_c) > 0$  is positive. The map  $s \mapsto s(1 - s)$  is strictly increasing on  $(0, 1/2]$  and strictly decreasing on  $[1/2, 1)$ . In the empirically relevant regime  $s_{S,j}^C, s_{N,j}^C \in (0, 1/2]$ —satisfied comfortably in our OECD ICIO and ADB MRIO panels, where the critical-sector Chinese-input share in swing-state production plateaus near 6%—the map is strictly increasing on the relevant interval, so  $\alpha_{S,j} > \alpha_{N,j}$  implies  $s_{S,j}^C > s_{N,j}^C$  and hence  $s_{S,j}^C(1 - s_{S,j}^C) > s_{N,j}^C(1 - s_{N,j}^C)$ . Together with the symmetric pass-through condition  $\eta_S \psi'_S \geq \eta_N \psi'_N$ —which holds whenever the Swing State holds the larger share of indirect C–U trade—the right-hand side of (38) is strictly positive. Hence

$$\left. \frac{\partial^2 \text{FVA}_{s,j}^C}{\partial \tau_j \partial \mathbf{1}\{s = S\}} \right|_{\sigma_c < 1} > 0,$$

which is the claim of Proposition 7. Crucially, this is a *steady-state* object: the proof does not require isolating a 2018 shock, and the cross-partial is identified off cross-sectional industry-level tariff variation interacted with swing-state identity.  $\square$

## B.9 Why Verticality Leads to Coercion: Three Cases

We contrast the vertical bottleneck model with two standard horizontal-rivalry structures to isolate the mechanism that generates coercion.

**Case 1: Horizontal rivalry (subsidy war).** Hegemon U and Hegemon C produce substitute final goods (e.g., competing EV manufacturers) and compete for the Swing State’s market through price subsidies. If Hegemon C lowers its price by  $\delta$ , Hegemon U can retain the Swing State simply by matching the subsidy ( $S_U \approx \delta$ ). Because the cost of matching is

linear, the optimal response to a rival’s efficiency shock is to increase subsidies—a “Race to the Bottom.” The security parameter  $\lambda$  remains low.

**Case 2: Horizontal sourcing rivalry (bidding war).** The Swing State is a common upstream supplier (e.g., a critical mineral exporter) sought by both powers. Rivalry takes the form of a common-agency bidding game where both hegemonies offer purchase contracts  $P_U$  and  $P_C$ . To secure the resource, the US simply needs to outbid China ( $P_U > P_C$ ). Since the cost of outbidding is linear, the equilibrium is defined by positive inducements; no coercion is needed.

**Case 3: Vertical bottleneck (coercion trap).** In the vertical model, the bottleneck technology (Assumption 2) makes the Decoupling Cost  $C(x; w_C)$  strictly convex and asymptotic on the decoupling range (Lemma 1). As China’s efficiency improves ( $w_C \downarrow$ ), the subsidy required to induce the Swing State to choose the alternative chain ( $x \rightarrow 1$ ) explodes ( $\partial^2 R(x)/\partial x^2 \gg 0$ ). The US cannot win a subsidy war against an efficient upstream monopoly because the cost of compensating the Swing State for the loss of the bottleneck input exceeds fiscal capacity ( $J(x) \ll 0$ ). Since the “Carrot” (subsidy) is mathematically inefficient against a bottleneck, the US is forced to use the “Stick” (coercion): instead of paying the Swing State to leave China ( $R \uparrow$ ), the US threatens to destroy its market access if it stays ( $R \downarrow$ ).

**Summary:** Horizontal competition leads to subsidies (paying for loyalty); vertical bottleneck competition leads to coercion (punishing disloyalty), specifically because the price of buying loyalty becomes infinite.

## B.10 Reconciling the Discrete Regime Switch with Smooth Empirical Dynamics

Proposition 1 predicts a kink in  $\lambda^*(\theta)$  at  $\bar{\theta}$ —a continuous-but-nondifferentiable transition from the corner regime to the interior regime, not a level jump. Yet the empirical dynamics

documented in Section 2 appear smooth. We offer three non-exclusive reconciliations.

(i) *Heterogeneous sectors crossing the threshold sequentially.* Different sectors have different bottleneck elasticities  $\sigma_k$  and different starting values of Upstream productivity  $\theta_k$ . If  $\theta_k$  rises at heterogeneous rates, the aggregate coercion intensity  $\sum_k \mathbf{1}\{\theta_k > \bar{\theta}_k\} \cdot \lambda_k^*$  is a sum of step functions triggered at different calendar dates, which smooths the aggregate into an apparently continuous process. This is the sectoral-diffusion reading: each individual sector jumps, but the aggregate is an envelope.

(ii) *Bayesian uncertainty about  $\sigma$ .* The Hegemon does not know  $\sigma$  perfectly; it updates based on realized supply disruptions and technological assessments. A posterior distribution over  $\sigma$  delivers a posterior distribution over  $\bar{\theta}$ , and the *expected* security doctrine  $\mathbb{E}[\lambda^*(\theta)]$  is a smooth function of  $\theta$  even when the realized  $\lambda^*(\theta; \sigma)$  is a step function for any given  $\sigma$ .

(iii) *Institutional adjustment costs.* Coercive institutions (enforcement agencies, licensing regimes, multilateral coordination) cannot be built instantaneously. Even after  $\theta$  crosses  $\bar{\theta}$ , capacity constraints on the adjustment rate of  $\lambda^*$  would translate the discrete target into a smooth realized path.

*Caveat on identification.* The stylized facts in Section 2 are descriptive rather than causal. We do not claim to identify the structural break in  $\bar{\theta}$  econometrically. A formal structural-break test on sector-specific coercion intensity is a natural follow-up project.

## C Multi-Sector Extension and Leakage Tests

This appendix develops a multi-sector extension of the baseline model and tests its two cross-sectional implications. The single-sector baseline of Sections 3.1–3.6 delivers the regime switch and the paradox of competitiveness; it does not, however, generate cross-sector or cross-origin moments. Subsection C.1 enriches the baseline on three margins (multi-sector with type-specific CES elasticities, a parallel direct  $C \rightarrow U$  sourcing channel, and an industry-level border tariff  $\tau_j$ ) and derives two propositions: a flow-margin leakage ratio (Proposition 6)

and a steady-state composition cross-partial on Chinese value-added (Proposition 7). Subsections C.2–C.4 test these predictions on an HS6 trade panel and an ICIO/MRIO value-added panel. The empirical results corroborate the same  $\sigma_c < 1$  primitive that drives the regime-switch evidence in Section 4 but are not load-bearing for the main-text validation; readers willing to take the multi-sector extension on faith may skip this appendix without loss of continuity for the central regime-switch + paradox argument.

## C.1 Multi-Sector Setup

Sections 3.5–3.6 characterized the Hegemon’s choice and welfare loss in a single-industry economy with a single sourcing route ( $C \rightarrow S \rightarrow U$ ). The cross-sectional moments tested in Appendix C require enriching this baseline on three margins: (i) *multi-sector*, with sector types  $k \in \{c, m\}$  differing in the CES substitution elasticity between Chinese and non-Chinese intermediates; (ii) *parallel sourcing channels*, admitting a direct  $C \rightarrow U$  flow alongside the  $C \rightarrow S \rightarrow U$  re-export route; and (iii) an *industry-level policy instrument*  $\tau_j$ , a border tariff levied on Chinese-origin shipments arriving directly. Together these enrichments deliver the headline asymmetry: a coercive shock fully displaces Chinese supply in high-elasticity (commodity) sectors but only partially in low-elasticity (critical) sectors, where Chinese value-added reaches the Hegemon indirectly via swing-state re-exports rather than directly. Under separability conditions made explicit below, the single-industry characterizations of Propositions 1–3 carry over as within-type statements (Appendix B.6).

The leakage asymmetry rests on a single observation: the Swing State’s capacity to substitute away from Chinese intermediates depends on the CES elasticity, which differs by type. Where  $\sigma > 1$  (gross substitutes), it switches suppliers cheaply when Chinese sourcing becomes more expensive; where  $\sigma < 1$  (gross complements), Chinese intermediates are nearly irreplaceable, and it continues sourcing them under coercive pressure, re-exporting their embodied content. We index sector *types* by  $k \in \{c, m\}$  (critical  $\sigma_c < 1$  vs. commodity  $\sigma_m > 1$ ) and industries within  $c$  by  $j$ ; the empirical *strategic* bucket of Section 2 is absorbed

into  $c$  since the propositions exploit only the binary  $\sigma \leq 1$  partition.

The Hegemon has two levers for displacing Chinese supply, operating at distinct interfaces of the chain. The Stage-2 doctrine  $\lambda_k$  and schedule  $R_k(x_k)$  act on the  $U$ -*Swing* interface, raising the Swing State's effective cost of China-alignment (Section 3.3). A direct-import tariff  $\tau_j$  acts on the  $U$ -*China* interface, raising the price  $U$  pays for Chinese-origin shipments arriving directly, without affecting Chinese inputs that enter via  $S$  and are re-exported as swing-origin goods after substantial transformation.<sup>18</sup> The two propositions below isolate one moment per lever: a *flow margin*, the response of bilateral trade volumes to a  $\lambda$ -increment (Proposition 6), and a *composition margin*, the cross-sectional structure of FVA shares conditional on  $\tau_j$  (Proposition 7). Both take  $\lambda$  (resp.  $\tau_j$ ) as the *operative shifter*—the single primitive whose comparative statics we trace, holding the others at their equilibrium values—rather than tracing  $\theta$  end-to-end as in Sections 3.5–3.6: cross-sectional asymmetries require fixing a Stage-2 shock and varying type or industry, since an aggregate  $\theta$ -shock is type-symmetric. Any primitive that triggers a regime switch—including the  $\theta \uparrow$  chain of Proposition 2—enters here through its induced  $\lambda$ -increment.

The analysis treats types as decoupled: (i) each type has its own CES input mix with no cross-type substitution; (ii) the Hegemon sets  $\lambda_k$  type by type, with no joint constraint linking  $\lambda_c$  and  $\lambda_m$ ; (iii)  $\tau_j$  and  $R_k(x_k)$  operate within a type, with no cross-type GE price spillovers. This is the minimum needed for a composition margin distinct from the flow margin. Relaxing (ii) by adding a shared coercion budget would push the Hegemon to concentrate  $\lambda$  on the steepest bottlenecks, strengthening the asymmetry; relaxing (i) or (iii) opens a multi-sector GE problem we leave to future work.

**Origins.** The cross-sectional comparison in Proposition 7 requires a contrast group beyond the three players of Sections 3.1–3.4. We index final-good exporting origins to Hegemon  $U$  by  $s \in \{S, N\}$ :

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<sup>18</sup> $\tau_j$  has no role in the single-industry baseline of Sections 3.1–3.6: the  $C \rightarrow S \rightarrow U$  chain admits no direct  $C \rightarrow U$  flow to tariff. The composition margin (Proposition 7) further requires industry-level variation that the type-level  $\lambda_k$  cannot provide.

$S$ : the *Swing State*, the midstream contracting party of Sections 3.2–3.4, with high baseline Chinese-input distribution parameter  $\alpha_{S,j}$  and engaged in the principal-agent contract  $(\lambda, R(x))$  with U. Empirically: Vietnam, Mexico, Malaysia.

$N$ : a representative *non-swing exporter*—another final-good supplier to U in industry  $j$  that is *not* party to U’s principal-agent contract (no  $\lambda_N$ , no  $R_N(x_N)$ ), takes Chinese and non-Chinese input prices parametrically, and has lower baseline Chinese-input distribution parameter  $\alpha_{N,j} < \alpha_{S,j}$ . Empirically: the 14 non-swing source countries in our ICIO/MRIO panel that export to U in critical sectors (Appendix C).

Crucially  $N \neq C$ : China is the upstream input supplier ( $w_C$  side), while non-swing origins are midstream final-good exporters parallel to  $S$  but with weaker Chinese-input dependence. The asymmetry that drives Proposition 7 is  $\alpha_{S,j} > \alpha_{N,j}$  (input dependence) combined with the principal-agent status of  $S$  alone (which renders  $S$  alignment-sensitive while  $N$  is not).

**Supply-chain primitives.** For each industry  $j$  within type  $k \in \{c, m\}$ , Hegemon U sources final goods through two channels: a *direct* flow  $q_j^{C \rightarrow U}$  from China at landed price  $p_{C,j}(1 + \tau_j)$ , and an *indirect* flow  $q_j^{s \rightarrow U}$  from each origin  $s \in \{S, N\}$ , which embeds a CES input mix at the  $C \rightarrow s$  interface (parallel to equation (2)) with type-specific elasticity  $\sigma_k$  between Chinese and non-Chinese intermediates and origin-specific share  $\alpha_{s,j}$ . Hegemon U’s value from  $j$  is additively separable across channels,

$$V_U^j = V_U^{\text{direct}}(q_j^{C \rightarrow U}; \tau_j) + \sum_{s \in \{S, N\}} V_U^{\text{indirect}}(q_j^{s \rightarrow U}, x_{s,j}; \lambda_k), \quad (39)$$

where  $x_{s,j} \in [0, 1]$  is origin  $s$ ’s industry- $j$  alignment with U (the within-industry counterpart of the scalar  $x$  in Section 3.2), and  $V_U^{\text{direct}}$  and  $V_U^{\text{indirect}}$  are differentiable in their arguments. Additivity across channels is a modelling assumption that rules out U-side cross-channel GE spillovers; we adopt it as the minimal structure preserving the within-type characterizations of Propositions 1–3 (Appendix B.6). By substantial-transformation rules of origin,  $q_j^{s \rightarrow U}$

clears U customs as origin- $s$  output and is exempt from  $\tau_j$ , which is why  $\tau_j$  enters only  $V_U^{\text{direct}}$ ; symmetrically,  $\lambda_k$  enters only  $V_U^{\text{indirect}}$  because the principal-agent schedule  $R_k(x_k)$  is administered at the U- $s$  interface. The two instruments link through Hegemon C's pricing: a higher  $\tau_j$  depresses C's revenue on the direct channel, prompting C in equilibrium to lower  $w_{C,j}$  at the  $C \rightarrow s$  interface to preserve indirect-channel volume. The induced shift in  $w_{C,j}$  has origin-specific pass-through  $\eta_s \psi'_s$  to the Swing State's effective Chinese-input shadow price—formally,

$$\frac{\partial p_{s,j}^C}{\partial \tau_j} = \eta_s \psi'_s > 0 \quad (40)$$

(Assumption A4 below)—and through the CES input mix raises the FVA-China share embodied in indirect flows, more steeply for the high-baseline-share Swing State than for non-swing origins. Substitution between Chinese and non-Chinese final goods reaching U therefore operates not at the U-side margin (where channels are additive) but at the C-side and S-side margins: C re-routes shipments through the lower-tariffed  $S$  chain by lowering  $w_{C,j}$ , and  $S$  absorbs the re-routed Chinese content through its CES input mix, with intensity governed by  $\sigma_k$ .

**Proposition 6** (Leakage Asymmetry (Flow Margin)). *Let two sectors  $k \in \{c, m\}$  differ only in their CES substitution elasticity, with  $\sigma_c < 1 < \sigma_m$  (critical vs. commodity). Treating  $\lambda$  as the operative shifter, let  $\Delta M_k^{C \rightarrow U}(\lambda)$  denote the equilibrium response of U's direct China imports and  $\Delta M_k^{C \rightarrow S}(\lambda)$  the corresponding response of Swing-State China imports. Then*

$$\left. \frac{\partial \Delta M_k^{C \rightarrow S} / \partial \lambda}{\partial |\Delta M_k^{C \rightarrow U}| / \partial \lambda} \right|_{k=c} > \left. \frac{\partial \Delta M_k^{C \rightarrow S} / \partial \lambda}{\partial |\Delta M_k^{C \rightarrow U}| / \partial \lambda} \right|_{k=m} :$$

*the leakage ratio is strictly positive in the bottlenecked sector ( $k = c$ )—where U's direct-import reduction is partially offset by re-exported Chinese content via  $S$ —and weakly negative in the commodity sector ( $k = m$ ), where substitution is near-complete; the sign flips across  $\sigma_k = 1$ . The claim is directional, not a magnitude comparison of two positive quantities.*

*Proof (Sketch).* A rise in  $\lambda$  raises the effective shadow cost of Chinese intermediates at the

U–S interface (via  $R(x)$ ), not at the C–S interface. The Swing State’s optimal input mix therefore responds with elasticity  $\sigma_k$ : for  $\sigma_c < 1$ , residual demand for Chinese intermediates is inelastic and the Swing State continues sourcing them, re-exporting the embodied content to U—the leakage channel; for  $\sigma_m > 1$ , supplier-switching is cheap and the leakage ratio shrinks toward zero. Algebraically, the leakage ratio reduces to

$$L_k = s_{k,S}^C (1 - s_{k,S}^C) (1 - \sigma_k) \phi_k, \quad (41)$$

where  $s_{k,S}^C \in (0, 1)$  is the Swing State’s expenditure share on Chinese intermediates in sector  $k$  and  $\phi_k \equiv X_k^{S \rightarrow U} / Y_k \in [0, 1]$  is its sector- $k$  re-export share to U. With  $s_{k,S}^C (1 - s_{k,S}^C) > 0$  and  $\phi_k > 0$  at any sector with positive swing-routed exports, the sign of  $L_k$  tracks  $(1 - \sigma_k)$ , flipping at  $\sigma_k = 1$ . The full derivation is in Appendix B.7.  $\square$

**Proposition 7** (Leakage Asymmetry (Composition Margin)). *Let  $\text{FVA}_{s,j}^C$  denote the Chinese value-added share in origin  $s$ ’s stationary-equilibrium gross exports to Hegemon U in industry  $j$ ,<sup>19</sup> let  $\tau_j \geq 0$  denote U’s tariff on direct Chinese exports in  $j$ , let  $s \in \{S, N\}$  index Swing versus Non-swing origins, and let  $s_{s,j}^C$  denote origin  $s$ ’s equilibrium Chinese-input expenditure share in  $j$ . Assume:*

(A1) *Complementarity.  $\sigma_c < 1$  in the critical production structure (same CES primitive as Proposition 6).*

(A2) *Baseline dependence.  $\alpha_{S,j} > \alpha_{N,j}$ , where  $\alpha_{s,j}$  is origin  $s$ ’s CES distribution parameter on Chinese inputs in industry  $j$ ; under baseline price symmetry across origins, this implies  $s_{S,j}^C > s_{N,j}^C$ .*

(A3) *Scope. The result is stated on the regime  $s_{S,j}^C, s_{N,j}^C \in (0, \frac{1}{2}]$ .*

(A4) *Ordered pass-through. The induced shift in origin  $s$ ’s effective Chinese-input price has weakly larger pass-through for the Swing State:  $\eta_S \psi'_S \geq \eta_N \psi'_N$ , where  $\eta_s$  is origin*

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<sup>19</sup>By *stationary equilibrium* we mean the cross-sectional pattern at fixed primitives  $(\tau_j, \alpha_{s,j}, \sigma_c, \lambda)$ ; the model has no explicit transitional dynamics.

$s$ 's Chinese-intermediate share and  $\psi'_s$  the marginal  $\tau_j$ -pass-through from Hegemon C's pricing to origin  $s$ 's Chinese-input shadow price (equation (40)).

Given (A1)–(A4), the difference-in-differences object

$$\frac{\partial \text{FVA}_{S,j}^C}{\partial \tau_j} - \frac{\partial \text{FVA}_{N,j}^C}{\partial \tau_j} \equiv \frac{\partial^2 \text{FVA}_{s,j}^C}{\partial \tau_j \partial \mathbf{1}\{s = S\}} > 0 :$$

the slope of the embodied-Chinese share on  $\tau_j$  is strictly steeper for swing-state exporters than for non-swing exporters.

*Remark on (A3)–(A4).* The (A3) range is load-bearing: the slope  $\partial s^C / \partial \ln p = (1 - \sigma_c) s^C (1 - s^C)$  is increasing on  $(0, 1/2]$  but decreasing on  $[1/2, 1)$ , so the cross-partial sign would flip outside  $(0, 1/2]$ . The range is empirically relevant: the swing-state FVA-China share in critical-sector exports plateaus near 6% in our OECD ICIO and ADB MRIO panels, well below the 50% inflection point. (A4) is stated as a hypothesis rather than derived from (A2) because the mapping from  $\alpha_{s,j}$  to  $\eta_s \psi'_s$  depends on origin  $s$ 's downstream bargaining structure vis-à-vis U, outside the scope of this paper.

*Proof (Sketch).* A higher  $\tau_j$  raises the shadow price of *direct* Chinese supply at the U–C interface; via Hegemon C's pricing response (equation (40)), origin  $s$ 's effective Chinese-input price rises with pass-through  $\eta_s \psi'_s$ . Differentiating the CES expenditure-share formula yields

$$\frac{\partial s_{s,j}^C}{\partial \tau_j} = (1 - \sigma_c) s_{s,j}^C (1 - s_{s,j}^C) \eta_s \psi'_s.$$

Under (A1) the leading factor  $(1 - \sigma_c)$  is positive; under (A3), both origins lie on the increasing arm of  $s \mapsto s(1 - s)$ , so (A2) places the Swing State on a steeper segment,  $s_{S,j}^C(1 - s_{S,j}^C) > s_{N,j}^C(1 - s_{N,j}^C)$ ; (A4) ensures the pass-through factor is also weakly larger for  $S$ . The product is therefore strictly larger at  $s = S$  than at  $s = N$ , delivering the difference-in-differences claim. Note the prediction concerns the *slope* of FVA on  $\tau_j$ , not its level (which (A2) already orders trivially). The full derivation is in Appendix B.8.  $\square$

Propositions 6 and 7 generate two complementary moments tested in Appendix C: a flow-margin event study exploiting the 2018 tariff shock as an identified  $\lambda$ -increment, and a cross-sectional triple-difference that identifies the steady-state cross-partial  $\partial^2 \text{FVA}_{s,j}^C / \partial \tau_j \partial \mathbf{1}\{s = S\}$ . The cross-sectional moment is the primary one: it does not require an isolated shock, and as a group-average object it is robust to borderline-country reclassification up to  $O(1/|S|)$ —a counter-fragility property our leave-one-out diagnostic confirms.

## C.2 Data and Specifications

We construct a balanced panel of bilateral US imports at the HS6 level over 2000–2024, assigning each code to one of three buckets—*critical* (rare earths, advanced semiconductors, semiconductor-manufacturing equipment), *strategic* (mature-node chips, batteries, EV components), and *commodity* (textiles, basic steel, basic chemicals). Appendix D details the HS6 concordance, the UN Comtrade pull (premium API, 63 targeted HS6 codes, 18 reporters), and the BIS Entity List panel built from the OpenSanctions Consolidated Screening List. For Test 2 we additionally use two separately maintained inter-country input–output tables—the OECD ICIO EXTENDED release and the ADB MRIO 72-economy release—to compute the swing-state Chinese-VA share variable that targets the cross-partial of Proposition 7; that construction is described in Subsection C.4.

The model’s three predictions map onto three regression-based tests, implemented through three specifications on the HS6 panel and one continuous tariff- $\times$ -swing specification on the value-added panel:

- **Spec 1 (triple-difference, single Post dummy).** The *average* post-2018 sector-asymmetric differential on US gross imports. A naive reading of Proposition 6:  $\beta_{\text{CHN} \times \text{Crit} \times \text{Post}} < 0$ —US imports from China fall more in critical than commodity sectors once  $\lambda$  rises. This is the “tariffs caused decoupling” sign that a casual reader would expect.

- **Spec 2 (paradox regression).** The *dose-response* of the sector-asymmetric effect to the Entity-List coercion stock  $EL_{t-1}^{CHN}$ . Under Proposition 2, the interaction with the critical-sector dummy should load negatively as a dose-response fingerprint of the paradox mechanism.
- **Spec 3 (event study).** The *dynamic* sector-asymmetric coefficient path year by year. Under Proposition 6, with swing-state intermediation already operating pre-2018, the  $(CHN \times \text{Critical})$  coefficient path should be *negative before 2018*—because the gap is a steady-state feature of the global production structure, not a 2018 policy break—and *flat or rising from below toward zero thereafter*, because additional coercion does not re-open a gap that leakage has already normalized.
- **Continuous tariff- $\times$ -swing triple-difference on Chinese-VA shares.** The *cross-partial* of Proposition 7, identified on swing-state-to-US gross exports decomposed by value-added origin: under  $\sigma_c < 1$ , the tariff-gradient of Chinese-VA content is steeper for swing-state exports than for non-swing-state exports. This is a steady-state object that does not require isolating a 2018 shock.

The two leakage tests partition these specifications by which proposition they identify. **Test 1** (flow-margin event study, Proposition 6) uses Specs 1 and 3 together: Spec 1 turns on whether the post period is below zero; Spec 3 turns on whether the pre period was below zero and the post period converged. The two can point in different directions, and in our HS6 panel they do—Spec 1 is mildly positive while Spec 3 recovers the closure-of-pre-existing-gap pattern Proposition 6 implies when leakage is a steady-state feature rather than a post-shock adjustment. **Test 2** (steady-state composition, Proposition 7) uses the continuous tariff- $\times$ -swing specification on the value-added panel; a binary HS6-level analogue would be uninformative because electronics does not vary continuously in  $\tau_j$  within-category.

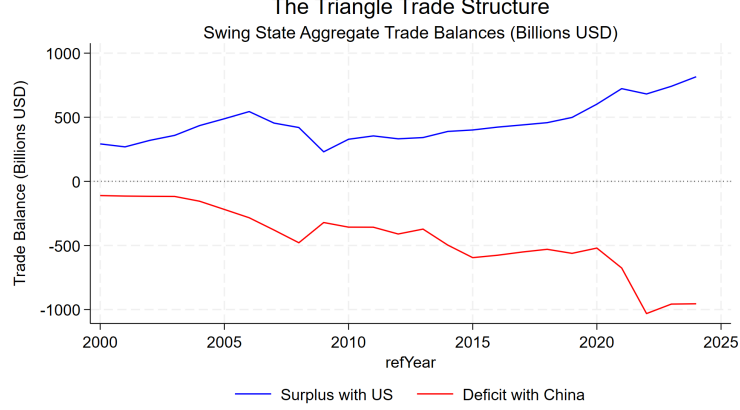


Figure 7: **The Triangle Trade.** Symmetrical deepening of the swing states’ aggregate trade surplus with the US and trade deficit with China over 2000–2024.

### C.3 Test 1: Flow-Margin Event Study (Proposition 6)

Before turning to the regression evidence, we note one further descriptive pattern that motivates the flow-margin specification: the triangle trade. As the swing states’ aggregate trade surplus with the US expands to roughly \$800 billion over 2000–2024, their trade deficit with China deepens proportionally to roughly \$1 trillion (Figure 7). This mirror image is consistent with a pass-through structure in which Chinese inputs are imported, processed, and re-exported to the US — a structure that the value-added evidence cited in the Introduction (roughly \$22 billion per year of Chinese content reaching the US through swing states by 2020–2022) confirms directly. Test 1 below targets the flow-margin signature of this pass-through.

Using this panel, we estimate an event-study specification for US imports from China in the critical bucket relative to a (swing-state  $\times$  non-critical) base, with 2017 as the omitted pre-treatment year. Identification is at the HS6  $\times$  partner  $\times$  year level ( $N \approx 17,600$ ). Fixed effects absorb partner-HS6, HS6-year, and partner-year variation, and standard errors are clustered at HS6 ( $\sim 60$  clusters):

$$\ln M_{ihst} = \alpha_{ih} + \gamma_{ht} + \delta_{it} + \sum_{\tau \neq 2017} \beta_{\tau} \cdot \mathbf{1}\{t = \tau\} \cdot \text{CHN}_i \cdot \text{Critical}_h + \varepsilon_{ihst},$$

where  $i$  indexes exporter (China + 17 candidate swing-state partners; the focal 3-state swing group {VNM, MEX, MYS} used in Test 2 is a subset),  $h$  indexes HS6 code, and  $t$  indexes years. HS6 resolution (relative to a 3-sector aggregate) raises sample size by a factor of 13 and permits cluster-robust inference on a genuine number of clusters.

The purpose of the event study is not to detect a 2018 break but to ask whether one *should* have existed. If Section-301 caused China–critical decoupling, the post-2017  $\beta_\tau$  path should turn sharply negative. Under the leakage reading of Proposition 6, the pre-2018 path was already negative because swing-state intermediation had absorbed the bilateral margin. The post-2018 path should then be flat against the 2017 baseline rather than deepen. The data match the latter. Pre-2018 coefficients sit at  $-0.3$  to  $-0.8$  log points, with 2012–2016 individually significant at 1–5%. Every post-2017 coefficient is statistically indistinguishable from the 2017 baseline. The joint pre-trend test  $F_{5,61} = 3.69$  ( $p = 0.006$ ) rejects a flat pre-period, which is exactly the pre-existing gap the leakage story predicts. See Appendix Figure 9 for the full coefficient path and Appendix Table 2 for the numerical values.

The headline single-coefficient triple-difference  $\beta_{\text{CHN} \times \text{Critical} \times \text{Post}} = +0.283$  (SE 0.245,  $p = 0.25$ ) is statistically insignificant with a positive sign. This mirrors the event study: the post-2018 coefficients cluster around zero while pre-2018 coefficients are negative, so the collapsed Post contrast loads weakly positive. The HS6 panel does not support a *widening* of the China–Critical gap after 2018; it supports the gap *closing*.

Column 2 of Table 2 adds group-specific linear trends interacting ( $\text{CHN} \times \text{Critical}$ ) and ( $\text{Swing} \times \text{Sector}$ ) with time. The absorbed China–Critical trend is  $-0.037$  log points per year (SE 0.028, not significant); the residual Post interaction is  $+0.684$  ( $p = 0.013$ ). Even with the trend taken out, there is no evidence of a *negative* post-2018 residual effect on China–critical imports. The interpretation most consistent with this evidence is that by 2018 the China–critical gap had already closed via swing-state intermediation, and subsequent coercion neither re-opened it nor slowed the convergence. This is consistent with the *leakage* mechanism of Proposition 6—swing states absorbed Chinese intermediate capacity into the

US supply chain—and inconsistent with a simple “tariffs caused decoupling” reading.

Because the pre-trend  $F$ -test rejects strict parallel trends, we report robust identified sets for the average post-treatment effect using the relative-magnitudes partial-identification procedure of Rambachan and Roth (2023), which bounds the post-period deviation as a multiple  $M$  of the worst pre-period deviation. Two features matter (Appendix Figure 10; numerical bounds in Appendix D.4). First, *zero lies inside the identified set at every  $M \leq 2$* , so the hypothesis that the post-2018 China–Critical gap equals the 2017 baseline is not rejected even under very weak parallel-trends assumptions. Second, the *upper* bound is positive throughout, ruling out any persistent post-2018 *widening* of the gap. The leakage reading—closure rather than widening—is thus not dependent on the parallel-trends assumption that the event-study literally rejects.

A final concern is selection on zeros. The log specification drops cells with nonpositive trade, a selection disproportionately concentrated in China $\times$ Critical $\times$ post-2020—exactly the cells where decoupling would generate new zeros. We therefore re-estimate on a balanced (HS6 $\times$ partner $\times$ year) panel with missing cells set to zero, using the Poisson pseudo-maximum-likelihood estimator of Santos Silva and Tenreyro (2006) via `ppmlhdfc` (Correia et al., 2020). On  $N = 22,305$  observations (a 26% increase over the log sample), the baseline triple-difference coefficient is +0.289 (SE 0.309) and the trend-robust variant is +0.531 ( $p = 0.050$ ). Both are close to their log-sample counterparts (+0.283 and +0.684) and, if anything, slightly weaker. The PPML event study (Appendix D, Table 4) reproduces the key qualitative pattern: negative pre-2018 coefficients (the largest,  $\tau_{2015} = -0.400$ , is individually significant) and a post-2018 band clustered around zero with no persistent negative drift. The zero-retention critique therefore does not rescue a “tariffs caused decoupling” reading; the gap-closing pattern is not an artifact of selecting on positive trade.

## C.4 Test 2: Steady-State Cross-Partial on Chinese Value-Added (Proposition 7)

The gross-trade test above ruled out a persistent post-2018 widening of the US–China gap but does not directly demonstrate Chinese content flowing through swing states. The value-added decomposition does. The triple-difference in gross trade is an *indirect* test of Proposition 6: a closing US–China bilateral gap in critical-sector imports can reflect leakage through swing-state intermediation, but only if swing-state exports to the US embed more Chinese value-added in critical sectors than in commodity sectors. We test this directly.

The target object is the cross-partial of Proposition 7: under  $\sigma_c < 1$ ,  $\partial^2 \text{FVA}_{s,j}^C / \partial \tau_j \partial \mathbf{1}\{s = S\}$  is strictly positive, i.e., the tariff-gradient of the Chinese-VA share in  $s \rightarrow$  US gross exports is steeper for swing-state origins than for non-swing origins. The identified moment is therefore a continuous tariff- $\times$ -swing interaction on the FVA-China-share panel. The binary analogue (post  $\times$  electronics  $\times$  swing) does not identify the same object because electronics does not vary continuously in  $\tau_j$  within-category, and the post-interaction with swing-state identity is absorbed by source-year and industry-year fixed effects once the binary electronics indicator is collinear with their tensor product; the continuous tariff-intensity specification is therefore the primary test, not a robustness check. Proposition 7 is a steady-state corollary of the same production-structure primitive ( $\sigma_c < 1$ ) that drives the flow-margin leakage of Proposition 6, but unlike that flow margin it does not require isolating a 2018 shock—a feature that becomes diagnostic when we read the placebo-year regressions below.

We use two separately maintained inter-country input–output tables: the OECD ICIO EXTENDED release (ISIC Rev. 4, 50 industries, 85 economies, 1995–2022) and the ADB MRIO 72-economy release (35 industries, 2017–2024). For each swing state  $s$ , industry  $j$ , and year  $t$ , we compute the share of gross  $s \rightarrow$  US exports accounted for by Chinese value-added:

$$\text{FVA}_{s,j,t}^{\text{CHN, share}} = \sum_{r \in \text{CHN}} v_r \cdot L_{r,(s,j)},$$

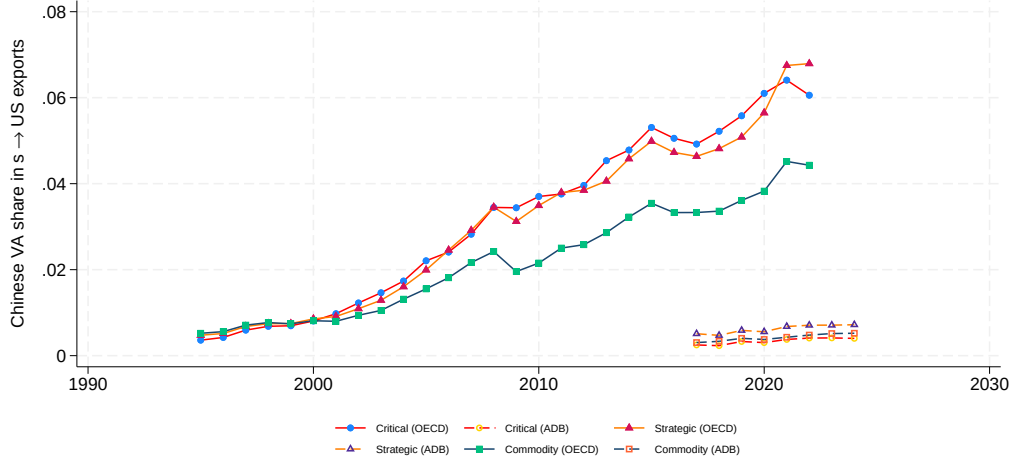


Figure 8: **Chinese value-added share in swing-state  $\rightarrow$  US exports, by sector bucket.** Cross-swing-state mean of  $FVA_{s,j,t}^{CHN}/E_{s,j,t}$ , computed from OECD ICIO EXTENDED (solid, 1995–2022) and ADB MRIO 72-economy release (dashed, 2017–2024). ADB levels are  $\approx 20\times$  lower because its 35-industry taxonomy aggregates semiconductors with household appliances in a single “electrical and optical” bucket, diluting the critical share. Levels and trends discussed in the text; full regression in Appendix Table 5.

where  $v_r$  is value-added per unit of gross output in Chinese source sector  $r$  and  $L = (I - A)^{-1}$  is the global Leontief inverse. This share is *multiplied by* gross bilateral exports  $E_{s \rightarrow US,j,t}$  to recover the dollar value of Chinese value-added embedded in those exports, which is what we use in the dollar-volume rescaling below. Figure 8 plots the across-swing-state mean by sector bucket and data source.

The level gap is the clearest evidence. In the OECD panel the critical-sector share rises from 0.4% in 1995 to a flat plateau of  $\approx 6.1\%$  over 2020–2022, while the commodity share rises from 0.5% to 4.4% over the same period. The critical–commodity gap opens in the mid-2000s and reaches  $\approx 1.8$  percentage points by 2015–2018. It then *widens* to  $\approx 2.3$  percentage points by 2020: Chinese value-added embedded in swing-state critical-sector exports to the US kept rising after the 2018 tariffs, even as gross US imports from China fell.<sup>20</sup>

<sup>20</sup>Scaled against 2022 swing-state  $\rightarrow$  US gross exports, the fitted shares imply roughly \$22 billion of Chinese value-added embedded in the critical bucket and \$23 billion in the commodity bucket. A back-of-envelope figure for the post-2018 descriptive-plot share acceleration ( $\approx 0.9$  percentage points in critical sectors) is approximately \$3 billion per year; this is a descriptive statistic, not a causal counterfactual, because the placebo-year regressions reported below show that approximately 47–87% of the post-2018 cross-sectional gap was already present before the 2018 tariffs (the earliest 2010 placebo recovers 87% of the 2018 magnitude).

In a stacked regression on  $N = 11,899$  (source- $\times$ -industry- $\times$ -dataset,  $\times$  year) observations with two-way fixed effects (Appendix Table 5), the post-2018 interaction on the critical bucket is  $+0.009$  ( $p < 0.05$  pooled;  $+0.012$ ,  $p < 0.05$  OECD only).

Disaggregating along Section-301 exposure, the post-2018 acceleration concentrates on *electronics*, the Section-301 flagship target (coded C26 in ISIC Rev. 4, D26 in the EXTENDED letter-prefix release):  $\hat{\beta}_{\text{elec}}^{\text{post}} = +0.030$  ( $p < 0.01$ ). The non-electronics critical sub-bucket—metal ores, coded B07 (D07T08 in the EXTENDED release)—moves in the opposite direction.

The sharpest specification is a continuous triple-difference. It absorbs source- $\times$ -year and industry- $\times$ -year shocks, and identifies off the interaction of (published) Section-301 tariff rates with swing-state identity and post-2018 timing (Appendix D.6). We estimate  $\hat{\beta}_{\text{tar} \times \text{swing}}^{\text{post}} = +0.102$  ( $p < 0.001$ ): a 25 pp Section-301 tariff increase on a swing-state-produced good is associated with a further  $+2.5$  pp rise in its Chinese-VA content *beyond* what industry-year and source-year effects absorb.

The  $\hat{\beta}_{\text{tar} \times \text{swing}}^{\text{post}} = +0.102$  is the sample analog of the cross-partial in Proposition 7: source- $\times$ -year and industry- $\times$ -year effects absorb the swing-specific aggregate dynamics and the industry-specific tariff dynamics, leaving the bilateral residual as the target object. The analogous binary triple-difference (post  $\times$  electronics  $\times$  swing, with metals as the within-industry control and the 14 non-swing sources as the geographic control) is, as anticipated, *not* identified under the same FE structure ( $\hat{\beta} = +0.014$ ,  $p = 0.36$ ).

Placebo-year regressions (Appendix Tables 6–7) with fake “post” cuts at 2010, 2013, and 2016 (restricted to year  $\leq 2017$ ) reveal that the continuous triple-interaction is *not* identified off a 2018 break. The tariff- $\times$ -swing coefficient is already  $+0.089$  ( $p = 0.001$ ) under a 2010 cut,  $+0.080$  ( $p = 0.004$ ) under a 2013 cut, and  $+0.048$  ( $p = 0.04$ ) under a 2016 cut, against  $+0.102$  ( $p < 0.001$ ) for the real 2018 cut. Pre-treatment placebos thus recover 47–87% of the 2018 magnitude (2010 cut: 87%; 2013 cut: 78%; 2016 cut: 47%). The monotonic decay from 2010→2016 suggests the trend was slowing before 2018 and that the 2018 coefficient

represents an acceleration of, not a departure from, a structural pattern. The binary triple-diff is uninformative in *both* the placebo and the real window (all four coefficients statistically indistinguishable from zero).

The continuous triple-diff’s  $p < 0.001$  headline clusters on the panel identifier (source  $\times$  industry  $\times$  dataset) and is conservative to alternative inference structures. Two-way clustering on (source, industry) gives  $\hat{\beta} = +0.102$  with  $SE = 0.040$  ( $t = 2.57$ ,  $p < 0.01$ ); here the conservative cluster count is  $K = \min(17, 28) = 17$ , with 17 source countries (3 swing + 14 non-swing) and 28 ISIC Rev. 4 industries. Both the panel-clustered and two-way-clustered variants reject zero at conventional levels. Leave-one-out of any single swing state leaves the coefficient significantly positive, though the point estimate ranges from +0.058 (dropping Vietnam) to +0.144 (dropping Malaysia).

A companion Honest-DiD exercise applied to the pooled FVA event study (Appendix Table 8) reinforces this reading. The robust 95% identified set for the average post-2018 critical effect has an always-positive upper bound (+0.025 at  $M = 0$ , rising to +0.066 at  $M = 2$ ). But the lower bound is exactly zero at  $M = 0$  and crosses into negative territory for any  $M > 0$ . So the post-2018 FVA effect does not reject zero even under strict parallel trends. We therefore read the FVA evidence as cross-sectional identification of Proposition 7 (the steady-state cross-partial) rather than as quasi-experimental identification of the Section-301 flow shock; the pre-trend pattern is an expected feature of a steady-state test, not a bug.

## D Empirical Methodology and Regression Results

This appendix documents the data construction and estimation underpinning the regression evidence in Appendix C (Test 1 flow event study and Test 2 FVA cross-partial) and the paradox dose–response regression in Section 4.2.

## D.1 Data Construction

**Trade flows.** Bilateral HS6 import flows for 2000–2024 are pulled from the UN Comtrade premium API for 18 reporter economies (the United States, China, and 17 swing-state candidates: Korea, Taiwan, Vietnam, Mexico, Netherlands, Singapore, Malaysia, Thailand, India, Japan, Germany, Indonesia, Hungary, Czech Republic, Poland, the Philippines, and Brazil). We restrict the commodity universe to a concordance of 63 HS6 codes, pre-classified into three sector buckets matching the model: *critical* ( $\sigma < 1$  in the empirical literature: rare-earth metals, advanced semiconductors, semiconductor manufacturing equipment, titanium, aerospace-grade parts), *strategic* (mature-node chips, batteries, EV components, solar), and *commodity* (textiles, basic steel, furniture, common chemicals). The concordance and assignment rules are reproducible from the pipeline in `empirical/scripts/11_build_hs_concordance.py`.

Reporter-year raw pulls are materialized as per-(reporter  $\times$  year) parquet files and stacked into a master HS6 panel via `12_build_comtrade_panel.py`. Because several reporters (Germany, Thailand, Mexico, Czech Republic, Hungary, Indonesia, Malaysia, Brazil) disaggregate trade by `motCode` (mode of transport) and `customsCode` (customs regime), we retain only the total rows `motCode=0` and `customsCode=C00`; verification against aggregate totals confirms this collapse recovers the national import totals reported in Comtrade’s aggregated files.

**Entity List coercion measure.** The BIS Entity List is assembled from the OpenSanctions `us_trade.cs1` (Consolidated Screening List) Follow-The-Money JSON feed, filtered to sanctions whose program descriptor contains “Entity List” (distinguishing BIS-EL from DPL, MEU, and UVL programs). This yields 3,205 listings spanning 2011–2025. The cumulative China-directed stock  $EL_t^{\text{CHN}}$  enters the paradox regression (Section 4) as a lagged variable  $EL_{t-1}^{\text{CHN}}/100$ .

## D.2 Estimation: $\sigma$ and the Limits of the Reduced Form

We estimate a robust unit-value double-difference in the spirit of Feenstra (1994), but without the full exporter-variety GMM identification:

$$\Delta^2 \ln p_{ikt} = \frac{1}{\sigma_k - 1} \Delta^2 \ln s_{ikt} + \nu_{ikt},$$

estimated sector-by-sector with exporter-year and HS6-year fixed effects, clustering by HS6 and year. We obtain  $\hat{\beta}_{\text{critical}} = -0.031$ ,  $\hat{\beta}_{\text{strategic}} = -0.012$ ,  $\hat{\beta}_{\text{commodity}} = -0.062$ . The sign pattern ( $\hat{\beta} < 0$ ) is consistent with  $\sigma < 1$  for all three buckets, but the magnitudes imply implausibly large negative  $\sigma$  values once  $\sigma = 1 + 1/\hat{\beta}$  is computed. This is the well-known problem with reduced-form implementations of Feenstra: identification of  $\sigma$  separately from supply-side shocks requires the heteroscedasticity-based GMM of Feenstra (1994), which exploits heteroscedasticity *across exporter varieties within* an HS code (a moment condition our median-demeaning specification does not exploit). We leave the full GMM implementation to a companion paper and, for the magnitudes used in the model, rely on the published sector-level elasticities of Atalay (2017) and Boehm et al. (2019), which place the relevant  $\sigma$  below unity for the critical and strategic buckets. Our own estimates are reported as directional evidence only.

## D.3 Leakage Triple-Difference Regression Table

Table 2 reports the full estimates from Appendix C.3. Column (1) is the triple-difference with a single `Post`×`Partner`×`Sector` interaction; the `swing_crit_post` term is absorbed by the partner-year and sector-year fixed effects (collinearity). Column (2) adds the group-specific linear trends interacting (`CHN` × `Critical`) and (`Swing` × `Sector`) with time. Column (3) is the event study whose coefficients are plotted in Figure 9.

Table 2: Leakage specification, US bilateral imports 2000–2024.

|                       | (1)<br>Baseline   | (2)<br>+Group trends | (3)<br>Event study   |
|-----------------------|-------------------|----------------------|----------------------|
| china_crit_post       | 0.283<br>(0.245)  | 0.684**<br>(0.262)   |                      |
| swing_crit_post       | 0.000<br>(.)      | 0.000<br>(.)         |                      |
| swing_comm_post       | -0.186<br>(0.405) | 0.272<br>(0.336)     |                      |
| chn_crit_trend        |                   | -0.037<br>(0.028)    |                      |
| swing_crit_trend      |                   | 0.000<br>(.)         |                      |
| swing_comm_trend      |                   | -0.038<br>(0.037)    |                      |
| tau2012               |                   |                      | -0.822***<br>(0.216) |
| tau2013               |                   |                      | -0.167<br>(0.271)    |
| tau2014               |                   |                      | -0.499<br>(0.309)    |
| tau2015               |                   |                      | -0.671***<br>(0.237) |
| tau2016               |                   |                      | -0.282<br>(0.327)    |
| tau2018               |                   |                      | 0.100<br>(0.330)     |
| tau2019               |                   |                      | -0.034<br>(0.373)    |
| tau2020               |                   |                      | -0.046<br>(0.336)    |
| tau2021               |                   |                      | 0.085<br>(0.321)     |
| tau2022               |                   |                      | -0.082<br>(0.361)    |
| tau2023               |                   |                      | -0.123<br>(0.380)    |
| tau2024               |                   |                      | 0.217<br>(0.394)     |
| N                     | 17649             | 17649                | 17649                |
| R2                    | 0.899             | 0.899                | 0.899                |
| Pre-trend F (2012-16) |                   |                      | 3.69                 |
| p-value               |                   |                      | 0.006                |

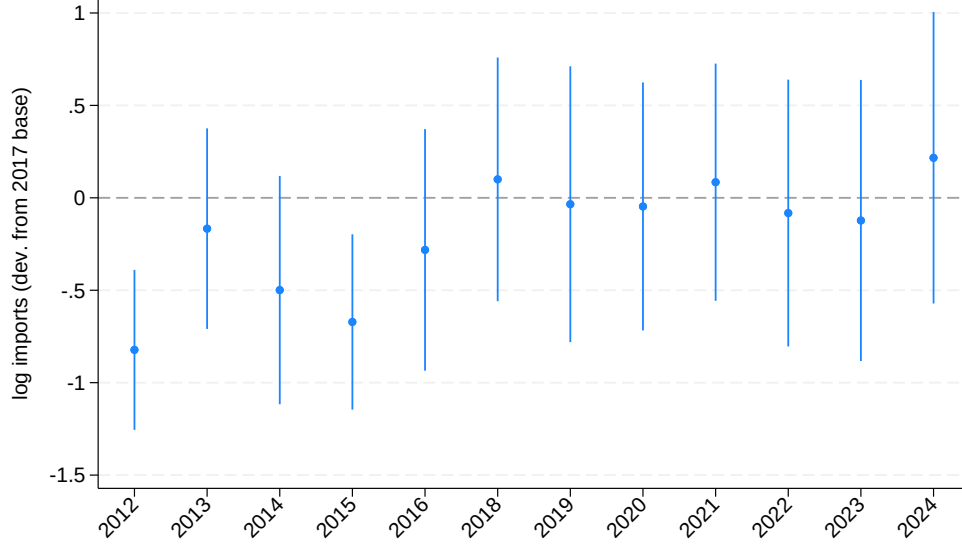


Figure 9: **Event study of US critical-sector imports from China.** Yearly triple-interaction coefficients  $\beta_\tau$  (China  $\times$  Critical  $\times$  year dummies) from the specification described in Appendix C.3, deviations from the 2017 base relative to a (swing-state  $\times$  non-critical) benchmark. Pre-2018 coefficients are significantly negative (the gap existed before the tariffs); every post-2017 coefficient is indistinguishable from the 2017 baseline (the gap did not deepen after Section-301). The pattern is the fingerprint of Proposition 6: leakage through swing states had already absorbed the bilateral China–Critical margin before the shock, leaving no post-2018 break to recover.

## D.4 Honest-DiD Sensitivity Bounds

Table 3 reports the 95% robust confidence intervals underlying Figure 10. Entries are the identified set for the average of the seven post-2018 event-study coefficients from column 3 of Table 2, computed under the Rambachan and Roth (2023) relative-magnitudes restriction  $\Delta^{RM}(M)$ .  $M$  is the maximum post-period second-difference deviation expressed as a multiple of the worst pre-period second-difference deviation; the “Original” row is the standard 95% CI without partial identification.

## D.5 PPML Zero-Retention Robustness

Table 4 re-estimates the triple-difference, trend-robust, and event-study specifications with the PPML estimator of Santos Silva and Tenreyro (2006) on a balanced HS6 $\times$ partner $\times$ year

Table 3: Honest-DiD sensitivity bounds for post-2018 average treatment effect.

| $M$      | Lower bound | Upper bound |
|----------|-------------|-------------|
| Original | -0.546      | +0.746      |
| 0.0      | -0.540      | +0.740      |
| 0.5      | -0.808      | +1.077      |
| 1.0      | -1.170      | +1.515      |
| 1.5      | -1.637      | +2.004      |
| 2.0      | -2.124      | +2.514      |

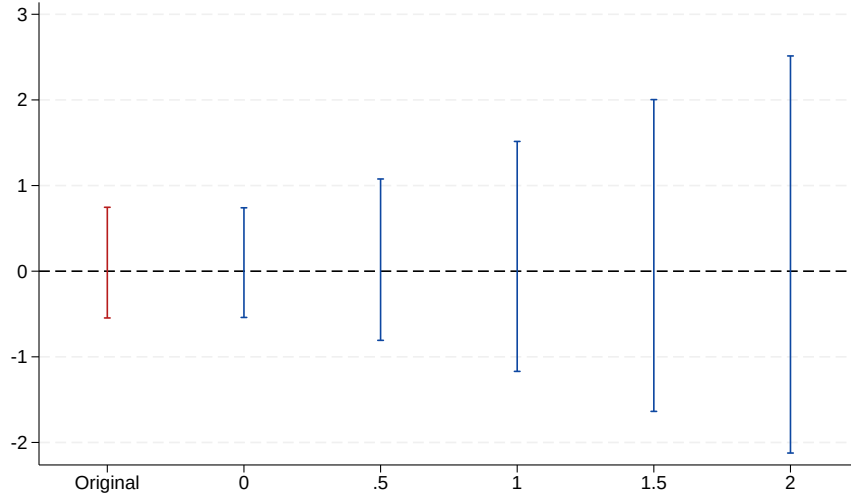


Figure 10: **Honest-DiD robust identified sets for the average post-2018 China×Critical effect in Test 1.**  $M$  bounds the maximum post-period second-difference deviation as a multiple of the worst pre-period second-difference deviation. Zero lies inside the identified set for all  $M \leq 2$ ; the upper bound remains positive throughout, ruling out a persistent post-2018 widening of the gap.

panel with zero-trade cells retained (`ppmlhdfe`, Correia et al., 2020). The sample grows from  $N = 17,649$  (log) to  $N = 22,305$ . Coefficients are semi-elasticities and thus directly comparable to the log-linear point estimates in Table 2. The pre-trend  $\chi^2$  still rejects flat pre-periods, and the honest-DiD sensitivity intervals would be qualitatively unchanged; we therefore do not re-report them.

Table 4: PPML robustness: leakage specification with zero-trade cells retained.

|                          | (1)<br>PPML Baseline | (2)<br>PPML +Group trends | (3)<br>PPML Event study |
|--------------------------|----------------------|---------------------------|-------------------------|
| china_crit_post          | 0.289<br>(0.309)     | 0.531**<br>(0.270)        |                         |
| swing_crit_post          | 0.000<br>(.)         | 0.000<br>(.)              |                         |
| swing_comm_post          | -0.169<br>(0.317)    | -0.056<br>(0.287)         |                         |
| chn_crit_trend           |                      | -0.031<br>(0.027)         |                         |
| swing_crit_trend         |                      | 0.000<br>(.)              |                         |
| swing_comm_trend         |                      | -0.012<br>(0.031)         |                         |
| tau2012                  |                      |                           | -0.014<br>(0.141)       |
| tau2013                  |                      |                           | 0.068<br>(0.118)        |
| tau2014                  |                      |                           | -0.294***<br>(0.110)    |
| tau2015                  |                      |                           | -0.400**<br>(0.164)     |
| tau2016                  |                      |                           | -0.147<br>(0.189)       |
| tau2018                  |                      |                           | 0.028<br>(0.200)        |
| tau2019                  |                      |                           | -0.134<br>(0.195)       |
| tau2020                  |                      |                           | 0.276<br>(0.270)        |
| tau2021                  |                      |                           | 0.384<br>(0.279)        |
| tau2022                  |                      |                           | 0.322<br>(0.331)        |
| tau2023                  |                      |                           | 0.104<br>(0.362)        |
| tau2024                  |                      |                           | -0.069<br>(0.358)       |
| N                        | 22305                | 22305                     | 22305                   |
| Pre-trend chi2 (2012-16) |                      |                           | 21.01                   |
| p-value                  |                      |                           | 0.001                   |

## D.6 Section-301 Tariff-Intensity Map

Columns (7)–(8) of Table 5 use a continuous industry-level Section-301 tariff rate,  $\tau_j$ , rather than a binary “critical” bucket. The rate is constructed from the combination of Fajgelbaum et al. (2020) HS-2-level average tariff increases on Chinese imports and the Bown (2020) HS-8 list-coverage tracker, aggregated via the UN Statistics Division HS2017→ISIC Rev. 4 correspondence. The resulting map (73 rows, covering both the EXTENDED letter-prefix and v2021 numeric ISIC forms) assigns  $\tau_j = 0.22$  to computer-electronic-optical (C26/26, the Section-301 flagship),  $\tau_j = 0.25$  to machinery NEC (C28/28, the heaviest list),  $\tau_j = 0.20$  to electrical equipment (C27/27) and motor vehicles (C29/29), and  $\tau_j = 0.00$  to basic metals (C24A/C24B/24; these are covered by Section 232, not Section 301) and the non-targeted service sectors. Full calibration and caveats are recorded in `empirical/processed/section301_isic4.README.md`; the values are calibrated averages rather than a line-by-line HS-8 pull, and should be treated as a defensible proxy rather than a research-grade tariff file.

## D.7 Direct FVA Decomposition: Two-Source Regression

Table 5 reports the eight-column fixed-effects regression underlying the value-added evidence in Appendix C.4 and Figure 8. The outcome is the Chinese value-added share in swing-state  $s$ ’s gross exports to the US in industry  $j$  at year  $t$ , computed from OECD ICIO EXTENDED (1995–2022) and ADB MRIO (2017–2024). Column (1) pools both sources; (2) and (3) report each source separately; (4) restricts to the three largest direct-passthrough swing states (Vietnam, Mexico, Malaysia); (5) splits the critical bucket on Section-301 exposure, separating computer-electronic-optical (C26/D26) from metal-ores (B07/D07T08); (6) is a binary triple-difference  $\text{post} \times \text{electronics} \times \text{swing}$  using VNM/MEX/MYS as the swing set and the remaining 14 source countries as within-industry control; (7) replaces the bucket dummies with a continuous industry-level Section-301 tariff rate  $\tau_j$  (Appendix D.6) interacted with  $\text{post}$ ; (8) extends (7) to a continuous triple-difference  $\text{post} \times \tau_j \times \text{swing}$ . Each panel is stacked with a `dataset` indicator. Columns (1)–(5) and (7) absorb  $(\text{source} \times \text{industry} \times$

dataset) and year; columns (6) and (8) additionally absorb source  $\times$  year and industry  $\times$  year, so that only the triple interaction is identified (all two-way interactions are swept into the fixed effects). Standard errors are clustered at the source- $\times$ -industry- $\times$ -dataset unit (i.e., the same grouping used for the `panel_id` fixed effect;  $\sim 354$  clusters in the OECD subsample).

Table 5: FVA regression: Chinese value-added share in swing-state  $\rightarrow$  US exports.

|                 | (1)<br>Pooled       | (2)<br>OECD         | (3)<br>ADB        | (4)<br>Direct-PT    | (5)<br>OECD-split    | (6)<br>Triple-diff | (7)<br>Tariff       | (8)<br>Tariff x Swing |
|-----------------|---------------------|---------------------|-------------------|---------------------|----------------------|--------------------|---------------------|-----------------------|
| post_crit       | 0.009**<br>(0.004)  | 0.012**<br>(0.005)  | -0.000<br>(0.000) | 0.009<br>(0.013)    |                      |                    |                     |                       |
| post_strat      | 0.011***<br>(0.002) | 0.012***<br>(0.002) | -0.000<br>(0.000) | 0.021***<br>(0.007) | 0.012***<br>(0.002)  |                    |                     |                       |
| post_elec       |                     |                     |                   |                     | 0.030***<br>(0.006)  |                    |                     |                       |
| post_cmet       |                     |                     |                   |                     | -0.011***<br>(0.002) |                    |                     |                       |
| post_elec_swing |                     |                     |                   |                     |                      | 0.014<br>(0.015)   |                     |                       |
| post_tariff     |                     |                     |                   |                     |                      |                    | 0.109***<br>(0.012) |                       |
| post_tar_swing  |                     |                     |                   |                     |                      |                    |                     | 0.102***<br>(0.025)   |
| N               | 11899               | 9886                | 2013              | 2111                | 9886                 | 9886               | 9886                | 9886                  |
| R2              | 0.792               | 0.775               | 0.864             | 0.824               | 0.783                | 0.929              | 0.784               | 0.931                 |

## Placebo-year robustness

Because the fixed-effects regressions in Table 5 identify post-2018 coefficients off a pre/post contrast, a referee will reasonably ask whether the same coefficient would emerge with a *fake* treatment year well before the tariffs. If so, the reported 2018 effect is loading on a pre-existing trend rather than on Section-301 itself. Tables 6 and 7 re-estimate the binary and continuous triple-difference specifications with placebo cuts at 2010, 2013, and 2016, restricting the sample to year  $\leq 2017$  to purely pre-date the actual tariffs.

For the binary spec the triple interaction is statistically indistinguishable from zero at all

four cuts (2010: +0.027,  $p = 0.10$ ; 2013: +0.025,  $p = 0.16$ ; 2016: +0.013,  $p = 0.45$ ; 2018: +0.014,  $p = 0.36$ ), so no 2018 break is identified—the binary spec is simply not powered by the within-electronics swing-vs-non-swing contrast under the full (source  $\times$  year, industry  $\times$  year) absorption. For the continuous spec the triple coefficient is strongly positive at all cuts (2010: +0.089,  $p = 0.001$ ; 2013: +0.080,  $p = 0.004$ ; 2016: +0.048,  $p = 0.04$ ; 2018: +0.102,  $p < 0.001$ ), with pre-2018 magnitudes recovering 47–87% of the 2018 estimate (2010: 87%; 2013: 78%; 2016: 47%). The monotonic decay from 2010 (+0.089) through 2013 (+0.080) to 2016 (+0.048), followed by a jump to +0.102 under the real 2018 cut, is consistent with a secular post-WTO restructuring of Chinese-swing-state-US production networks that was slowing as swing states approached a saturation point, then accelerated again after Section-301. Appendix C.4 accordingly treats the FVA regressions as descriptive corroboration of the production-structure parameters underlying Proposition 6 rather than as quasi-experimental identification of the Section-301 shock.

Table 6: Placebo-year robustness (binary spec): post  $\times$  electronics  $\times$  swing at fake cuts vs. the real 2018 cut.

|                                    | (1)<br>2010       | (2)<br>2013      | (3)<br>2016      | (4)<br>2018 (real) |
|------------------------------------|-------------------|------------------|------------------|--------------------|
| Placebo 2010 (post x elec x swing) | 0.027*<br>(0.016) |                  |                  |                    |
| Placebo 2013 (post x elec x swing) |                   | 0.025<br>(0.017) |                  |                    |
| Placebo 2016 (post x elec x swing) |                   |                  | 0.013<br>(0.016) |                    |
| Actual 2018 (post x elec x swing)  |                   |                  |                  | 0.014<br>(0.015)   |
| N                                  | 8121              | 8121             | 8121             | 9886               |
| R2                                 | 0.921             | 0.920            | 0.919            | 0.929              |

### Honest-DiD sensitivity on the FVA event study

Table 8 reports the Rambachan and Roth (2023) robust identified set for the average post-2018 critical-bucket effect from the FVA event study (OECD ICIO EXTENDED,  $t = 2005$ –

Table 7: Placebo-year robustness (continuous spec):  $\text{post} \times \tau_j \times \text{swing}$  at fake cuts vs. the real 2018 cut.

|                                      | (1)<br>2010         | (2)<br>2013         | (3)<br>2016        | (4)<br>2018 (real)  |
|--------------------------------------|---------------------|---------------------|--------------------|---------------------|
| Placebo 2010 (post x tariff x swing) | 0.089***<br>(0.026) |                     |                    |                     |
| Placebo 2013 (post x tariff x swing) |                     | 0.080***<br>(0.027) |                    |                     |
| Placebo 2016 (post x tariff x swing) |                     |                     | 0.048**<br>(0.023) |                     |
| Actual 2018 (post x tariff x swing)  |                     |                     |                    | 0.102***<br>(0.025) |
| N                                    | 8121                | 8121                | 8121               | 9886                |
| R2                                   | 0.922               | 0.920               | 0.919              | 0.931               |

2022,  $t = 2017$  omitted as base). The parameter  $M$  bounds the maximum post-period second-difference deviation as a multiple of the worst pre-period second-difference deviation;  $M = 0$  reproduces the standard OLS identified set and  $M = 2$  corresponds to a considerable weakening of parallel trends. Figure 11 plots the sensitivity curve. The  $M = 0$  identified set is *already*  $[0.000, 0.025]$ : even under the standard parallel-trends restriction, the 95% lower bound is exactly zero, so the pooled post-2018 FVA effect is at best marginally significant before any smoothness relaxation is imposed. For any  $M > 0$  the lower bound crosses into negative territory ( $-0.001$  at  $M = 0.5$ ;  $-0.021$  at  $M = 2$ ), while the upper bound remains positive throughout and rises from  $+0.025$  to  $+0.066$ . The reading is asymmetric: a persistent post-2018 *widening* of the Chinese-VA-share gap cannot be ruled out even under weak parallel-trends assumptions (the upper bound is always positive), but a persistent post-2018 *contraction* cannot be rejected either. The pooled event-study mean is therefore not a quasi-experimental identifier of the Section-301 shock; the sharper cross-sectional evidence on swing-state-tariff heterogeneity (Table 5, columns 7–8) does more identifying work than the pooled pre/post contrast.

Table 8: Honest-DiD sensitivity: robust 95% identified set for average post-2018 critical effect in the FVA event study.

| $M$ | Lower bound | Upper bound |
|-----|-------------|-------------|
| 0.0 | 0.000       | 0.025       |
| 0.5 | -0.001      | 0.033       |
| 1.0 | -0.002      | 0.044       |
| 1.5 | -0.011      | 0.055       |
| 2.0 | -0.021      | 0.066       |

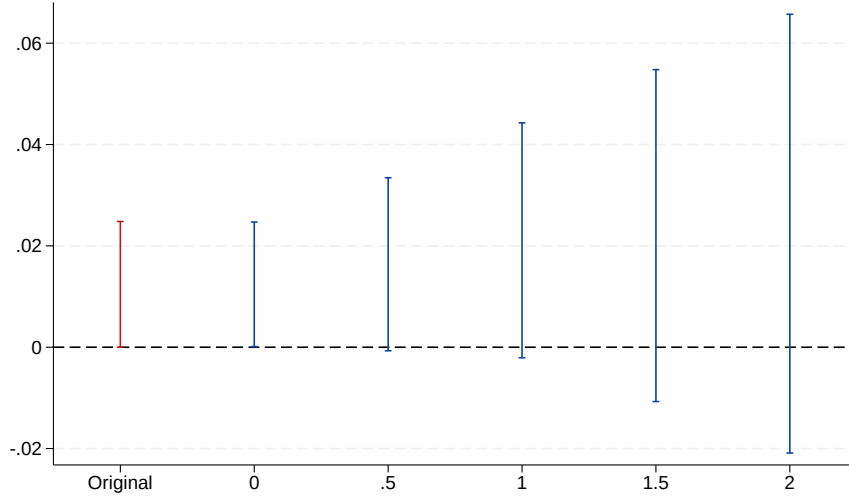
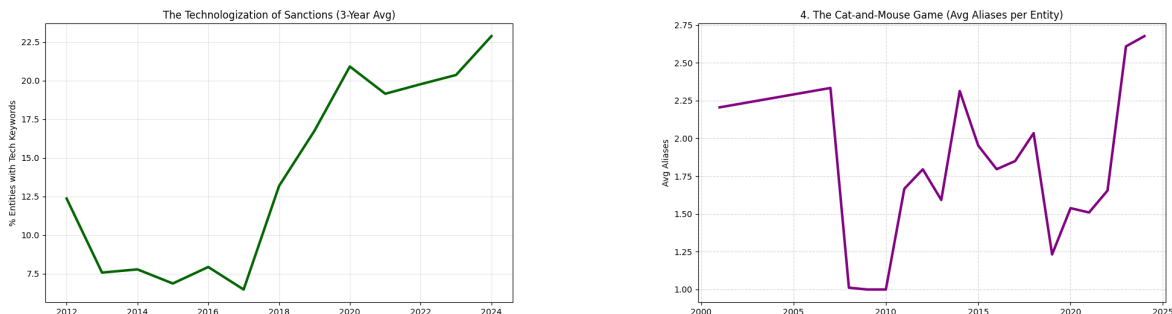


Figure 11: **Honest-DiD robust identified sets for the average post-2018 critical effect in the FVA event study.**  $M$  bounds the max post-period second-difference deviation as a multiple of the max pre-period second-difference deviation. The upper bound is always positive; the lower bound crosses zero for  $M > 0$ .

## D.8 Supplementary Descriptive Figures

This subsection collects two descriptive series referenced in Section 4. (The event-study coefficient path for Test 1 is placed with Table 2 in Appendix D.3; see Figure 9.) Figure 12 documents the sectoral narrowing of the Entity List toward high-technology targets (share of new entries containing “Semiconductor,” “AI,” or “Quantum” in the entry title) and the rising administrative effort per listing (mean aliases per sanctioned entity, a rough proxy for the “Cat-and-Mouse” dynamic). Figure 13 plots the sector-aggregate series underlying the dose-response regression of Spec 2 (paradox regression); the within-HS6 analog of that slope

dissolves at HS6 resolution and is reported in the text for reference only.



(a) Precision targeting: tech-keyword share of new Entity List entries.

(b) Evasion resistance: mean aliases per sanctioned entity.

Figure 12: **Precision targeting and evasion resistance on the Entity List.** Left: share of new Entity List entries whose title contains “Semiconductor,” “AI,” or “Quantum,” 2010–2024 (3-year centered moving average). Right: mean number of aliases per sanctioned entity. Descriptive series; not load-bearing for the formal analysis.

## D.9 Robustness Panels for Section 2 Motivational Figures

This subsection collects the two robustness panels referenced from the Section 2 discussion of Figures 1 and 2. Figure 14 addresses the “selection in swing-state set” pushback: re-running the aggregate Chinese-input share excluding the three largest post-2018 contributors (Vietnam, Mexico, Korea) leaves the level lower by about one percentage point but preserves the trajectory and the post-2018 acceleration, confirming the pattern is broadly distributed across the 17-country set rather than driven by one or two outliers. Figure 15 is the ADB MRIO cross-validation of footnote 9: the OECD TiVA Leontief construction is the headline series in Figure 2(b), and an analogous decomposition built from the ADB Asia-Pacific MRIO yields the same direction (post-2018 rise in indirect Chinese value-added through swing states) at a smaller magnitude, the level gap reflecting ADB’s narrower country coverage (no European swing states).

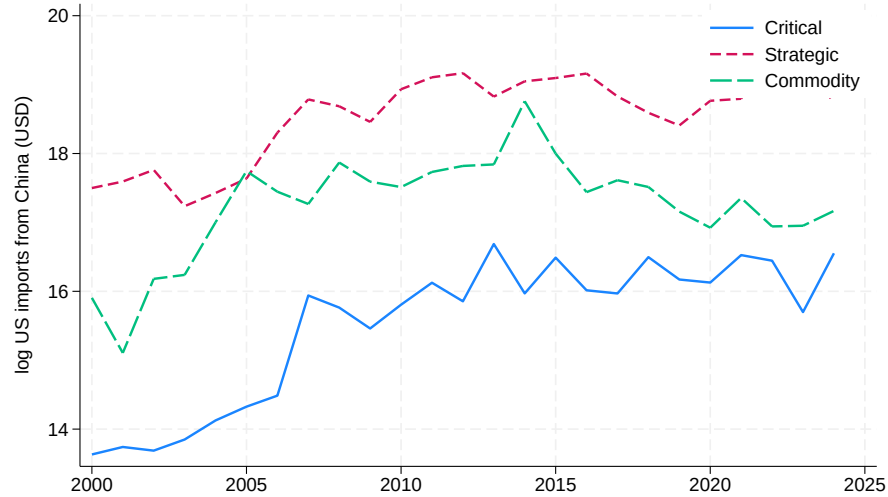


Figure 13: **Paradox regression: sector-aggregate series.** Sector-bucket trajectories of US imports from China plotted against the lagged Entity-List China-listings stock  $EL_{t-1}^{CHN}$ . The 3-sector aggregate loads sharply negative on the critical bucket; the within-HS6 analogue (Spec 2 in the text) is a precise null, so the series is reported for reference only.

## References

- Acemoglu, D., Carvalho, V. M., Ozdaglar, A., and Tahbaz-Salehi, A. (2012). The network origins of aggregate fluctuations. *Econometrica*, 80(5):1977–2016.
- Aghion, P. and Tirole, J. (1997). Formal and real authority in organizations. *Journal of Political Economy*, 105(1):1–29.
- Aiyar, S., Chen, J., Ebeke, C. H., Garcia-Saltos, R., Gudmundsson, T., Ilyina, A., Kangur, A., Kunaratskul, T., Rodriguez, S. L., Ruta, M., Schulze, T., Soderberg, G., and Trevino, J. P. (2023). Geo-economic fragmentation and the future of multilateralism. IMF Staff Discussion Note SDN/2023/001, International Monetary Fund.
- Alfaro, L. and Chor, D. (2023). Global supply chains: The looming “great reallocation”. NBER Working Paper 31661, National Bureau of Economic Research. Presented at Jackson Hole Economic Symposium.

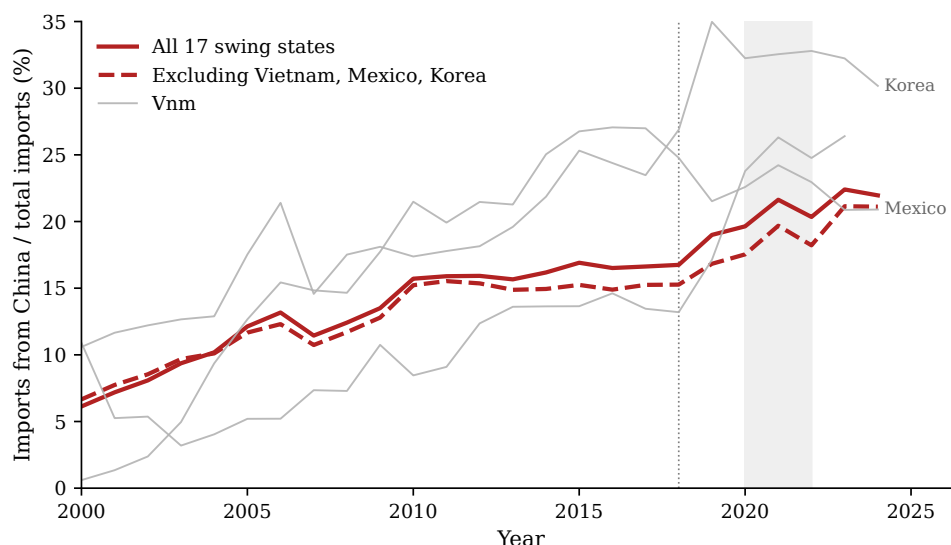


Figure 14: **Vertical Trap, drop-top-3 robustness.** Solid red: the swing-state aggregate Chinese-input share from Figure 1(a). Dashed red: the same aggregate excluding Vietnam, Mexico, and Korea—the three largest post-2018 contributors. Thin grey lines plot each excluded country’s individual share. The level falls by  $\approx 1$  percentage point in 2024 (21.96%  $\rightarrow$  21.11%); the trajectory and the post-2018 acceleration survive. Vertical dotted line marks the 2018 Section-301 onset; grey shading marks the COVID period (2020–2022).

Antràs, P. (2016). *Global Production: Firms, Contracts, and Trade Structure*. CREI Lectures in Macroeconomics. Princeton University Press, Princeton, NJ.

Atalay, E. (2017). How important are sectoral shocks? *American Economic Journal: Macroeconomics*, 9(4):254–280.

Autor, D., Dorn, D., Hanson, G., and Majlesi, K. (2020). Importing political polarization? the electoral consequences of rising trade exposure. *American Economic Review*, 110(10):3139–3183.

Bagwell, K. and Staiger, R. W. (1999). An economic theory of gatt. *American Economic Review*, 89(1):215–248.

Ballard, C. L., Shoven, J. B., and Whalley, J. (1985). General equilibrium computations of the marginal welfare costs of taxes in the United States. *American Economic Review*, 75(1):128–138.

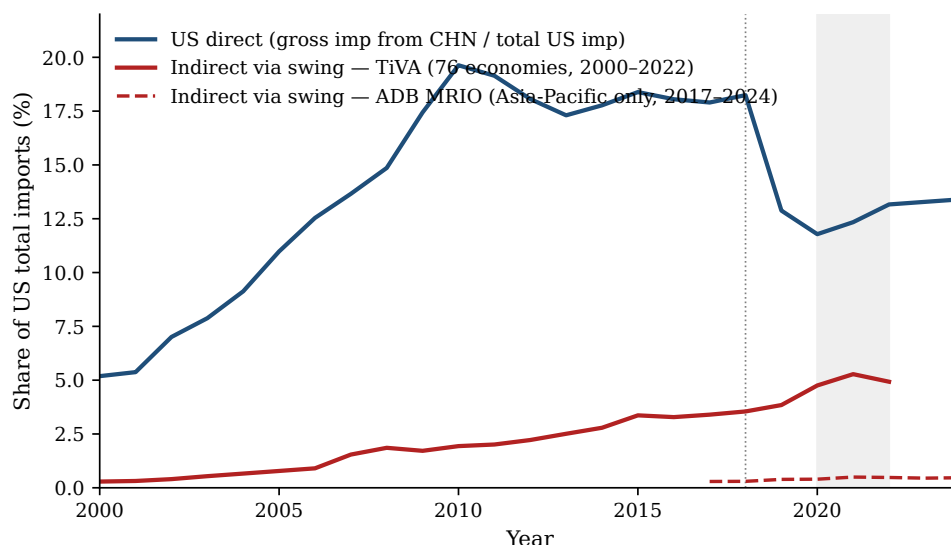


Figure 15: **Decoupling Divergence, ADB MRIO cross-validation.** Solid blue: US direct Chinese-import share (US imports from China / US total imports), identical to the headline series in Figure 2. Solid red: indirect channel constructed via OECD TiVA (76 economies, 2000–2022), reproducing Figure 2(b). Dashed red: indirect channel constructed via the ADB Asia-Pacific MRIO (2017–2024), restricted to the ten Asia-Pacific swing states present in the ADB panel. The direction matches the TiVA construction; the level gap reflects the narrower country coverage. Vertical dotted line marks 2018; grey shading marks COVID (2020–2022).

Baqae, D. R. and Farhi, E. (2024). Networks, barriers, and trade. *Econometrica*, 92(2):505–541.

Bernheim, B. D. and Whinston, M. D. (1986). Menu auctions, resource allocation, and economic influence. *Quarterly Journal of Economics*, 101(1):1–31.

Boehm, C. E., Flaaen, A., and Pandalai-Nayar, N. (2019). Input linkages and the transmission of shocks: Firm-level evidence from the 2011 tōhoku earthquake. *Review of Economics and Statistics*, 101(1):60–75.

Bown, C. P. (2020). The US–China trade war after two years: Insights for the Biden administration. PIIE Policy Brief 20–15, Peterson Institute for International Economics.

Brander, J. A. and Spencer, B. J. (1985). Export subsidies and international market share rivalry. *Journal of International Economics*, 18(1–2):83–100.

- Broda, C. and Weinstein, D. E. (2006). Globalization and the gains from variety. *Quarterly Journal of Economics*, 121(2):541–585.
- Caliendo, L. and Parro, F. (2015). Estimates of the trade and welfare effects of nafta. *Review of Economic Studies*, 82(1):1–44.
- Clayton, C., Maggiori, M., and Schreger, J. (2024). A theory of economic coercion and fragmentation. NBER Working Paper 33309, National Bureau of Economic Research. Revise-and-resubmit, *Journal of Political Economy*.
- Clayton, C., Maggiori, M., and Schreger, J. (2026). A framework for geoeconomics. *Econometrica*, 94(1):105–136.
- Correia, S., Guimarães, P., and Zylkin, T. (2020). Fast Poisson estimation with high-dimensional fixed effects. *Stata Journal*, 20(1):95–115.
- Dixit, A., Grossman, G. M., and Helpman, E. (1997). Common agency and coordination: General theory and application to government policy making. *Journal of Political Economy*, 105(4):752–769.
- Dixit, A. K. (1996). *The Making of Economic Policy: A Transaction-Cost Politics Perspective*. Munich Lectures in Economics. MIT Press, Cambridge, MA.
- Fajgelbaum, P. D., Goldberg, P. K., Kennedy, P. J., and Khandelwal, A. K. (2020). The return to protectionism. *Quarterly Journal of Economics*, 135(1):1–55.
- Farrell, H. and Newman, A. L. (2019). Weaponized interdependence: How global economic networks shape state coercion. *International Security*, 44(1):42–79.
- Feenstra, R. C. (1994). New product varieties and the measurement of international prices. *American Economic Review*, 84(1):157–177.
- Grossman, G. M. and Helpman, E. (1994). Protection for sale. *American Economic Review*, 84(4):833–850.

- Grossman, S. J. and Hart, O. D. (1983). An analysis of the principal-agent problem. *Econometrica*, 51(1):7–45.
- Handley, K. and Limão, N. (2017). Policy uncertainty, trade, and welfare: Theory and evidence for china and the united states. *American Economic Review*, 107(9):2731–2783.
- International Monetary Fund (2023). Geoeconomic fragmentation and foreign direct investment. In *World Economic Outlook: Navigating Global Divergences*, chapter 4. International Monetary Fund, Washington, DC.
- Johnson, R. C. and Noguera, G. (2012). Accounting for intermediates: Production sharing and trade in value added. *Journal of International Economics*, 86(2):224–236.
- Khan, S. M., Mann, A., and Peterson, D. (2021). The semiconductor supply chain: Assessing national competitiveness. CSET Issue Brief, Center for Security and Emerging Technology, Georgetown University.
- Koopman, R., Wang, Z., and Wei, S.-J. (2014). Tracing value-added and double counting in gross exports. *American Economic Review*, 104(2):459–494.
- Maskin, E. and Tirole, J. (1999). Unforeseen contingencies and incomplete contracts. *Review of Economic Studies*, 66(1):83–114.
- Mattoo, A., Freund, C., Mulabdic, A., and Ruta, M. (2023). Is US trade policy reshaping global supply chains? Policy Research Working Paper 10593, World Bank.
- Rambachan, A. and Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, 90(5):2555–2591.
- Rey, P. and Tirole, J. (2007). A primer on foreclosure. In Armstrong, M. and Porter, R., editors, *Handbook of Industrial Organization*, volume 3, chapter 33, pages 2145–2220. Elsevier.

- Rodrik, D. (2018). Populism and the economics of globalization. *Journal of International Business Policy*, 1(1–2):12–33.
- Rogoff, K. (1985). The optimal degree of commitment to an intermediate monetary target. *Quarterly Journal of Economics*, 100(4):1169–1189.
- Santos Silva, J. M. C. and Tenreyro, S. (2006). The log of gravity. *Review of Economics and Statistics*, 88(4):641–658.
- Schulze, T. and Xin, W. (2025). Demystifying trade patterns in a fragmenting world. IMF Working Paper 2025/129, International Monetary Fund.
- Segal, I. (1999). Contracting with externalities. *Quarterly Journal of Economics*, 114(2):337–388.
- Yi, K.-M. (2003). Can vertical specialization explain the growth of world trade? *Journal of Political Economy*, 111(1):52–102.